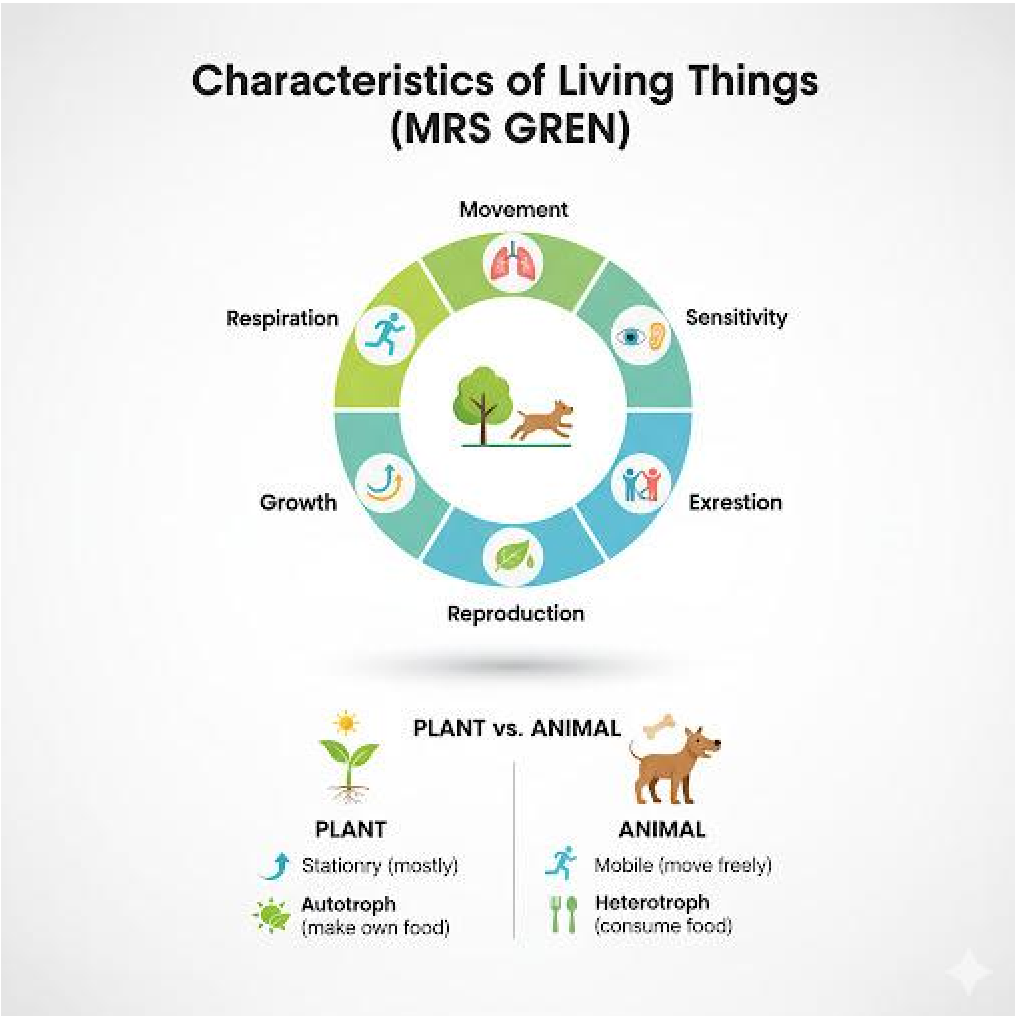
SS1 BIOLOGY LESSON NOTES

FIRST TERM

1. Recognizing Living Things
2. Classification of Living Things
3. Classification of living things: Monera, Protista, Fungi, Plantae, animalia
4. The cell.
5. **Midterm Examination**
6. Cell and its environment
7. **Midterm break**
8. Some Properties and Functions of the Cell
9. Respiration
10. Growth and irritability

WEEK 1: RECOGNIZING LIVING THINGS



Meaning of living things

In biology, living things—also known as organisms—are entities that have life and exhibit certain characteristics.

Characteristics of living things

Everything in nature can be classified into two groups: living and non -living things.

The living things can be distinguished from their nonliving counterparts through the following characteristics observable in all living things:

1. **Movement:** Animals can move from place to place on their own in search of food. Higher plants move certain parts of their body in response to growth or external stimuli
2. **Respiration:** In order to perform the numerous life processes, living things need much energy. The energy can only be obtained when the organism respires. Therefore, respiration is the oxidation of food substances in the presence of oxygen to produce energy with carbon (iv) oxide and water released as by products.
3. **Nutrition:** The act of feeding is called nutrition. All organisms need food to carry out their biological activities. Green plants can manufacture their food. Hence, they are autotrophs while animals are dependent on plants for their food, so they are heterotrophs.
4. **Irritability**: Is the ability of living things to respond to external and internal stimuli in order to survive. External stimuli may be light, heat, water, sound or chemical substances.
5. **Growth:** this is the tendency of organisms to increase irreversibly and rapidly in length and size and in mass. The essence of growth is to enable organisms to repair or replace damaged or old tissues in their bodies. The food eaten by an organism provides the basis of growth.
6. **Excretion:** Toxic waste products of metabolism & other unwanted materials have to be eliminated to ensure proper functioning of the bodies of organisms. Such wastes include water, carbon (iv) oxide etc.
7. **Reproduction:** Is the ability of a living organism to give birth to young ones (off springs). The essence is to ensure life continuity. Reproduction can be sexual (involving two organisms) or asexual (involving one organism)
8. **Life span / death:** Every organism has a definite and limited period of existence. Life, for all organism has five main stages, namely: origin (birth), growth, maturity, decline and death.
9. **Competition:** Living things tend to struggle for the basic things of life in order to survive. Hence, they compete for food, water, light, mates and space.
10. **Adaptation:** To survive, every organism possesses ability to get used to change in its environment.

DIFFERENCES BETWEEN PLANTS AND ANIMALS

	PLANTS	ANIMALS
1.	Undergo passive movement.	Undergo free or active movement with well-developed organs of locomotion.
2.	Gaseous exchange takes place through the entire body.	Gaseous exchange is through special organs.
3.	Green plants photosynthesize i. e. they are autotrophs.	Animals do not photosynthesize i. e. they are heterotrophs.
4.	They exhibit slow response to stimuli.	They exhibit fast response to stimuli.
5.	Growth is apical and indefinite (continuous).	Growth is uniform and definite (limited).
6.	No specialized sense organs.	Possess specialized sense organs.
7.	No specialized excretory systems.	They have special and well developed excretory systems.
8.	Cell has rigid non-living cellulose cell wall which provides mechanical support.	Have thin, flexible cell membrane. Mechanical support is provided by external exoskeleton or internal endoskeleton.
9.	They store food (carbohydrates) as starch except fungi which store food as glycogen.	They store carbohydrates as glycogen

All living organisms can be generally classified as plants or animals. However, plants can be distinguished from animals in the following ways:

Organization of life

Life on Earth is organized in a hierarchy of complexity, ranging from the tiniest molecules to the vast ecosystems. Understanding the levels of organization in life helps us appreciate the intricate interactions that sustain living organisms. In this lesson, we will explore the various levels of organization with relevant examples.

Cells: Cells are the basic structural and functional units of life. Example: Human body cells, such as muscle cells and nerve cells.

Tissues: Tissues are groups of similar cells that work together to perform a specific function. Example: Muscle tissue contracts to create movement, while nervous tissue transmits signals.

Organs: Organs are composed of different tissues working together to carry out specific functions. Example: The heart, composed of cardiac muscle tissue, blood vessels, and connective tissue, pumps blood.

Systems: Organ systems consist of multiple organs working together to perform complex functions. Example: The digestive system includes organs like the stomach, liver, and intestines, which work together to process and absorb nutrients.

Assignment

1. Explain the differences between growth in **plants** and animals
2. Define the following characteristics of living things (a) respiration (b) irritability (c) excretion
3. What are the similarities between living and non-living thing?

WEEK 2: CLASSIFICATION OF LIVING THINGS 1

MEANING AND IMPORTANCE OF CLASSIFICATION

Classification is the process of grouping things based on shared characteristics. In Biology, classification is used to organize and understand the vast diversity of life on Earth.

Importance of classification

- Makes it easier to study and understand living things.
- Helps scientists communicate effectively about organisms.
- Reveals evolutionary relationships between organisms.

Meaning of taxonomy

Taxonomy is the science of classifying living things. It involves identifying, naming, and grouping organisms. The term "taxonomy" comes from the Greek words "taxos" (arrangement) and "nomos" (law or order). The system of classification of living things used today is based on that introduced by a Swiss scientist called Carolus Linnaeus. He arranged living things into seven major groups. This are:

Kingdom

Phylum (animal) or division (plant)

Class

Order

Family

Genus

Species

The basic unit of classification of living things is the species.

Binomial system of nomenclature

Carolus Linnaeus also introduced a system of naming living things which is popularly used by scientists today. In this system, each organism is given two names. The first name is the generic name and the second name is the specific name . This scientific names are written in italics or are underlined. Examples scientific names of some organisms are given below.

Man _____ *Homo sapiens*

Lion _____ *panthera leo*

Maize _____ *zea Mays.*

Kingdoms

Carolus Linnaeus classified all living things into two major kingdoms namely plant kingdom and animal kingdom. Under this arrangement, lot of one celled organisms could not fit in properly. Many biologist then decided to place living things into five kingdoms. These are: **monera, protista, fungi, plantae and animalia.** In classification of living things, virus specifically could not fit into any of the five kingdoms. As a result of this, it is treated separately.

Virus

Virus is a microscopic organism which cannot be seen by an ordinary microscope but with electron microscope.

Characteristics of virus

- Virus is microscopic in nature
- It possesses either RNA or DNA.
- It cannot reproduce by binary fission
- It does not have structures used in synthesis of protein.
- It does not respire, geed, excrete, etc.

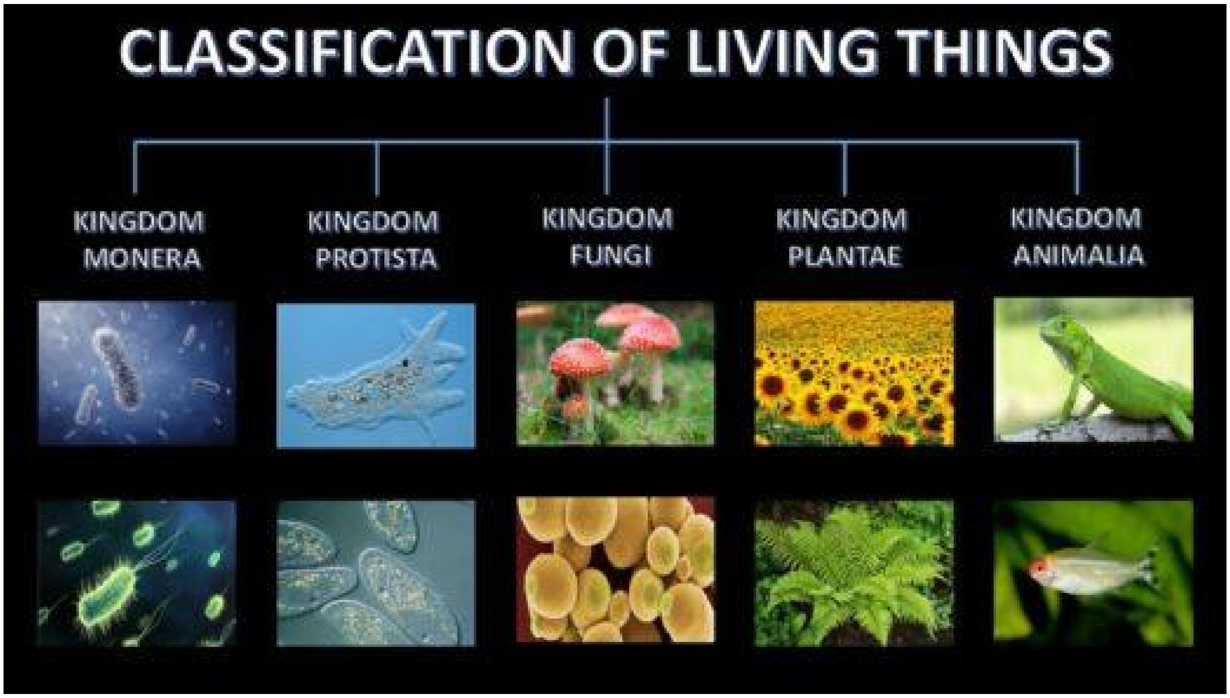
Virus as living things

- Virus can reproduce when present in another living cell.
- It possesses characteristics which can be transmitted from one generation to the next.

Viruses as a non-living things

- When a virus is extracted from a living cell and placed in a non-living medium, it assumes a crystalline form and thus becomes non-living things
- Virus cannot respire, excrete or responds to stimuli.

KINGDOM MONERA, PROTISTA, AND FUNGI.



Kingdom Monera

This group consists of simplest living organisms (bacteria, blue-green anabaena).

- They are microscopic single-celled.
- The cell wall does not contain cellulose. It is made up of protein and fatty materials.
- They have no definite nucleus. Nucleus lack nuclear membrane and DNA are scattered in the cytoplasm.
- They lack most cell organelles except the ribosome.
- Reproduction is asexual by binary fission.

Kingdom Protista

- They are unicellular organisms.

- The organisms are all eukaryotes i. e. cell have definite nucleus.
- Most protists are aquatic organisms.
- They move either by cilia, flagella or pseudopodia.
- Some are free living while few are parasitic. Protists can be broadly divided into two groups;

    1. Protozoa: animal-like protists e.g. amoeba, paramecium, plasmodium, trypanosome.
    2. Protophyta: Plant-like protists e. g. Diatoms, chlorella, chlamydomonas.

Note: Euglena is a protist with plant and animal like features.

Kingdom Fungi

- They are non-green organisms which do not photosynthesize (lack chlorophyll).
- All fungi except slime moulds are non-motile.
- They have rigid cell wall made up of chitin and polysaccharides.
- They reproduce asexually by producing spores and sexually as well.
- Most of them are saprophytes while some are parasites.
- They lack true roots, stem and leaves.
- Few are unicellular (yeast) while most are multicellular (rhizopus, mushroom).
- Here's a lesson evaluation, assignment, and conclusion based on the provided lesson note:

Kingdom Plantae

The familiar green plants belong to this kingdom. Plant is non- motile multicellular organism whose cells have cellulose cell and chloroplast. The chlorophyll in the chloroplast enables plant to make their own food by photosynthesis. The plant kingdom include

1. Thallophyta
2. Bryophyte

3. Tracheophyta

Thallophyta

Example of a thallopyta is Algae

Characteristics

1. They have no roots, stem and leaves
2. Chlorophyll is present

Bryophyte: Example of a bryophyte are moss and liverwort

Characteristics

1. No true root stem or leaf
2. Have rhizoids

Tracheophytes: The Tracheophytes is made up of;

1. **Pteridophyte (fern):**Presence of true roots, stem and leaves

2. **Spermatophytes:** Produce pollens and form seeds.

The spermatophyte is divided into

A. **Gymnosperms:** Have cones and most members possess needle leaves. They do not have flowers, they have naked seeds

B. **Angiosperms:** Have flowers and produce seeds inside fruits.
They are flowering plants. Angiosperm is divided into

Monocotyledonous plants

1) They possess one seed leaf or cotyledons, possess

2) parallel venation with narrow leaved plants, no tree except the palms

Dicotyledonous plants

1. They possess two seed leaves or cotyledons,
2. possess net veined and they broad leaved plants,
3. They are herbs, shrubs and tree plants

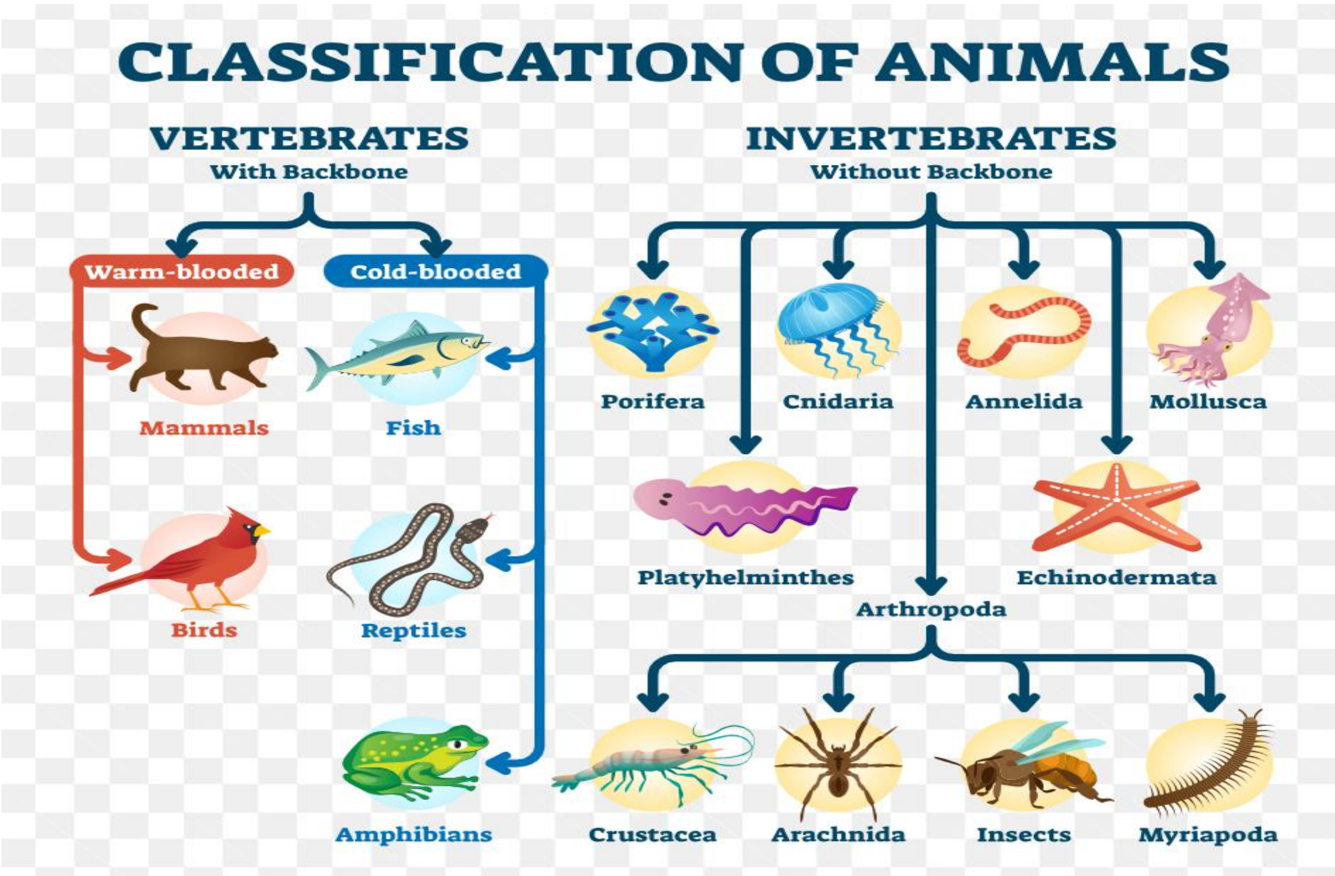
Lesson Evaluation

- What is classification in biology?
- State the importance of classification.
- Who introduced the system of classification used today?
- Use a flow chart and describe kingdom Plantae.
- What are the seven major groups of classification?
- What is the basic unit of classification?

Assignment

- Explain the binomial system of nomenclature with examples.
- Describe the characteristics of viruses and why they are considered both living and non-living things.
- Compare and contrast the kingdoms Monera and Protista.

WEEK 3: CLASSIFICATION OF LIVING THINGS II



Kingdom Animalia

The kingdom Animalia is divided into vertebrate and invertebrate

The invertebrate include:

Phylum Protozoa e.g. Amoeba, Plasmodium, Trypanosome

Characteristics

1. Unicellular, acellular or non-cellular
2. They are mostly microscopic
3. Some are free -living in water while some are parasitic in animals

Phylum Porifera e.g. Sponges

Characteristics

1. Multicellular, Sessile, and largely aquatic
2. Single body cavity with several holes
3. Body with two layers of cells
4. Skeleton is calcareous or siliceous
5. No nervous system 6. Reproduction is asexual by budding

Phylum Coelenterata e.g. Hydra, Sea anemone, Jelly fish

Characteristics

1. They are simple and Multicellular.
2. They are aquatic
3. They have two body layers
4. Organs are absent
5. Possession of nematoblast called stinging cells.
6. Asexual reproduction is by budding while sexual reproduction is by fusion of gametes.

Phylum Platyhelminthes e.g. the flat worms Examples include liver fluke, tape worms, planarian

Characteristics

1. Flat and thin bodies
2. They are bilaterally symmetrical
3. They possess three body layers
4. Nervous, excretory and reproductive systems are present
5. Members are mostly hermaphrodites.
6. Members are both free -living and parasitic e.g. blood fluke, liver fluke planaria
7. The body has a single opening –the mouth

Phylum: Nematoda e.g. (The round worm)

Characteristics

1. Long, slender and cylindrical bodies
2. Members have pointed ends

3. Presence of three body layers
4. Mouth and anus present
5. Body covered with cuticle 6. They are bilaterally symmetrical
7. Sexes are separate
8. Members are free living and parasitic Examples: Ascaris, Hookworm, Tapeworm, Guinea worm, Filaria worm, Onchocerca

Phylum: Annelida

Characteristics

1. Cylindrically elongated body
2. Three body layers
3. Body divided into smaller segments
4. Two openings of the alimentary canal
5. **Presence of nervous, excretory, reproductive, and circulatory systems.**
   Examples: Nigerian earthworms, *Hirudo*, *Lumbricus*.

Phylum: Mollusca

Characteristics

1. Soft and unsegmented body
2. Three body layers
3. Body covered with calcareous shell
4. Prominent head and foot
5. Eyes and tentacles present on the head
6. The body is short and composed of a head, a ventral muscular foot and visceral humps

Examples: Snail slugs

Phylum: Arthropoda

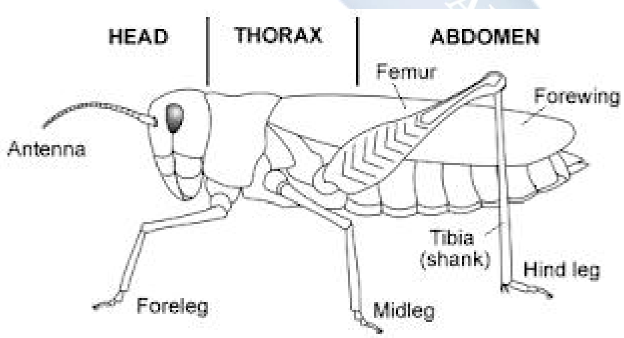
Kingdom arthropoda is sub divided into five classes

- Insecta
- Myriapoda
- Arachnida
- Annelida
- Crustacea

Class Insecta:

Characteristics

1. A pair of jointed antennae
2. A pair of compound eyes
3. Body divisions, head, thorax and abdomen
4. Thoracic division – prothorax, mesothorax and metathorax
5. A pair of legs per thoracic division
6. Respiration by means of trachea
7. Insect s may have wings
8. They undergo metamorphosis Examples: Cockroach, Housefly, Butterfly, Grasshopper



Class Myriopoda:

1. Characteristics They have two body regions – Head with fused trunk and abdomen
2. The abdomen consists of many limb bearing segments
3. A pair of simple eye, jaws and short antennae on the head

4. Breathing by means of trachea The class has two sub classes i.e

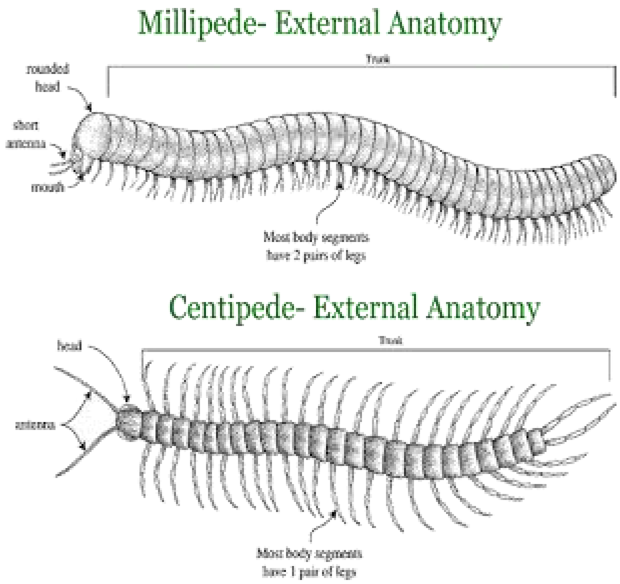
a. Chiplopoda -Centipede

i. A pair of appendages on the segment

b. Sub class – Diplopoda – Millipede

i. 2 Pairs of appendages per segment

ii. They are herbivorous



Structure of millipede and centipede

Class Arachnida

Characteristics

1. Body divisions -prostoma and opisthosoma (abdomen)
2. Presence of chelicerae and pedipals
3. Eight simple eyes
4. Eight walking legs
5. Respiration by means of lung book Examples: Spider, Scorpions.

Class Crustacea

Characteristics

1. Body regions cephalothorax and abdomen
2. A pair of antennae for feeling and antennules for smelling
3. pairs of walking legs
4. A pair of stalked compound eyes
5. Respiration by means of gills. Examples: Prawns, Shrimps, Crab, Water flea, Barnacle

Phylum Echinodermata

Characteristics

1. Aquatic in habitat
2. Radially symmetrical
3. Triploblastic
4. They are coelomate
5. Calcareous exoskeleton
6. Suckers and tube feet for locomotion
7. Separate sexes Examples – Star fishes

VERTEBRATES

Vertebrate are divided into five classes namely.

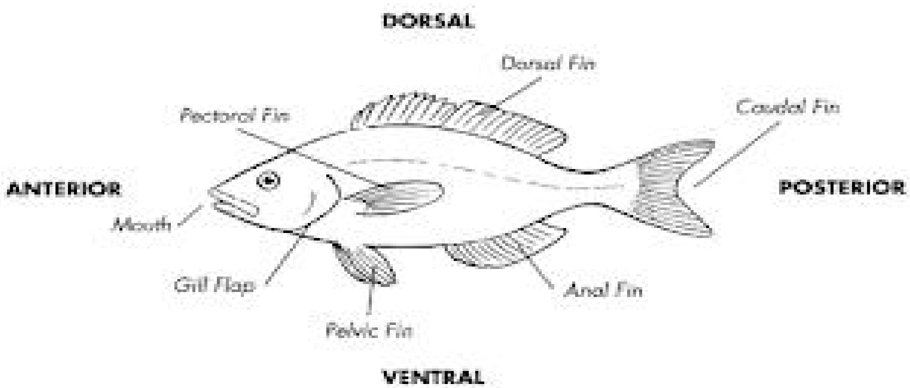
1. Pisces
2. Eves
3. Amphibians
4. Reptiles
5. Mammals

Class Pisces- Fishes

Characteristics

1. Visceral cleft persist as gill in adult
2. Paired limb i.e pectoral and pelvic fins

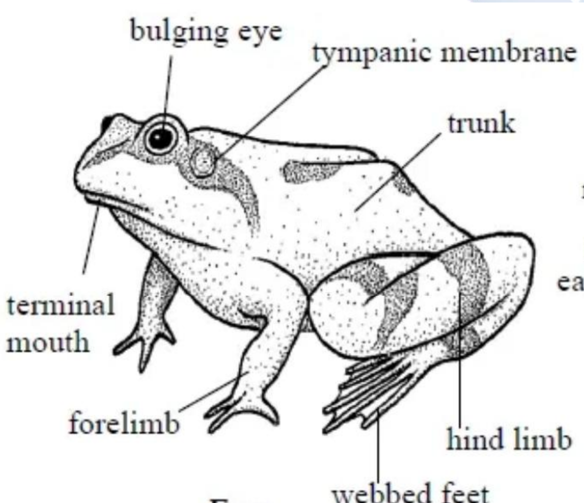
3. Well-developed lateral lines
4. Endo and exoskeleton with cycloid and placoid scales
5. External fertilization Example: Tilapia, Dog fish



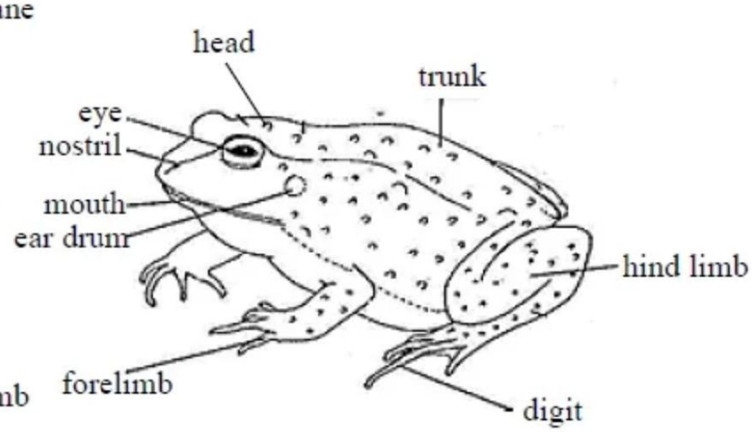
Class Amphibians

Characteristics

1. Presence of pentadactyl limbs
2. Soft skin without scale
3. Gills at tadpole stage but lungs in the adult
4. Presence of external middle ear without external pinna
5. Fertilization is external Examples – Toad, frogs, strew and salamander



Frog

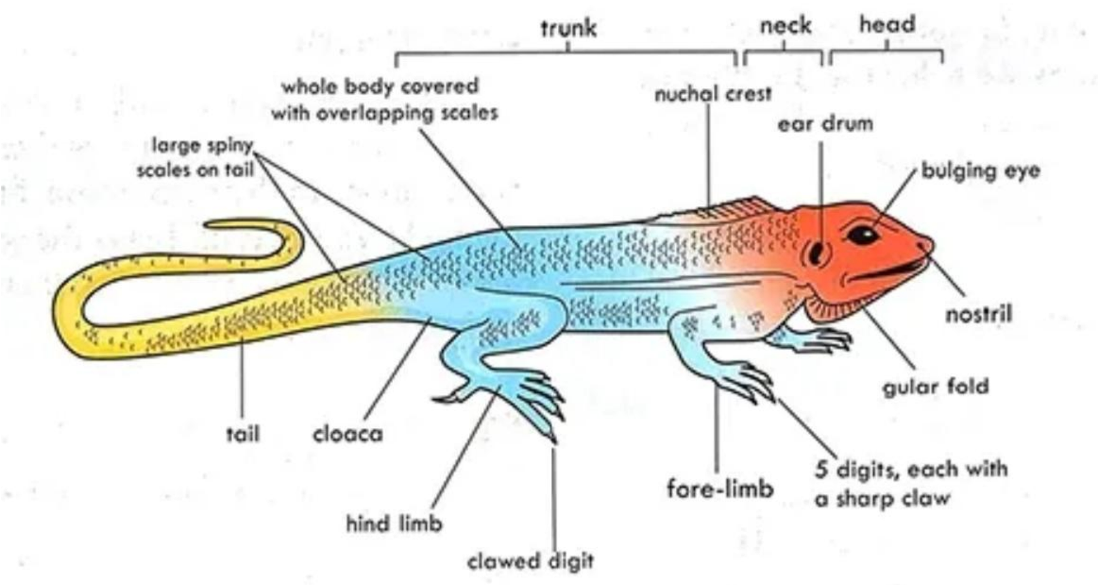


Toad (Bufo regularis)

Class Reptalia (Reptiles)

Characteristics

1. Pentadactyl limb
2. Dry skin with scale
3. Lungs as the respiratory organs
4. Eggs with yolk and no larva stages
5. Fertilization is internal Examples – Lizards, Snakes, Crocodile, Tortoise

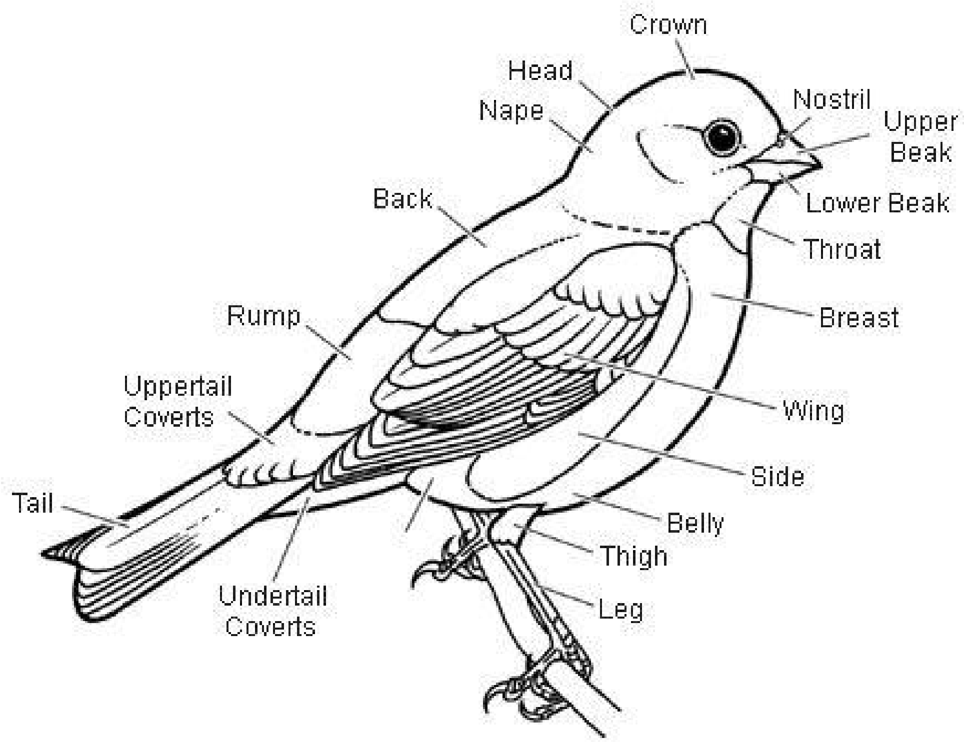


Aves (Birds)

Characteristics

1. Pentadactyl limb with the fore limb modified into wings
2. Warm blooded animal
3. Feathers on the skin
4. Presence of beak but teeth is absent
5. Scales on the legs and feet

6. Lungs as respiratory organ
7. Eggs with much yolk
8. No larva stage
9. Fertilization is internal Examples – Pigeon, Domestic fowl



Class: Mammalia

Characteristics

1. Pentadactyl limbs
2. Warm -blooded animal
3. Hair with sweat gland on their skin
4. Presence of mammary gland that produce milk for the young ones
5. Four- chambered hearts
6. Eggs are small and young ones develop inside their mother
7. Fertilization is internal

8. They are viviparous

Evaluation:

1. Describe kingdom animalia
2. Explain different phyla in kingdom animalia.
3. Describe the class under kingdom animalia.

Assignment:

Draw the structure of a tilapia fish, lizard, grasshopper and earth worm

WEEK 4: THE CELL

A cell is the basic structural and functional unit of all living organisms. It is the smallest unit of life that can carry out the fundamental processes necessary for an organism's existence. Cells can vary widely in structure, function, and size, but they share common features such as a cell membrane, genetic material (DNA or RNA), and the ability to undergo reproduction and metabolism. The diversity of cells contributes to the complexity and variety of living organisms.

Forms in which cells exist

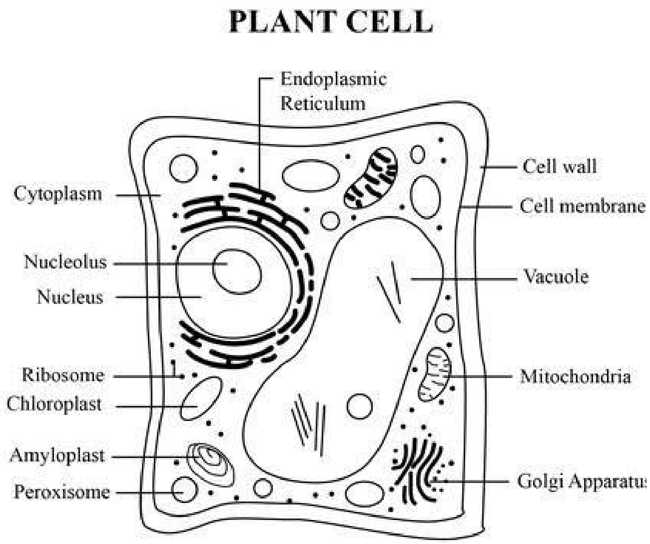
Cells of living organisms exist in different forms, they include

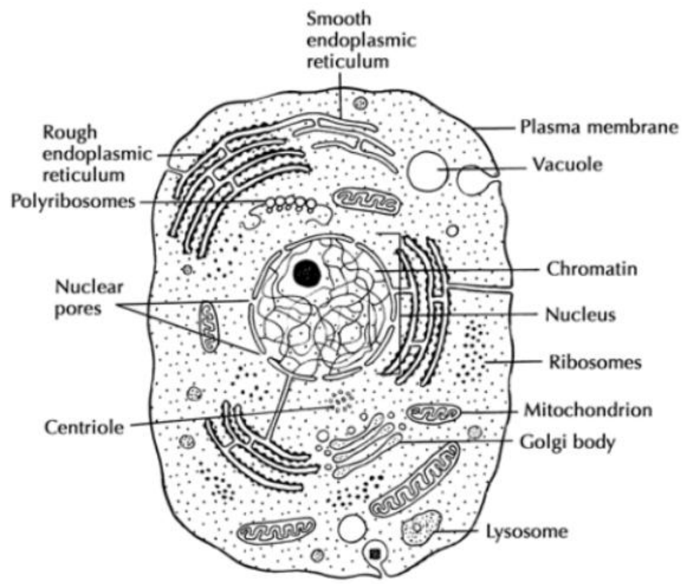
1. **Unicellular/single or free-living:** Unicellular organisms consist of a single cell as their entire body. These organisms carry out all the necessary life processes within a single, microscopic cell. Examples of unicellular organisms include: Bacteria, Archaea, Protists, Yeasts
2. **As colony:** A colony of cells refers to a group of individual cells that live together but may not be structurally connected. Each cell in the colony is capable of functioning independently and carries out its own life processes. This is common in certain microorganisms. Examples include, Bacteria, Algae, Protozoa
3. **As filament:** Cells organized in filaments typically form a chain or thread-like structure, and this arrangement is commonly observed in certain types of organisms. Examples include: Cyanobacteria, Spirogyra (Green Algae), Streptomyces (Bacteria)
4. **As part of a living organism:** Cells are the fundamental units of life and are organized into tissues, organs, and systems within living organisms. The following is an overview of how cells function as part of a multicellular organism:

Plants and Animal Cell

Structure of plants and animals cell

Plant and animal cells share similarities in basic structure but have key differences. Both contain a cell membrane, nucleus, and cytoplasm. Plant cells have a rigid cell wall and chloroplasts for photosynthesis, while animal cells lack these features. Animal cells may have centrioles, absent in most plant cells, aiding in cell division.





Function of the cell components

Cells have various components, each with specific functions:

1. Cell Membrane: Regulates what enters and exits the cell, providing a protective barrier.
2. Nucleus: Contains genetic material (DNA) and controls cellular activities.
3. Cytoplasm: Gel-like substance where organelles are suspended and where many cellular processes occur.
4. Endoplasmic Reticulum (ER): Rough ER synthesizes proteins, while smooth ER is involved in lipid synthesis and detoxification.
5. Golgi apparatus: Modifies, packages, and transports proteins and lipids.
6. Mitochondria: Generates energy through cellular respiration.
7. Vacuole (in plant cells): Stores nutrients and waste products.
8. Ribosomes: Synthesize proteins using information from the DNA.
9. Lysosomes: Contain enzymes for digestion and waste removal.
10. Cytoskeleton: Provides structural support and helps in cell movement.

11. Centrioles (in animal cells): Aids in cell division.

SIMILARITIES AND DIFFERENCES BETWEEN PLANTS AND ANIMAL CELL

Similarities between plant and animal cells

Plants and animal cells share similarities in basic structures such as

1. The presence of a cell membrane, nucleus, and organelles like mitochondria and endoplasmic reticulum.
2. Both cell types also have a similar genetic material, DNA, housed in the nucleus. However, plant cells have unique features like a cell wall, chloroplasts for photosynthesis, and a large central vacuole, which are not present in animal cells.

Differences between plants and animals cell

The following are some major difference between plant and animal cells.

Feature	Plant Cell	Animal Cell
Cell wall	Present (composed of cellulose)	Absent
Chloroplasts	Present (for photosynthesis)	Absent
Vacuoles	Large central vacuole	Smaller and fewer vacuoles
Shape	Fixed, often rectangular	Variable in shape
Lysosomes	Less common, may have similar structures	Common, present as true lysosomes
Centrioles	Usually absent	Present, involved in cell division
Morality structure	Typically absent	Cilia, flagella, or other motility structures

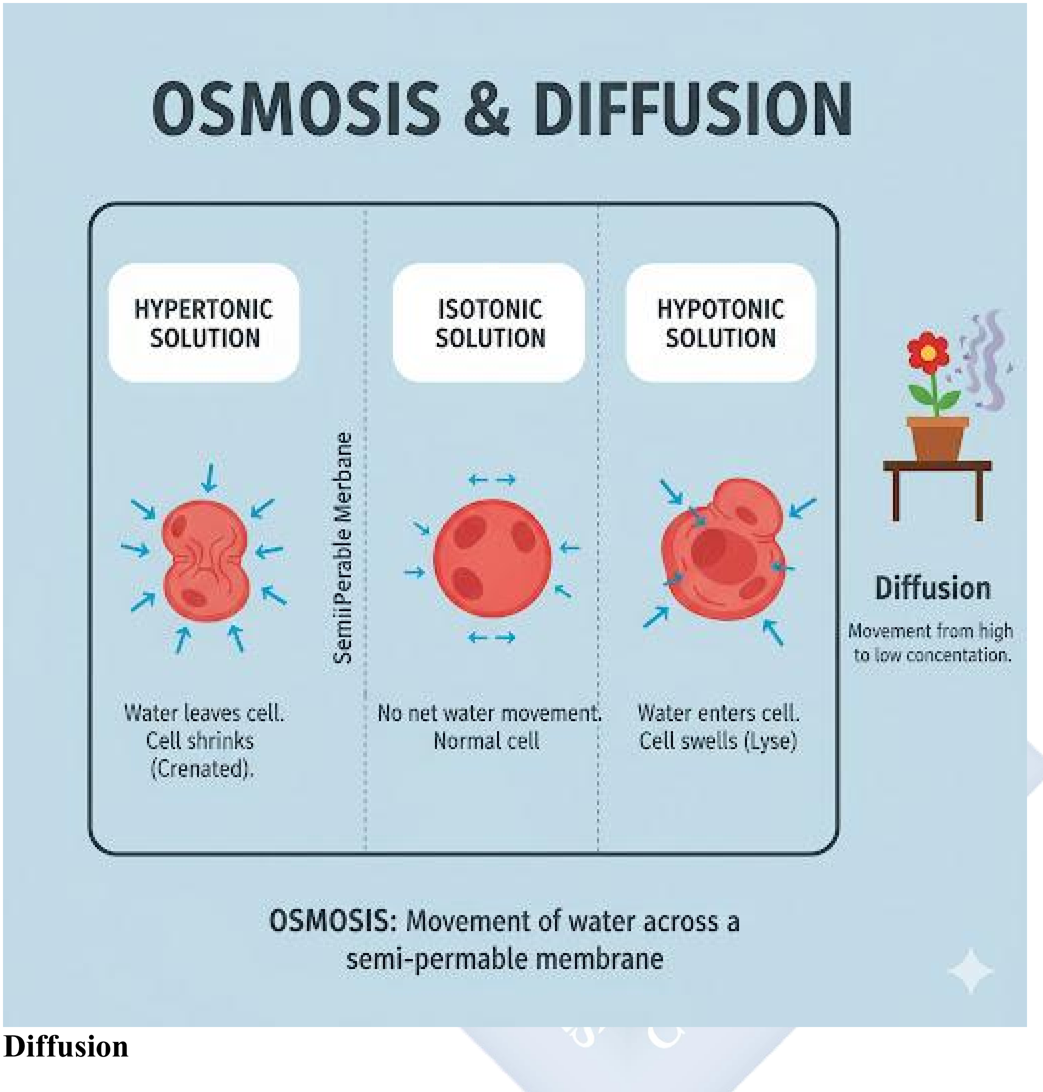
Nucleus position	Usually centrally located	Can be centrally or peripherally located

Assignment:

1. Draw and label the following

- Euglena and amoeba
- Chlamydomonas and Volvox
- Spirogyra

WEEK 6: CELL AND ITS ENVIRONMENT



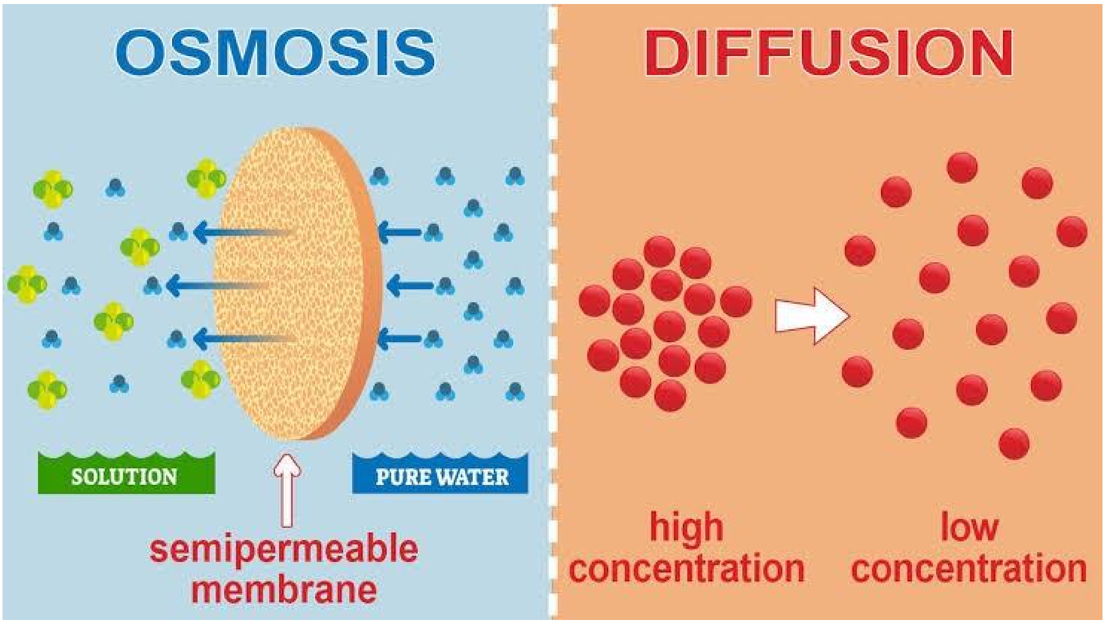
Diffusion

Diffusion is the spontaneous movement of particles (atoms, ions, or molecules) from an area of higher concentration to an area of lower concentration. This movement occurs down a concentration gradient, with the goal of achieving equilibrium, where the concentration becomes uniform throughout the available space.

Process of Diffusion

The process of diffusion involves the random motion of particles. It can occur in gases, liquids, or solids. The speed of diffusion depends on factors such as temperature, particle size, and the medium through which diffusion is taking place.

In gases and liquids, diffusion is facilitated by the constant motion of particles. In solids, diffusion can occur through the movement of atoms or molecules within the structure of the material.



Significance of Diffusion

1. Nutrient and Gas Exchange: Diffusion is crucial for the exchange of gases (like oxygen and carbon dioxide) in biological systems. It also facilitates the movement of nutrients into cells and waste products out of cells.
2. Biological Processes: Diffusion plays a fundamental role in biological processes such as respiration and photosynthesis. In these processes, gases move in and out of cells to support metabolic activities.
3. Cellular Uptake of Substances: Cells utilize diffusion to take up essential molecules. For example, nutrients can diffuse across cell membranes to reach cells.
4. Chemical Reactions: Diffusion helps in mixing substances, allowing for chemical reactions to occur. In a homogeneous mixture, reactants can come into contact more frequently, facilitating reactions.
5. Reversibility

Osmosis

Osmosis can be defined as the movement of water molecules from a region of lower concentration to the region of higher concentration through a semi-permeable membrane.

A permeable membrane allows molecules to pass through it freely while a selectively permeable membrane only allows certain molecules to pass through it. Osmosis will only occur when a semi-permeable membrane separates weak and strong solutions.

Living cells may find themselves in any of the following situation:

- When the fluid surrounding the cell is more concentrated than the inside of the cell, the surrounding fluid is said to be **hypertonic** to the content of the cells. A net movement of water molecules out of the cell into the surrounding fluid occurs and causes the cell to shrink. This process is known as Exosmosis.
- When the fluid surrounding the cell is less concentrated than the inside of the cell, the surrounding fluid is said to be **hypotonic** to the content of the cell. There is a net movement of water molecules from the surrounding fluid into the cells. This process is known as Endosmosis.
- When the surrounding fluid and the cell concentration have the same concentration, they are said to be **Isotonic**. A net movement of water molecule in and out of the cells does not occur.

Importance of osmosis

1. It aids the absorption of water from the soil into the vacuole of the root hairs
2. It aids the movement of water from the root hairs into the cells of other parts of the plants
3. It helps to control the opening and closing of the stomata pores
4. It gives turgidity to the plant cells i.e. it gives support.
5. It aids intracellular movement of water in animals
6. It aids reabsorption of water from the kidney tubules into the blood
7. It causes haemolysis of red blood cells.

Plasmolysis

Plasmolysis is generally reversible if the plant cell is placed in a hypotonic (lower solute concentration) environment. In a hypotonic solution, water will move back into the cell, and the protoplast will expand, restoring turgor pressure and the cell's normal state.

Hemolysis

Hemolysis is the rupture or destruction of red blood cells. This can occur when red blood cells are exposed to a hypotonic solution, causing water to enter the cells and potentially burst them.

Turgidity: Turgidity refers to the state of a cell when it becomes swollen and rigid due to the absorption of water. This is often observed in plant cells when they are in a hypotonic environment.

Flaccidity: Flaccidity is the opposite of turgidity. It occurs when a cell loses water and becomes limp, causing it to lose its rigidity. This often happens in plant cells in a hypertonic environment.

Osmoregulation: Osmoregulation is the process by which organisms regulate the concentration of water and solutes within their cells to maintain internal balance. It involves various mechanisms, including osmosis and active transport.

Active Transport: Active transport is a process that requires energy (usually ATP) to move molecules against their concentration gradient, from an area of lower concentration to an area of higher concentration. This mechanism is essential for maintaining concentration gradients in cells.

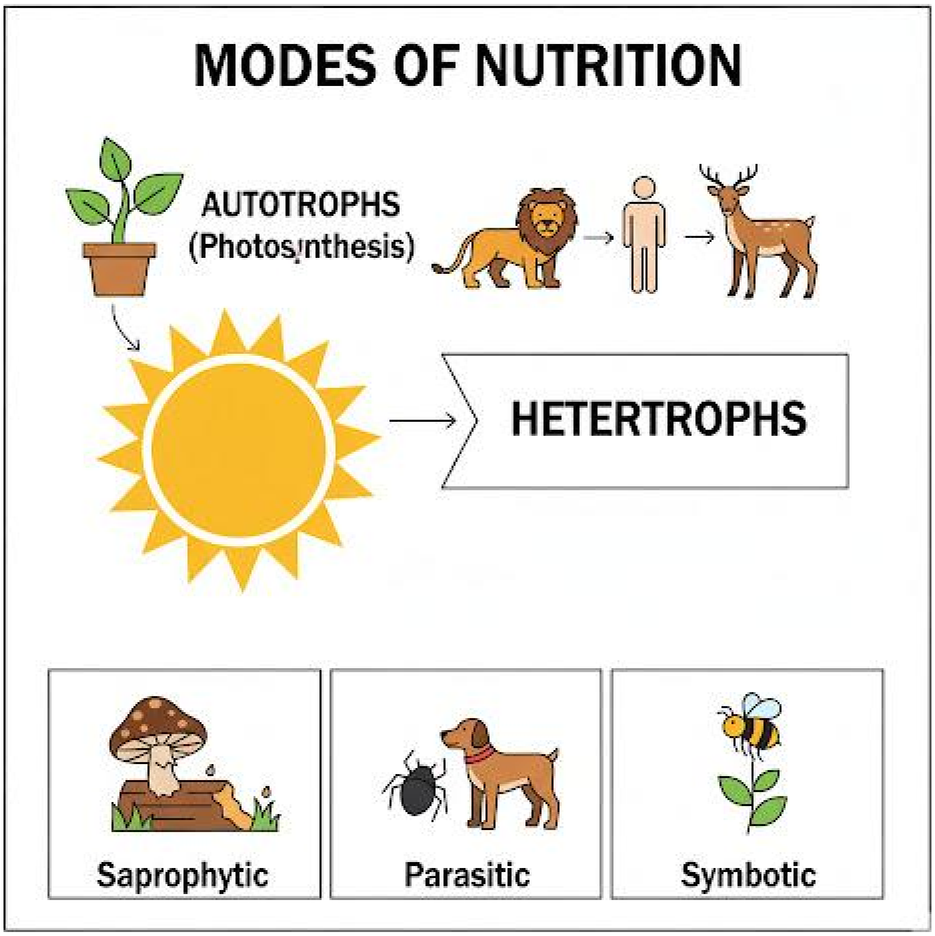
Evaluation

1. Explain the following terms: Hypertonic, Hypotonic and Isotonic solution
2. State the differences between osmosis and diffusion
3. Define diffusion and state factors that can affect it.
4. How does endosmosis lead to turgidity?
5. Plasmolysis results from exosmosis. Explain.
6. Of what importance is diffusion to life?

ASSIGNMENT

1. Which structures must be present in a cell for osmosis to take place? A. cell (sap) vacuole and cell wall B. cell wall and cell membrane C. chloroplast and cytoplasm D. cytoplasm and cell membrane
2. The scent from a bunch of flowers spreads throughout a room. How does the scent spread? A. by conduction B. by diffusion C. by osmosis D. by transpiration
3. Which of the following environmental conditions is ideal for plant cells to remain turgid? A. Hot, dry weather B. Cold, dry weather C. Cool, humid weather D. Windy weather
4. Osmosis occurs through a membrane that can be _____ permeable A. Fully B. slowly C. differentially D. freely
5. Which of the following processes takes place when a plant cell is put in a hypotonic solution? (a) water moves into the cell and the cell bursts (b) water leaves the cell and the cell becomes flabby (c) water moves into the cell and the cell becomes turgid (d) the cell becomes plasmolysed.

WEEK 8: PROPERTIES AND FUNCTIONS OF CELLS (NUTRITION)



Macro and micro nutrients

Macro nutrients are essential nutrients that the body requires in relatively large amounts, including carbohydrates, proteins, and fats. They provide the energy necessary for bodily functions.

Micro nutrients are nutrients needed in smaller quantities, including vitamins and minerals. These play crucial roles in various physiological processes and overall health.

Mode of nutrition

The mode of nutrition refers to how organisms obtain and consume nutrients for their growth and metabolism. There are two main modes of nutrition:

1. Autotrophic Nutrition: Organisms capable of synthesizing their own food from inorganic substances use autotrophic nutrition. Plants, algae, and some bacteria perform photosynthesis or chemosynthesis to produce their own organic molecules.

Photosynthesis is the biological process by which green plants, algae, and some bacteria convert light energy into chemical energy stored in the form of glucose or other organic compounds. It takes place in the chloroplasts of plant cells and involves the use of sunlight, carbon dioxide, and water.

Processes:

1. Light Absorption: Pigments in chloroplasts, such as chlorophyll, absorb light energy.
2. Water Splitting: Water molecules are split, releasing oxygen and providing electrons and protons.
3. Carbon Fixation: Carbon dioxide from the air is incorporated into organic molecules (initially into a three-carbon compound during the Calvin cycle).
4. Glucose Synthesis: Through a series of chemical reactions, glucose and other organic compounds are synthesized.

Conditions:

1. Light: Photosynthesis requires light, usually provided by sunlight. Light is absorbed by chlorophyll to initiate the process.
2. Carbon Dioxide: Plants take in carbon dioxide from the air during photosynthesis.
3. Water: Water is crucial for the initial reactions, providing electrons and protons. It is absorbed by plant roots from the soil.

Significance:

i. Oxygen Production: Photosynthesis is a primary source of oxygen, crucial for the survival of many organisms.

ii. Energy Storage: The process stores solar energy in the form of chemical energy (glucose), which is later used by organisms for metabolic activities.
iii. Food Production: Photosynthesis forms the basis of the food chain by producing organic compounds that serve as a source of nutrition for heterotrophic organisms.
iv. Carbon Cycle: It plays a key role in the carbon cycle by fixing carbon dioxide from the atmosphere.

Chemosynthesis: is a process by which certain organisms produce carbohydrates from chemicals rather than using sunlight as an energy source, as in photosynthesis. This process is typically found in environments with limited or no sunlight, such as deep-sea hydrothermal vents, certain caves, and some extreme environments.

Heterotrophic Nutrition: Organisms that cannot produce their own food and rely on external sources for nutrients practice heterotrophic nutrition. Animals, fungi, and some bacteria are examples. They obtain organic compounds by consuming other organisms or organic matter.

I. **Holozoic Nutrition**: Holozoic nutrition is a type of nutrition in which organisms ingest complex organic substances, then digest and absorb the nutrients. It is characteristic of most animals, including humans.

- Process: In holozoic nutrition, organisms consume solid or liquid organic matter, digest it internally through processes like mechanical and chemical digestion, and absorb the resulting nutrients for energy and growth.

II. P**arasitic Nutrition:** Parasitic nutrition is a type of heterotrophic nutrition where an organism (parasite) lives on or inside another organism (host) and derives nutrients at the expense of the host. This relationship is usually harmful to the host. Examples include, Tapeworms in the human intestine or fleas on a dog are examples of parasites that obtain their nutrients from the host organism.

III. **Symbiotic Relationship**: Symbiotic relationships involve a close and often long-term interaction between two different species, where at least one of the species benefits. It can be classified into mutualism, commensalism, and examples include

- **Mutualism:** Bees and flowers - bees get nectar, while flowers get pollination.

- **Commensalism:** Barnacles on a whale - barnacles benefit by having a surface to attach to, while the whale is unaffected.

3. **Saprophytic Nutrition:** saprophytic nutrition is a type of nutrition in which organisms feed on dead and decaying organic matter. These organisms, called saprophytes or decomposers, play a crucial role in breaking down organic material and recycling nutrients. Examples include Fungi like mushrooms and certain bacteria are saprophytes that obtain nutrients from decaying organic material.

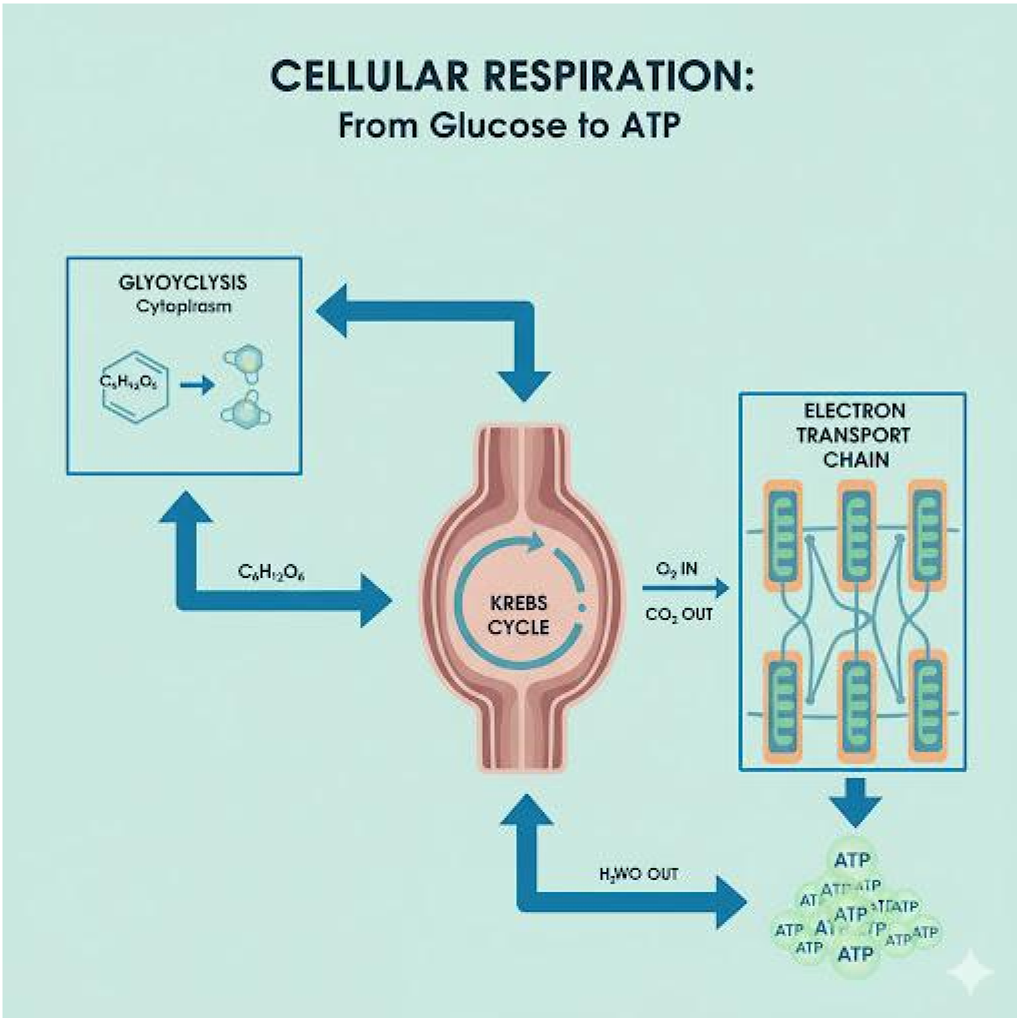
Evaluation:

1. Identify the two modes of nutrition
2. What is photosynthesis, describe the process of photosynthesis
3. What is heterotrophic mode of nutrition
4. Differentiate between photosynthesis and chemosynthesis

Assignment:

Explain the process of feeding in amoeba and paramecium

WEEK 9: SOME PROPERTIES AND FUNCTION OF THE CELL II



Cellular (Internal/Tissue) Respiration

The oxidation of glucose in the cell to release energy is known as cellular respiration and it occurs in the mitochondria of all living cells. There are two types of cellular respiration i.e. aerobic and anaerobic respiration

Aerobic respiration

When cellular respiration takes place in the presence of oxygen is known as aerobic respiration. The largest amount of ATP possible is generated through it from one molecule of glucose with the release of carbon (IV) oxide and water as by product.

$C_6 H_{12} O_6 + 6O_2 \rightarrow 6CO_2 + 6H2O + 38ATP$

Anaerobic respiration

In some organisms such as bacteria, fungi and endoparasites, the cells gets energy from breaking down glucose in the absence of oxygen this is known as anaerobic respiration. In this type of respiration, lesser amount of ATP is produced. The pyruvic acid produced is converted to alcohol in plants (alcoholic fermentation) while in animals, lactic acid is produced which leads to muscle fatigue in athletes

In plants

$C_6 H_{12} O_6 \quad + \quad CH_2COCOOH \rightarrow C_2H_5OH + CO_2 + 2ATP$

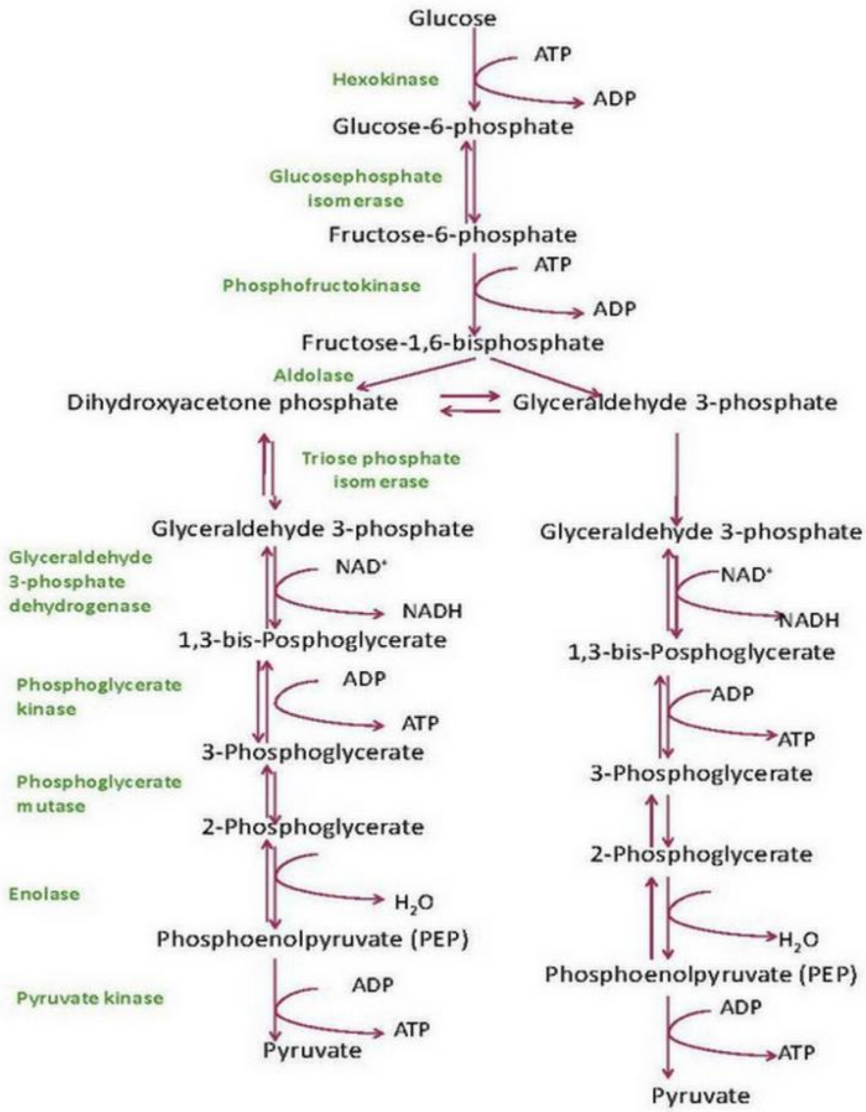
In animals

$C_6 H_{12} O_6 + CH2COCOOH \rightarrow CH_2\ CHOHCOOH + CO_2 + 2ATP$

Mechanism of respiration

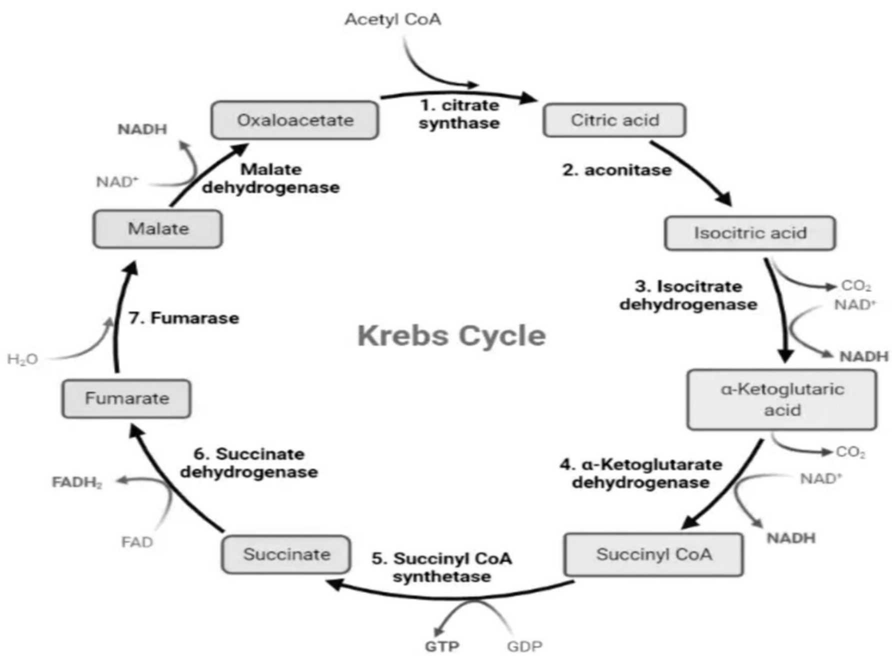
Cellular respiration involve two main stages

1. **Glycolysis:** this occurs in the cytoplasm of the cells. First the glucose molecule is phosphorylated by the addition of a phosphate group to the glucose to become glucose-6-phosphate which is carried out by an enzyme known as hexose kinase. There are ten steps in this pathway that leads to the breakdown of one molecule of 6 – carbon into two molecules of the 3 carbon pyruvic acid by the enzymes in the cytoplasm. This process does not require oxygen. Glycolysis takes place in the cytoplasm. At the end of the pathway, 2 ATP is used to produce 4 ATP (2 ATPs from each pyruvate) so that a net energy of 2 ATP is generated in glycolysis, this is called “substrate level phosphorylation”.



1. **Krebs's cycle** (also known as citric acid cycle CAC or Tricarboxylic acid cycle TCA): Here, each pyruvic acid is further oxidized completely to carbon dioxide and water in the mitochondria. The pyruvic acid from the glycolysis is converted to acetic acid through the removal of one molecule of CO_2. The acetic is carried into the Krebs's cycle by co enzyme A. The combination of acetic acid and co enzyme A forms acetyl-coA. Acetyl-coA combines with oxaloacetatic acid to form citric acid which is an important started of the Krebs's cycle. Krebs's cycle takes place in the matrix of the mitochondrion. Most of the ATP is generated in the cycle. The oxidation process in Krebs's cycle leads to the production of 36 ATPs (18 ATP from each pyruvate). The process used in the production

of ATP in Krebs's cycle is called Oxidative phosphorylation. A total of 38 ATP is generated from the aerobic breakdown of glucose. The Krebs's cycle is particularly important because it is key pathway that connects protein, fats and carbohydrates.



EVALUATION

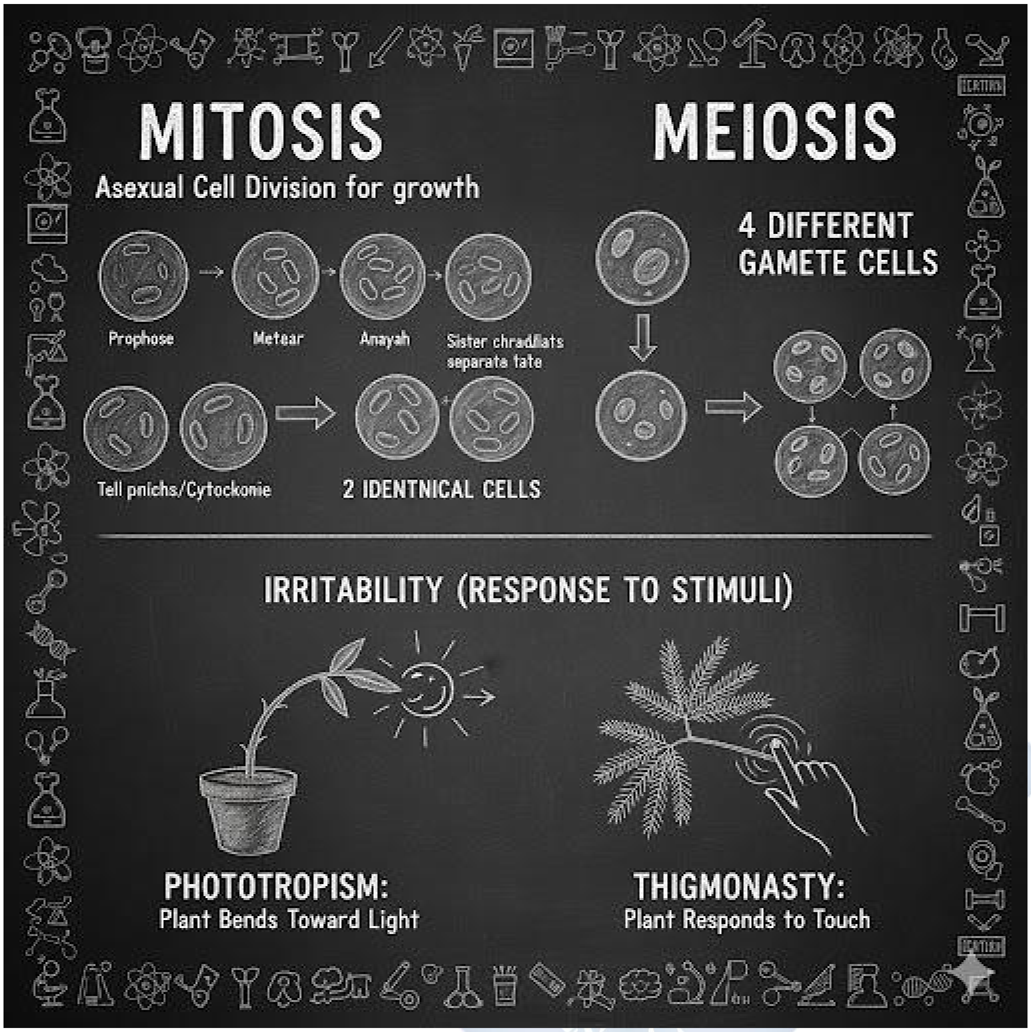
1. Differentiate between micro and macro elements
2. State four importance of macro elements in plants
3. Differentiate between external and internal respiration
4. Explain briefly 'the Krebs's cycle
5. What do you understand by (a) muscle fatigue (b) oxygen debt

ASSIGNMENT

1. The organelle involved in tissue respiration is the A. endoplasmic reticulum B. Golgi body C. mitochondrion D. ribosome

2. In the absence of oxygen, the pyruvic acid produced during glycolysis is converted to CO_2 and A. water B. glycerol C. ethanol D. citric acid
3. Glycolysis takes place in A. lysosome B. Mitochondrion C. Nucleus D. ribosome
4. The starting substance in Krebs's cycle is A. Citric acid B. Acetic acid C. oxalic acid D. Malic acid
5. A total of __ ATP is produced from one glucose during aerobic respiration A. 34 B. 36 C. 38 D.40

WEEK 10: GROWTH AND IRRITABILITY



Meaning of Growth

Growth is the irreversible increase in size, dry mass, and complexity of an organism due to cell division (mitosis) and cell enlargement.

Key Points:

- Growth involves both cell multiplication (mitosis) and cell expansion (enlargement).
- It is irreversible—once an organism grows, it cannot return to its previous size.
- Growth is measured by changes in height, weight, or volume.

Basis of Growth

Growth occurs through two main processes:

Cell Enlargement: after cell division, new cells absorb water and nutrients, increasing in size. This is due to the synthesis of protoplasm (cytoplasm and nucleus). Example: A seedling growing into a tall plant due to cell expansion.

Cell Differentiation: Cells become specialized in structure and function. Undifferentiated cells (stem cells) develop into tissues and organs (e.g., nerve cells, muscle cells).Example: A zygote divides and differentiates into various tissues in a baby.

Mitosis (Cell Division for Growth and Repair)

Mitosis is the process where a single cell divides into two genetically identical daughter cells.

Stages of Mitosis

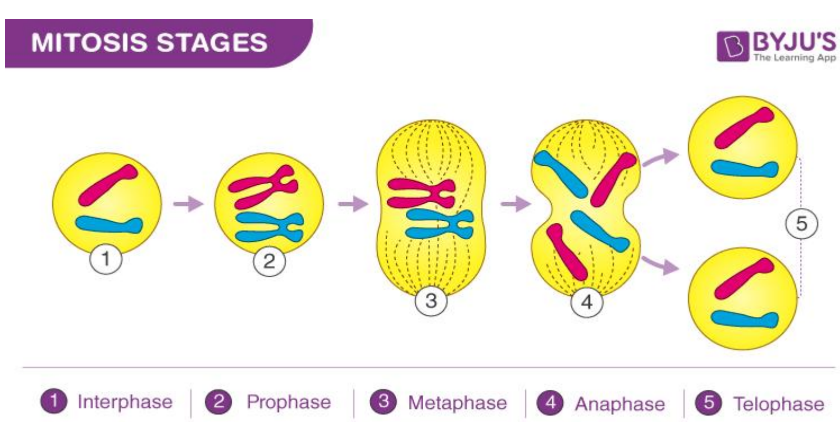
Interphase (Resting Phase): Cell prepares for division. DNA replicates, organelles double.

Prophase: Chromatin condenses into chromosomes, nuclear membrane begins to disappear and Spindle fibers form.

Metaphase: Chromosomes align at the equator (middle) of the cell. Spindle fibers attach to centromeres.

Anaphase: Sister Chromatids separate and move to opposite poles

Telophase: Chromatids reach poles and form new nuclei, cytokinesis occurs—cytoplasm divides into two cells.



Importance of Mitosis:

1. Growth of multicellular organisms.
2. Repair of damaged tissues (e.g., wound healing).
3. Asexual reproduction (e.g., budding in yeast).

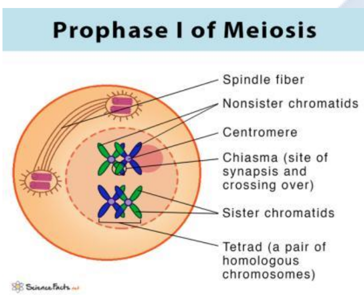
Meiosis (Cell Division for Reproduction)

Meiosis produces gametes (sperm and egg cells) with half the chromosome number (haploid).

Stages of Meiosis

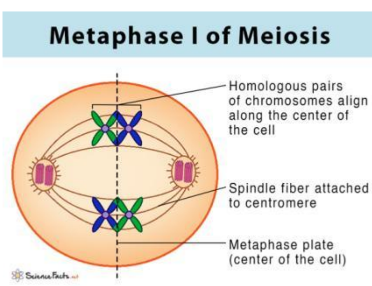
Meiosis I (Reduction Division):

Prophase I: Homologous chromosomes pair up (synapsis).Crossing over occurs, increasing

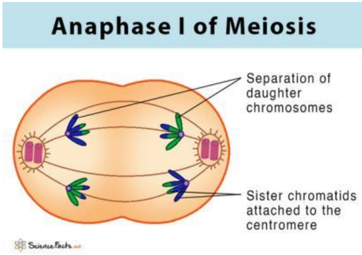


genetic variation.

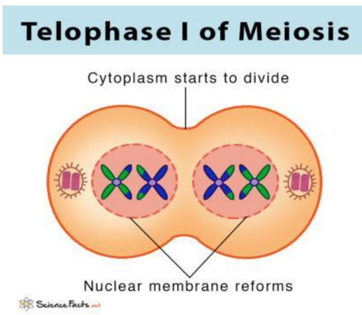
Metaphase I: Homologous pairs align at the equator.



Anaphase I: Homologous chromosomes separate (not chromatids).



Telophase I: Two haploid cells form.



Meiosis II (Similar to Mitosis):

1. Prophase II:

This stage follows a brief inter kinesis period where DNA replication doesn't occur. The chromosomes, which are already condensed from meiosis 1, prepare for the second division.

2. Metaphase II:

The chromosomes line up in a single file at the cell's equator, similar to metaphase in mitosis. Spindle fibers attach to the kinetochores (protein structures on the centromeres of each sister chromatid).

3. Anaphase II:

Sister chromatids are pulled apart by the spindle fibers and move towards opposite poles of the cell.

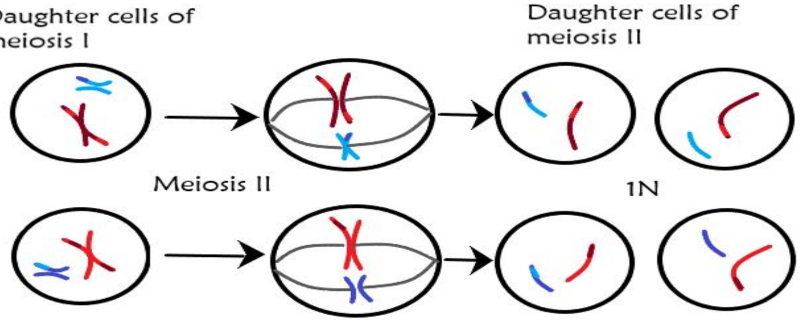
4. Telophase II:

The separated sister chromatids reach the poles, and nuclear membranes form around them.

5. Cytokinesis II:

The cytoplasm divides, resulting in four daughter cells, each containing a single set of chromosomes (haploid).

This process ensures genetic diversity by separating sister chromatids, creating cells with different genetic combinations, which is essential for reproduction.



Importance of Meiosis:
Produces genetic variation (due to crossing over).
Ensures offspring have the correct chromosome number
Essential for sexual reproduction.

IRRITABILITY AND TYPES OF RESPONSES

Irritability is the ability of organisms to respond to stimuli. A stimulus is any change in external or internal environmental condition which can bring about a change in the activity of the whole or part of the organism.

Response is the term used for the change in activity of the organism. There are three major types of responses, these include tactic, nastic and tropic movements.

1. TAXIS OR TACTIC MOVEMENT: is a directional movement or response of a whole organism from one place to another in response to external stimuli such as light, temperature, water and certain chemicals. Examples of tactic movement include;

- Euglena or chlamydomonas swimming away from high light intensity (Negative phototaxis). In a moss plant, sperm swim towards the chemical produced by the egg cell (positive chemotaxis).

2. Nastism or nastic movement: is a non-directional sleep movement or response of a part of a plant in response to non-directional stimuli such as light intensity, temperature and humidity. Example of nastic movement include;

- The folding of the leaflets of mimosa plant when touched.
- Closing of the morning glory flower when light intensity is low.

3. Tropism or tropic movement: is a unilateral growth and directional movement of a part of a plant in response to directional stimuli. These responses are experienced in growth regions (root and shoot apices) and are controlled by certain plant hormones known as auxins. Tropic movement are named according to the stimuli e.g.

- Shoots bend towards light (positive phototropism) while roots bend away from light (negative phototropism).
- Shoots bend away from gravity (negative geotropism) while roots bend toward gravity (positive geotropism).
- Tendrils of climbing plants twine around a support (positive thigmotropism) while root tips grow away from it (negative thigmotropism).

Movement

Organisms moves from one place to the other in search of food, water, mates and escaping predator or harsh weather conditions.

Cyclosis in Cell

Cyclosis (cytoplasmic streaming) is the mass rotational movement of the cytoplasm and its contents in cells. Cyclosis brings about the transportation of substances from one part of the cell to the other and the exchange of materials between the cell organelles. Cyclosis occur in

1. Protozoa like amoeba known as amoeboid movement.
2. Chloroplasts of some plants where they move independently to place their broad surface parallel to the surface of the leaf to receive sufficient sunlight for photosynthesis.

Evaluation:

1. Define growth in living organisms
2. Differentiate between cell enlargement and differentiation.
3. List the stages of mitosis and explain what happens in each.
4. How does meiosis contribute to genetic variation?
5. Compare mitosis and meiosis in a tabular form.

Assignment:

Draw and label the stages of mitosis and meiosis in your notebook.

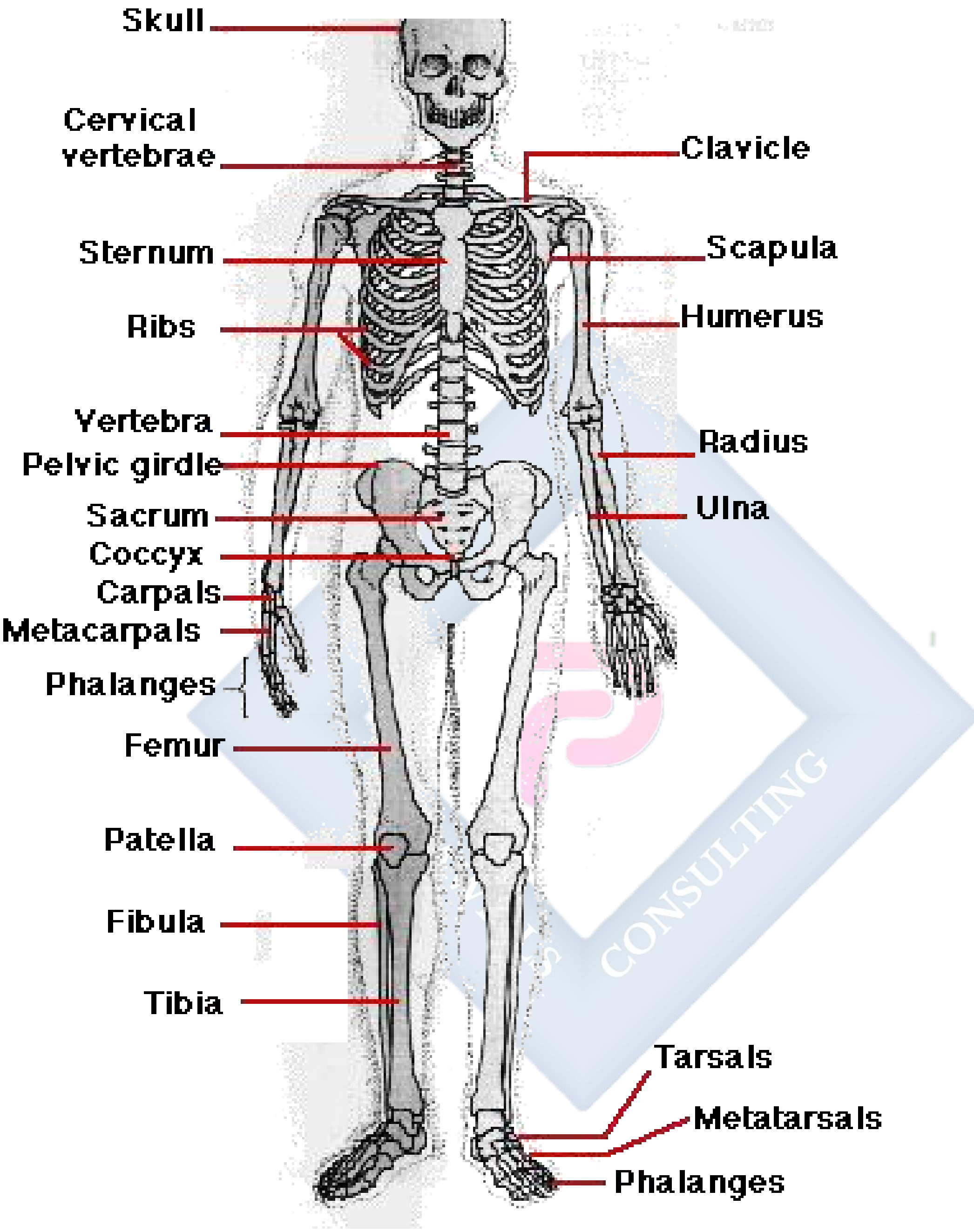
In a tabular form, give at least five differences between mitosis and meiosis.

SS 1 BIOLOGY LESSON NOTES

SECOND TERM

1. Tissue and supporting system
2. Nutrition
3. Basic Ecological Concepts
4. Population studies.
5. **Midterm examination**
6. Functioning Ecosystem, Autotrophy, and Heterotrophy - Food Webs and Trophic Levels
7. **Midterm break**
8. Energy Transformation in Nature
9. Relevance of Biology to Agriculture
10. Micro-organisms Around Us

WEEK 1: TISSUE AND SUPPORTING SYSTEM



Introduction

To carry out life processes, all organisms (plants and animals) need tissues. Tissues are group of similar cells that carry out specific functions. Skeleton is the framework of the body which provides support, shape and protection to the soft tissues and organs in animals. It forms the central core of human body and it is covered by muscles and blood vessels and skin.

Forms of skeletal materials

There are 3 forms of skeletal materials found in animals .These are

1. Chitin
2. Cartilage
3. Bones

Chitin

It is a tough non-living material present in arthropods (invertebrates). It acts as a hard outer covering to the animal and is made up of series of plates covering or surrounding organisms. Chitin is very tough, light and flexible. However, it can be strengthened by impregnation with 'tanned' proteins and particularly in the aquatic crustaceans like crabs, by calcium carbonate.

Cartilage

This is a tissue present in skeleton of complex vertebrates. Cartilage consists of a hard matrix penetrated by numerous connective tissue fibres. The matrix is secreted by living cells called **chondroblasts**. These later become enclosed in spaces (lacunae) scattered throughout the matrix. In this condition the cells are termed **chondrocytes**. It acts as a shock absorber in between bones during movement because it is tough and flexible with a great tensile strength. It is found predominantly in mammals and cartilaginous fishes e.g. shark.

Evaluation

1. What is skeleton?

2. State two main components of skeleton. (b) Differentiate between cartilage and chitin.

Types of cartilage

Cartilages are of three main types in mammals and they are

Hyaline cartilage

This contains a dense meshwork is the most common type and can be found on surface of moveable joint, trachea and bronchi (for ease of respiration) and also in protruding parts of the nose.

White fibrous cartilage

Tougher than the hyaline cartilage and can be found in the intervertebral disc of vertebral column.

Yellow elastic cartilage

Found in the external ear (pinna) and epiglottis (*cartilaginous flap covering the trachea active during food swallowing).

Bone

This is the major component of skeletal system and it consist of living cells (osteocytes), protein fibers (collagen), and minerals such as calcium carbonate and calcium phosphate. These minerals (the non- living constituent) makes up two-third of a mass of bone .Hence, bone is strong and very rigid unlike cartilage. Bones are highly vascularised.

The skeleton of a young vertebrate embryo is made up of cartilage. As the embryo grows bone cells (osteocyte) replaces cartilage cells. Hence, the cartilage tissue becomes hardened into bone through the addition of minerals in a process called **OSSIFICATION**

DIFFERENCES BETWEEN BONES AND CARTILAGE

	Bone	**Cartilage**
1	Bones produce red and white blood cells	Cartilages do not
2	Made up of both living cells and dead cells	Made up of mainly living cells.
3	Bones are often rigid	Cartilage are often flexible
4	They are made up mainly of mineral substance such as calcium	Mineral substance are absent
5	Can never be replaced by cartilage	Can be replaced by bones
6	Flexible only in young ones	Cartilage both in young ones and adult is flexible.

TYPES OF SKELETON

The three main types of skeleton in animals are

1. **Hydrostatic skeleton:** This is the type present in soft bodied animals e.g. earthworm, sea anemones etc**.** Such animal use pressure to support itself**.** They also have a muscular body wall which is filled with fluid. The fluid presses against the muscular wall causing them to contract and exerting force against the fluid.

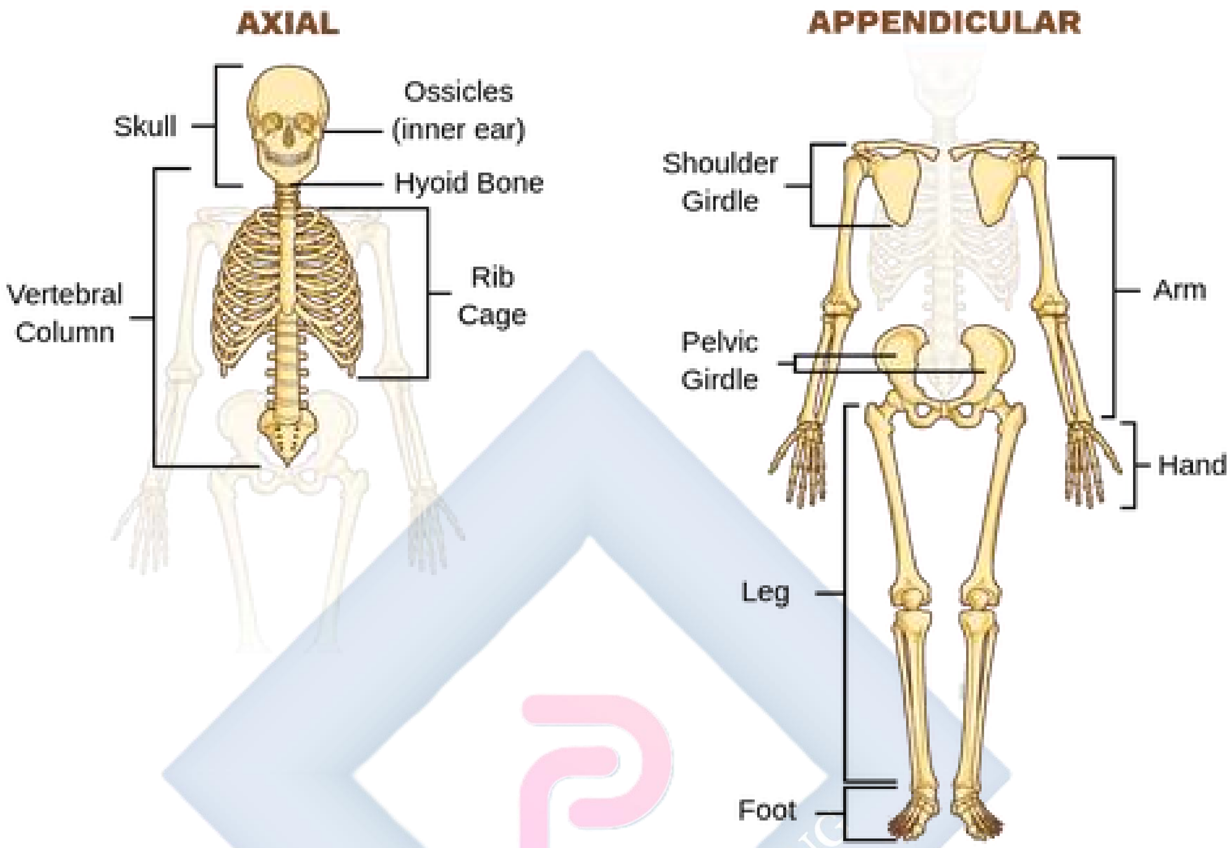
2. **Exoskeleton:** This is the outer skeleton present in arthropods. It is secreted by the cells covering the body of the animals and the main component is chitin (non living substance). Exoskeleton also supports animals against gravity and enables them to move about. Animals with these skeleton types periodically shed the old skeleton; grow rapidly in size when the new exoskeleton is still soft and extensible. The shedding process is called MOULTING or ECDYSIS.

3. **Endoskeleton**: This is an internal skeleton present in all vertebrates. Endoskeleton of vertebrates are composed mainly of bones and the bones grow steadily as the animal grows **(hence no need** for moulting). Bones of many sizes and shapes make up the endoskeleton of vertebrates. These bones are attached together as moveable joints by tough flexible fibers called ligaments hence the skeleton is flexible. Muscles are also attached to the bones usually by tendons to provide posture and bring about body movement.

Function of skeleton

- It supports the body of organisms.
- Skeleton acts as the framework of the body
- Protection of delicate organs e.g. heart, brain, etc.
- Used for locomotion through the limbs in action.
- Important component of respiration e.g. breathing involve active movement of the ribs.
- Production of blood via bone marrows.

BONES OF THE AXIAL AND APPENDICULAR SKELETON



The skeleton of vertebrate such as fish, frog, lizard, bird and man consist of bones and cartilages. It can be classified into two.

1. **Axial skeleton:** which consists of the skull, ribs, sternum and the vertebral column.
2. **Appendicular skeleton** : it is made up of limb girdles (pectoral and pelvic girdles), and the limbs (the fore and hind limbs).

AXIAL SKELETON

The skull

The Skull is made up of flat bones joined together by suture joint which has three parts: Cranium (brain-box), facial skeleton and the jaws; including maxilla (upper) and mandible.

Functions

- It protects the brain.

- Also protects the olfactory organ, eyes, middle and inner ear.
- Gives shape to the head.
- Bears the teeth.

The vertebral column

It forms the back bone, protecting the spinal cord. It is made up of 5 groups of bones called the vertebrate each of which is built on similar basic pattern. The vertebrate are held together with strong ligament and comprehensible cartilage pads called into intervertibratal disc.

Types of Vertebrae and Location

Vertebrae	Location	Rat	Rabbit	Cat	Cow	Humans
Cervical	Neck	7	7	7	7	7
Thoracic	Chest	13	12-13	13	13	12
Lumber	Upper trunk	6	6-7	7	6	5
Sacral	Lower trunk	4	4	3	5	5
Caudal	Tail	30	16	18-25	18-20	4

A typical vertebra

A typical vertebra has the following structural features

- **Neural canal** which is for the passage of the spinal cord.
- **Neural spine** which projects upward and backward for the attachment of muscle.
- **Transverse processes** for the attachment of muscles and ligaments.
- **Centrum;** a solid bony pieces below the neural canal
- **Zygapophyses** are the particular surfaces for joining together of successive vertebrate.

This could be pre-zygapophysis (facing inward and upwards) or post-zygapophysis (facing outward and downwards

Cervical Vertebrae

The first cervical vertebra is called the **atlas** while the second is called the **axis.**

The Atlas It has a large neural canal, flat and broad transverse processes, short neural spine which could be absent at times. It also has a vertebrarterial canal for the passage of blood vessels. Centrum is absent.

Function of Atlas

Permits the nodding of the head.

The Axis

It has a broad and flat Centrum, a large and flat neural spine, reduce transverse processes and a vertebrarterial canal. It articulates with the atlas through odontoid process

Functions

- It permits the turning or twisting of the head.
- Forms pivot joint with the atlas.

Thoracic Vertebra

Have a long and prominent neural spine, a pair of short transverse processes, a large neural canal and neural arc and large cylindrical centrums .They also have particular surfaces for attachment of the ribs.

Function

- Aids attachment of ribs
- Assist in breathing
- Attachment of muscles at the shoulder and back

Lumbar Vertebrae

Each has large and flat transverse processes, broad and flat neural spine, large and thick centrums and well developed zygapophyses. It has extra paired projections namely

1. Anapophysis
2. Metapophysis

Functions of Lumbar

- It provides attachment for abdominal muscles
- It bears considerable weight of the body

Sacral Vertebrae

This fuse together to form a singular bony mass called sacrum. Each sacral vertebrate has a narrow neural canal, reduced neural spine and large centrums. The first differs from the remaining four by

1. Having a pair of transverse processes which is large and wing-like while the others are attached to the muscles of the back.
2. Presence of a small neural canal which generally becomes narrower in the lower four vertebrae.

Function

- Joins the pelvic girdle to provide support, rigidity and strength.

Caudal vertebrae

These are joined together to form a singular bony mass called **coccyx.** Each has no neural spine, no neural canal and no transverse process. It appears as a solid rectangular mass of bone

Functions

- Supports the tail
- Provides attachment for tail muscle

The Appendicular Skeleton

Pectoral girdle: found around the shoulder in man and it consists of two halves which are held by muscles. Each halve is made up of three bones

- Scapula
- Clavicle
- Coracoids

The scapula and coracoids are fixed together as the scapula is flat and triangular with a hollow called GLENOID CAVITY at its tip. This cavity articulates or joins with the head of humerus to form the shoulder joint. The clavicle is a small rod of bone attached to a ligament joining the sternum to the scapula

Functions

- The pectoral girdle gives attachment to muscles and ligaments.
- It provides firm support to the fore limbs.

Pelvic girdle: found around the waist in man and it consists of two halves which are joined to each other ventrally and to the sacrum dorsally. Each halve of the pelvic girdle is made up of 3threebones. They are

- Ilium
- Ischium
- Pubis

These three bones form a depression (on their outer surface) called ACETABULUM which articulates with the head of the femur to form the hip joint.

Limbs

The limbs include the fore (upper) and the hind (lower) limbs. In most vertebrates, both limbs have the same basic plan i.e. each limb has a long bone followed by a pair of two long bones next to this is a set of small bones terminating with five digits.

The fore limbs- This is made up of an upper arm bone called **humerus** which joins with two other long bones at its lower end (radius and ulna) to form the elbow joints. Radius and ulna (the ulna is longer) are the bones of the fore arm, next are the wrist bones called **carpals** which are a small bones. These are followed by the digit bones called **metacarpals** which terminate in the **phalanges** (finger bones). In man, each digit has three phalanges except the thumb which has two phalanges.

The hind limbs-This is made up of thigh bones called femur (which is the largest and longest bone in the body). Its round upper end is the end that terminates at two rounded projections called **condyles** which forms the knee joint together with tibia. A small flat bone called **patella** is found in front of the knee joint. Next to the femur are tibia and fibula-Tibia is longer and larger. These are followed by bones of the ankle called **tarsals**. The lower limb terminates as at the digit bone metatarsals and each digit is made up of three phalanges

The ribs

These are long semicircular rods which connects the thoracic vertebrates to the sternum. They are found in the chest region of the body. In man, they are 12 pairs

Function

- They form a cage protecting the lungs and the heart
- They assist in breathing.

A typical rib

A typical rib has a head, a neck and a body. The **first seven** ribs are connected directly to the sternum through coastal cartilages. They are therefore called **true ribs**. The **next five** are called **false ribs.** The eighth to tenth ribs have a common articulation to the sternum, each one attached to the coastal cartilage to the one above .The **eleventh and twelfth pairs** of ribs are called **floating ribs** because they have no connection to the sternum.

SUPPORTING TISSUES IN PLANTS

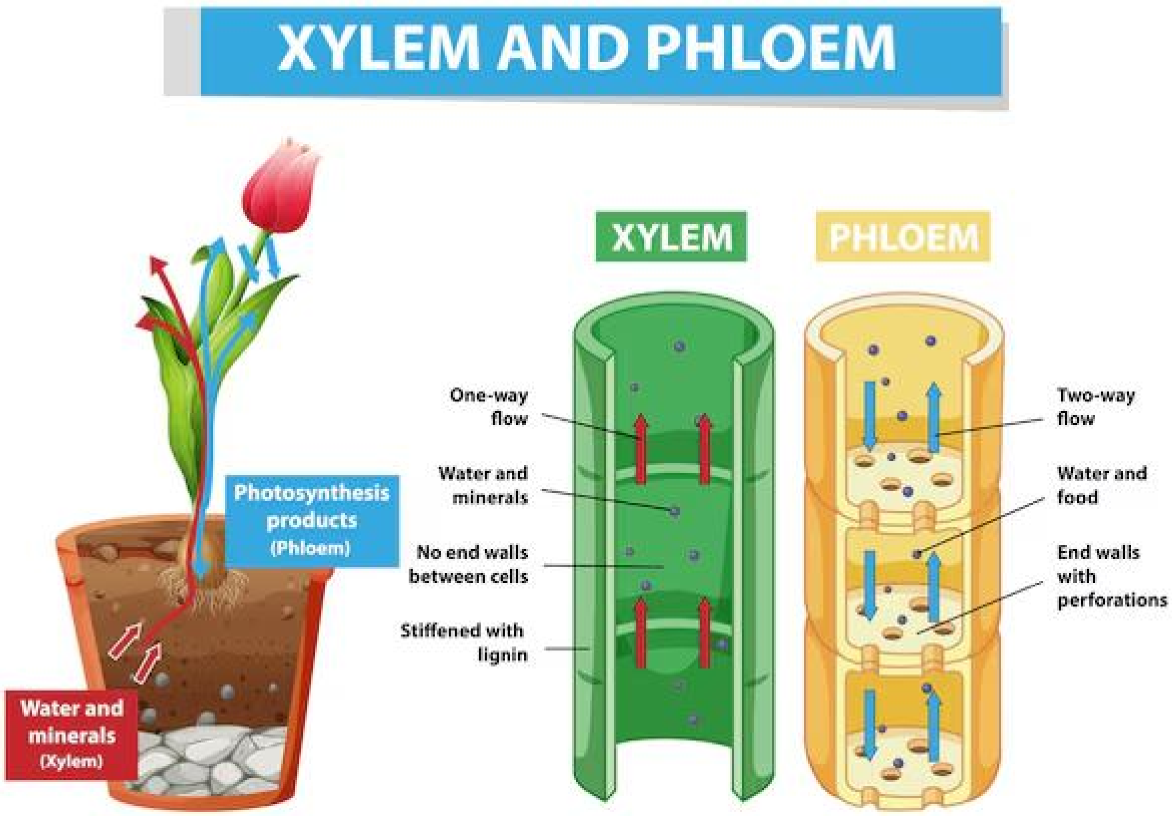
The needs for supporting tissues in plant are for:

1. definite shape;
2. strength;
3. rigidity;
4. resistance against external force such as wind and water.

Types of supporting tissues

- Parenchyma tissues
- Collenchyma tissues
- Sclerenchyma tissues

- Xylem tissues
- Phloem tissues



Parenchyma Tissues

They are made up living cells with cellulose and many air spaces between within them. This is the most common and abundant plant tissue.

Functions

- It gives firms and turgidity to the stems of hibiscus
- stores food and water
- takes part in food synthesis in leaf mesophyll

Collenchyma Tissues

Made up of living cells which are elongated and thickened at the corners.

Functions

- Provides strength and support in young growing plant.

- Gives flexibility and resilience to plant.

Sclerenchyma Tissues

Sclerenchyma tissues are made up of thick-walled cells containing cellulose and **lignin**. These tissues are rich in **fibres** and provide strength and support to the plant.

Functions

- gives flexibility to plant
- provides strength, rigidity ,hardness and support to plant

Xylem

Xylem tissues are found in vascular tissues of stems, roots and leaves

Functions

- provides support strength and shape to the plant
- Helps to conduct water and mineral salt from the roots to leaves.

Phloem tissues

Also located in the vascular bundles of all plants in their roots, stems and leaves

Functions

- Conduction of manufactured food from site of production to site of consumption and storage.
- Assist to provide support to the entire plant.

Evaluation

1. Define a skeleton

2. Explain the different forms of skeletal materials

3. Describe the different types of skeleton

4. Describe the bones of the axial and appendicular skeleton.

Assignment

Section A

1. Draw and label the human skeleton
2. Draw and label the different bone of the vertebrate column

Section B

1. ………… is a non-living skeletal material A. chondroblasts B. osteocyte C. elastic cartilage D. chitin
2. The articulating surface for joining together successive vertebrates is called A. neural spine B. zygapophyses C. transverse processes D. neural canal
3. The canal for the passage of blood vessels in vertebrae is called A. neural canal B. cervical canal C. vertebraterial canal D. zygaphosis
4. Endo- skeleton is present in the following animals except A. dog B. snake C. shark D. lizard
5. The most abundant supporting tissue in plants is A. sclerenchyma B. parenchyma C. xylem D. phloem.

WEEK 2: ANIMAL NUTRITION

Animal nutrition

Animals generally cannot manufacture their food. Rather, they depend directly or indirectly on plants for their food. Hence they are called heterotrophs. Based on their food types, animals are grouped into three:

1. Carnivores which feed on flesh or other animals e.g. lion.
2. Herbivores which feed on plants e.g. goat.
3. Omnivores, which feed on both plants and animals e.g. man.

Classes of food substances

Foods eaten by animals are generally classified into seven i.e.

i. Carbohydrate
ii. Proteins
iii. Fat and oil
iv. Mineral Salt
v. Vitamins
vi. Water
vii. Roughages

Carbohydrate: This is got from food like bread, yam rice etc. It basically consists of carbon, hydrogen and oxygen. Carbohydrates are of three types:

1. Monosaccharides (Simple sugars) which include glucose, fructose and galactose
2. Disaccharides (Reducing sugars) which include maltose, sucrose and lactose.
3. Polysaccharides (Complex sugars) e.g. starch, cellulose, chitin under the action of enzymes like ptyalin, maltase, lactase etc., and starch yields glucose as product of its digestion. Excess carbohydrate is stored in the body in form of glycogen in muscles and liver. This can be reconverted to glucose during starvation.

Importance of Carbohydrates

1. It gives animals energy.
2. It provides heat needed to maintain body temperature
3. It can be used for lubrication e.g. mucus
4. It provides the body with a strong framework e.g. exoskeleton in insects.

Proteins: These are complex molecules made up of smaller units called amino acids. Protein is made up of carbon, hydrogen, oxygen, nitrogen and sometimes phosphorus and Sulphur. Food like egg, meat, fish, beans etc. gives you protein. Proteins are broken down into amino acids under the action of enzymes like pepsin, rennin, trypsin and erepsin.

Importance of Proteins

1. Growth in young ones.
2. Repair of worn-out tissues.
3. Production of enzymes.
4. Production of hormones.
5. It supports reproduction.
6. It is for tissue and all formation i.e. body building.

Fats & oil (lipids)

Fats are solid lipids at room temperature while oil is the liquid. Fat and oil consist of carbon, hydrogen and little oxygen. When digested, it gives rise to fatty acids and glycerol. Foods like palm oil, groundnut, and Soya beans give fat and oil. Lipids are broken down to fatty acids and glycerol when acted upon by lipase enzymes.

Importance of Fat and Oil

1. It gives you energy even more than carbohydrates
2. It supplies essential fatty acids to the body.
3. It helps in the maintenance of body temperature
4. It provides the body with fat-soluble vitamins

Mineral salt

These are usually taken in very small quantity in the food we eat except sodium chloride (table salt) and iron tablet, which can be taken directly by man. The lack of these salts results in nutritional deficiency. The minerals include calcium, magnesium, potassium, Phosphorus, Sulphur, chlorine, iron, Iodine, fluorine, manganese, copper, cobalt and sodium.

Importance of Mineral Salts

1. Regulate body metabolisms
2. Components of bones and teeth
3. Aids blood formation
4. Control chemical reactions in the body

5. Aids the formation of enzymes and pigment

These are organic food substances needed by man and other animals in small quantity for normal growth and development. Lack of or inadequate supply of any of these vitamins results in nutritional deficiency.

Vitamins can be grouped into two:

1. Water-soluble vitamins
2. Fat – soluble vitamins

The water-soluble vitamins include: vitamins B complex and vitamin C. Vitamin B complex include vitamin, B2, B3, B5, B6 and B12.

Fat-soluble vitamins include vitamins A, D, E and K.

Vitamins, source functions and deficiency symptoms

	SOURCE	**FUNCTION**	**DEFICIENCY SYMPTOMS**
Vitamin A	Liver, eggs, fish milk, palm oil, fish Vegetables	(i) Normal growth of body cells and skin (ii) Proper vision of the eye	(i) Night blindness (ii) Reduced resistance to disease
Vitamin B1	Yeast, milk, beans, Ground nut	(i) Normal growth (ii) Proper functioning of heart and nervous system	Beri-beri (wasting of Muscles), paralysis

Vitamin B2	Yeast, soya beans, egg, milk, green Vegetables	(i) Growth, proper functioning of the eye (ii) Formation of co-enzymes	(i) Slow growth (ii) Dermatitis
Vitamin B3	Yeast, beans, milk, Vegetables	Formation of co-enzymes for cellular respiration	Pellagra
Vitamin B12	Kidney, liver, fish Milk	Formation of red blood Cells	Pernicious Anaemia
Vitamin C	Fresh fruits and Green vegetables	(i) Aids wound healing (ii) Helps to resist infection	Scurvy
Vitamin D	Fish, milk, egg, Liver, sun's Ultraviolet rays	(i) Increases absorption Of calcium and phosphorus. (ii) Calcification and hardening Of bones	Ricket; Osteomalacia
Vitamin E	Green vegetables, Egg, butter, liver	Promotion of fertility In animals	Sterility Premature abortion
_Vitamin K	Fresh green vegetables, liver	Aids blood clotting	Hemorrhage

Water

This is of utmost importance to all organisms and it is made up of two elements, hydrogen and oxygen. Water can be got from food, river, stream, pond etc. water makes up 75% of the human body.

Importance of water

1. Metabolic activities of the body of animals.
2. Digestion of food.
3. Maintenance of body temperature.
4. It is a medium of transportation for all nutrients.
5. It helps to maintain the osmotic balance in body tissues.
6. It helps in excretion of metabolic waste from the body e.g urine.

Roughages

These are indigestible fibrous materials got from vegetables, fruit, carbohydrates and proteins. Roughages aid digestion, lack of which can lead to constipation.

Balanced diet

Balanced diet is a diet containing a correct proportion of all the food substances. On a general note, a balanced diet contains 15% protein, 15% fat and oil, 10% vitamin, minerals and water and 60% carbohydrate. Once a food is taken at these proportions, there is a normal growth and development in the body.

Functions of balanced diet

1. It makes us healthy.
2. It gives ability to be resistant to diseases
3. It makes available energy needed to carry out all biological activities.
4. It prevents malnutrition and deficiency symptoms. For examples, a diet that lacks protein results into a nutritional disease called kwashiorkor in children.

The protein deficient child has the following features

1. Retarded growth.
2. Loss of weight.
3. Swollen legs effect (oedema).
4. Cracked / split stomach and thin legs etc.

Digestive enzymes

Enzymes are organic (protein) catalysts produced by living cells which help to speed up and slow down the rate of chemical reactions. Digestive enzymes aid the breaking down of complex food substances into simple, soluble and diffusible form. Enzymes have the following characteristics.

1. Enzymes are soluble
2. Enzymes are protein
3. They are specific in their actions
4. Enzymes are sensitive to temperature i. e. they work best between 35oC to 40oC
5. Enzymes are PH specific
6. Enzymes brings about reversible reactions
7. Enzymes needs co-enzymes to activate them and can be inactivated by inhibitors such as mercury and cyanide

Classes and functions of enzymes

Digestive enzymes are classified based on the type of food they act upon. These include

1. Proteases e. g. pepsin, rennin, trypsin and erepsin. They act on protein.
2. Amylases e. g. ptyalin, lactase, maltase, sucrose. They act on carbohydrates
3. Lipases which act on lipids (fats and oils)

Evaluation

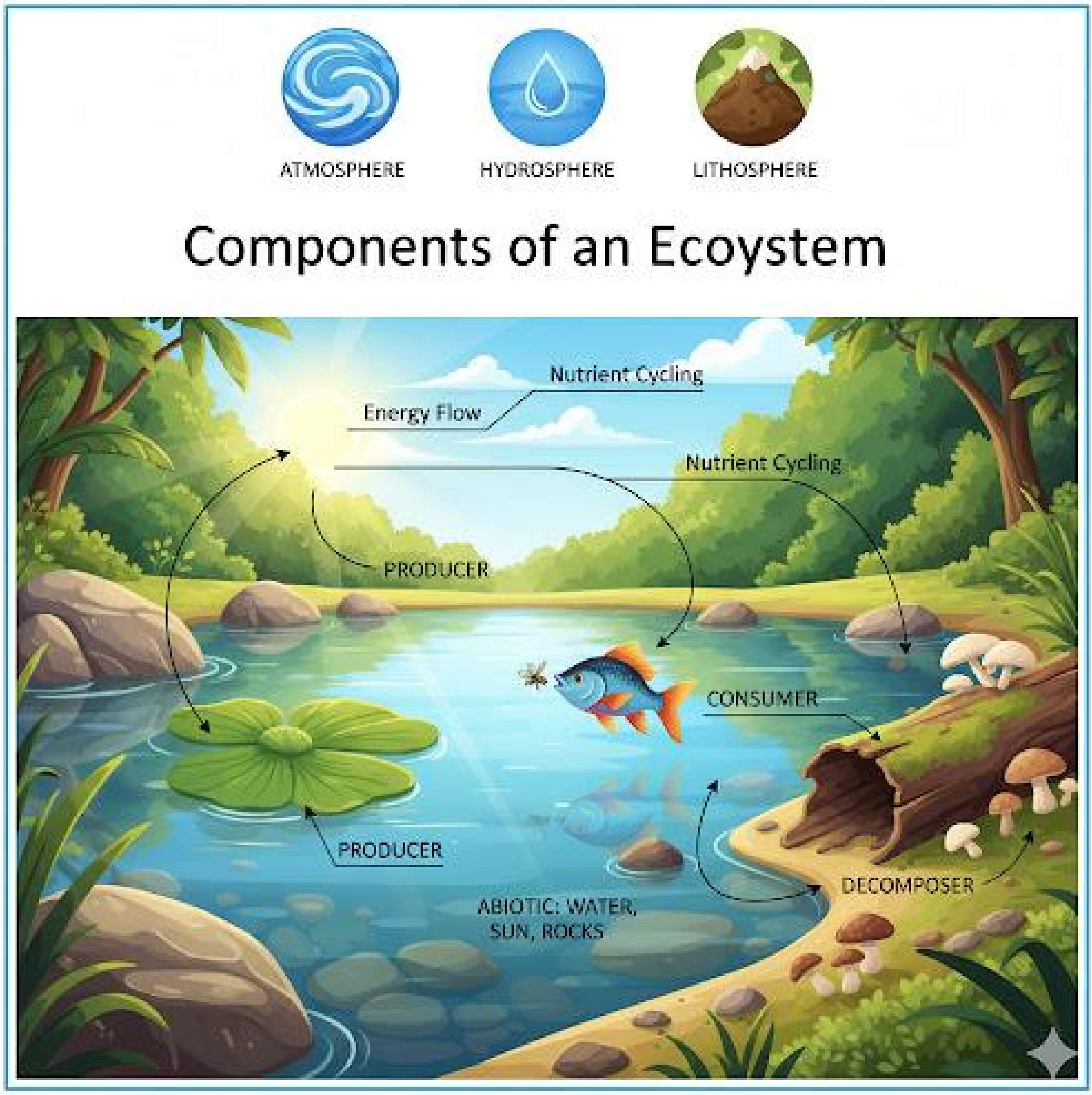
1. State two food items each that supply A. Protein B. Lipids C. Mineral salts

2. List two functions each for protein and lipids.
3. State the functions of A. Chlorine B. Magnesium C. Iodine.
4. What are the diseases resulting from deficiency of: A. Sodium B. Calcium C. Iron
5. State the functions and deficiency symptoms of all the water soluble vitamin

Assignment

1. Vitamins are organic food substances required by animals in ____ quantity A. no B. small C. large D. high
2. The following except one are fat-soluble vitamins A. vitamin A B. Vitamin B C. Vitamin K D. Vitamin E
3. One of the following food substances is indigestible in man A. protein B. lipids C. roughages D. carbohydrates
4. The highest source of energy is from _______ A. carbohydrate B. proteins C. lipids D. vitamins
5. Rickets (poor bone formation) in children is a deficiency symptom of ____ A. potassium B. calcium C. chlorine D. manganese.

WEEK 3: DEFINITION OF ECOLOGY



Ecology is the study of plant and animals (as well as microorganism) in relation to their environment. As a practical science, ecological studies involve:-

- Studying the distribution of living organisms
- Finding out how living organisms depend on themselves and their non-living environment for survival.
- Measuring factors affecting the environment.

Branches of ecology

Depending on whether the organisms are studied alone or in groups, ecology is divided into two:

1. **AUTECOLOGY:** This is the study of an individual organism or a single species of organism and its environment e.g. the study of a student and his school environment.
2. **SYNECOLOGY:** This involves studying the inter-relationships between groups of organisms or different species of organism living together in an area e.g. study of fish, crabs, seaweeds, etc. in a pond.

BASIC ECOLOGICAL CONCEPTS

The various concepts closely associated with ecology include;

Environment

This includes external and internal factors, living or nonliving which affects an organism or a group of organisms. These include the habitat (with its peculiar physical conditions e.g. light, food, water, air), the animals preying on other animals or the diseases affecting the organisms.

Biosphere (ecosphere)

This is the zone of the earth occupied by living organisms so as to carry out their biochemical activities.

The ecosphere consists of 3 major portions:

- Lithosphere: This is the solid portion (the outer-most zone) of the earth which is made up of rocks and minerals. This zone forms 30% of the earth surface and it is the basis of human settlement.
- Hydrosphere: This is the liquid (aquatic) part of the biosphere. It covers about 70% of the earth's crust. It consist of water in various forms; solid, liquid or gas (water vapour), hydrosphere includes lakes, pools, spring, oceans, ponds, rivers, etc.

- Atmosphere: This is the gaseous portion of the earth. It consists of three main gases; nitrogen (78%), oxygen (21%) and carbon (iv) oxide (0.03%). There are also 0.77% rare gases.

Habitat

This is a place where an organism is naturally found. Habitat is always affected by environmental factors. Habitat can be divided into three:-

- Aquatic habitat: this is where plants and animals (as well as micro organisms) live in water e.g. Sea, ocean, lagoons, streams, etc.
- Terrestrial habitat: this is where the organisms live on land e.g. forests, grassland, (savanna), desert etc.
- Arboreal habitat: These include tree trunks and tree tops where some organisms (usually animals) are naturally found.

Ecological niche

This refers to the habitat and the entire habit (behavioural, feeding, breeding) of an organism. It is the physical space occupied by an organism and its functional role in the community.

Population

This is the total number of organisms of the same species living together in a habitat e.g. population of cockroaches in Biology laboratory.

Community

This is made up of all the populations of living organisms that exists together in a habitat e.g. a community of decomposer, insects and birds on a decaying log of wood.

Ecological system (ecosystem)

This refers to the basic functional unit in nature, which consists of all living factors and their interaction with non-living factors of the environment. An ecosystem can be natural or artificial.

Biomes

This is a large natural terrestrial ecosystem. It is a plant and animal community produced and maintained by the climate.

Components of an ecosystem

Biosphere comprises of various ecosystems. There are two main components of the ecosystem, the abiotic and the biotic components.

Abiotic components

These are non-living components and they are basic elements and compounds of the environment in which an organism lives.

Abiotic components include organic substances (e.g. carbohydrates, lipids, and proteins), inorganic substances (e.g. CO2, H2O), climatic factors (e.g. light, temperature, rainfall) as well as edaphic factors (e.g. soil types, texture, topography etc).

Biotic components

These are the living things in the ecosystem. These include the producers, the consumers and the decomposers.

1. Producer: These are autotrophs in that they synthesize their food from simple inorganic substances e. g. green plants, protophytes and chemosynthetic bacteria.
2. Consumers: These are heterotrophs which feed on the producers or one another. They may be primary, secondary or tertiary consumers e.g. non-green plants, animals protozoa and some bacteria
3. Decomposers: These are saprophytes. They break down remains of plants and animals and release usable nutrients to the soil. These nutrients are used by plants to make food e.g. fungi and some bacteria

Generally living things influence other living things in many ways such as in feeding (parasitism), shading from sunlight, pollination and dispersal of seeds, competition. Some of these factors are favourable while others are unfavourable.

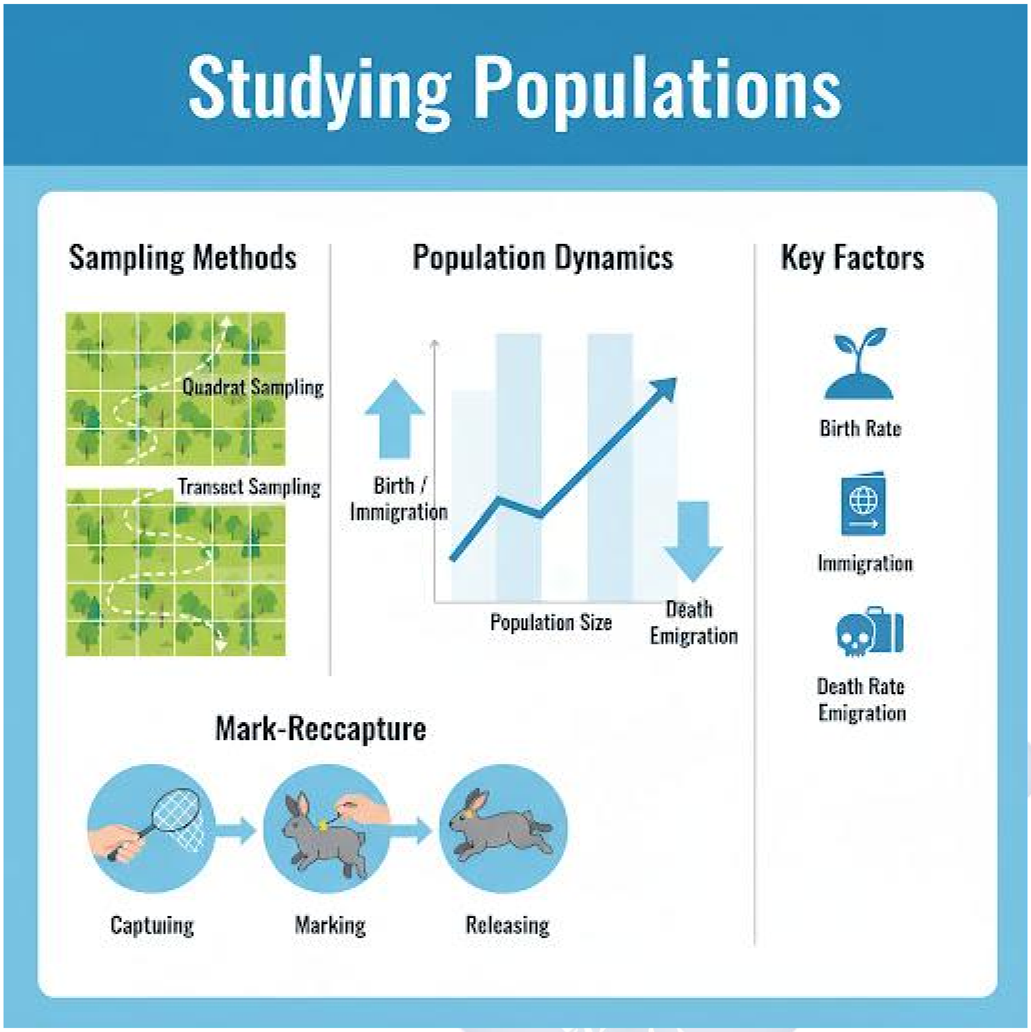
Evaluation

i. Define and state the branches of Ecology.
ii. Describe the basic ecological concepts.
iii. Describe the ecological system and state its components.

Assignment

1. Which of the following is not classified as a terrestrial habitat? A. forest B. guinea savanna C. literal zone D. desert
2. The activities of an organism which affect the survival of another organism in a habitat can be described as A. biotic factors. B. Abiotic factors C. climatic factor D. Edaphic factors.
3. The number of individuals of the same species interacting in a habitat at a particular time is best described as A. community B. ecosystem C. population D. biome.
4. A biotic community with its physical environment (abiotic factors) defines A. ecosystem B. population C. habitat D. biosphere.
5. The most abundant gas in the atmosphere is A. oxygen B. nitrogen C. carbon dioxide D. a rare gas.

WEEK 4: POPULATION STUDIES



Population Size: This is the total number of individuals of a species in a defined area at a specific time. The population size can be estimated through direct counting, mark and recapture methods, or aerial surveys.

Population Density: This is the number of individuals of a species per unit area or volume. The population density is calculated by dividing the population size by the total area or volume of the habitat.

It provides insights into how closely individuals are packed in a given space, influencing resource availability and interactions.

Dominance*:* This is the degree to which a particular species or individual has a disproportionately large impact on its community, influencing the distribution and abundance of other species. Dominance is often assessed through species abundance and their impact on community structure. In plant communities, dominant species may occupy a large portion of the habitat and influence the composition of associated species

Methods of Population Studies

Census: This is the direct counting of all individuals in a population. It provides an accurate count, although it's Impractical for large, mobile, or elusive populations.

Mark and Recapture: It is a subset of individuals is captured, marked, released, and later recaptured. The proportion of marked individuals in the second capture is used to estimate the total population size. It is effective for mobile or elusive populations.

1. Quadrat Sampling: Sampling units (quadrats) of known size are randomly placed in a habitat, and the number of individuals within each quadrat is counted. It is useful for immobile or sessile organisms, however, assumes homogeneity within quadrats, and may not capture spatial variations.
2. Transect Sampling: Here a line or path is established through a habitat, and the number of individuals or species intersecting the line is recorded. It is useful for studying distribution along a gradient.
3. Remote Sensing: The use of satellite or aerial imagery to estimate population size or vegetation cover. It is useful for large-scale studies.

FACTORS AFFECTING POPULATION

Birth Rate: The number of births per unit of population over a specified period.

Death Rate: The number of deaths per unit of population over a specified period. High death rates decrease population size, while low death rates support population growth.

Migration: The movement of individuals into (immigration) or out of (emigration) a population or geographic area. Immigration increases population size, while emigration decreases it. Migration can alter population composition.

Resource Availability: The availability of essential resources such as food, water, shelter, and space. However, limited resources can lead to competition, impacting population size and distribution.

Environmental Conditions: Factors like climate, weather, and habitat conditions.

Disease and Parasitism: Disease: Pathogens that affect the health of individuals.

Parasitism: Organisms living on or in another organism, benefiting at the host's expense.

Disease and parasitism can decrease population size by causing illness or death.

N**atural Disasters:** Catastrophic events like floods, wildfires, earthquakes, and hurricanes. Natural disasters can have significant impacts on population size and distribution.

Human Activities: Activities like Hunting and Fishing leads overharvesting of species for food or trade, while habitat Destruction leads to deforestation, urbanization, and pollution. All these human activities can cause population decline or endangerment.

Climate Change: Long-term changes in temperature, precipitation, and weather patterns. Climate change alters habitat suitability, affecting the distribution and abundance of populations.

Measurement of ecological factors

	INSTRUMENTS	**USES**
a.	Photometer	Light intensity
b.	Hydrometer	Light intensity in water
c.	Wind vane	Direction of wind
d.	Anemometer	Speed of wind
e.	Rainguage	Amount of rainfall
f.	Hygrometer	Relative humidity
g.	Barometer	Pressure

h.	Glass thermometer	Temperature
i.	Colorimeter or pH scale	Acidity / Alkalinity
j.	Secchi disc	Turbidity
k.	Sweep insect net	Catching insects

RELATIONSHIP BETWEEN SOIL TYPES AND THE WATER-HOLDING CAPACITY

The relationship between soil types and the water-holding capacity of soil is influenced by several factors:

1. Texture: Soil texture, determined by the proportion of sand, silt, and clay, affects water retention. Clay soils have smaller particles and can hold more water than sandy soils.
2. Porosity: Soil porosity, the amount of pore space between soil particles, influences water-holding capacity. Well-structured soils with good porosity can retain more water.
3. Organic Matter: The presence of organic matter enhances soil structure and water retention. Organic matter acts like a sponge, holding water and making it available to plants.
4. Depth of Soil Profile: Deeper soil profiles generally have more space to store water, benefiting vegetation during dry periods.
5. Compaction: Compacted soils have reduced pore spaces, limiting water retention. Uncompacted soils allow for better water infiltration and storage.
6. Topography: Sloped areas may experience faster water runoff, reducing water availability for plants. Flat or gently sloping terrain allows for better water retention.

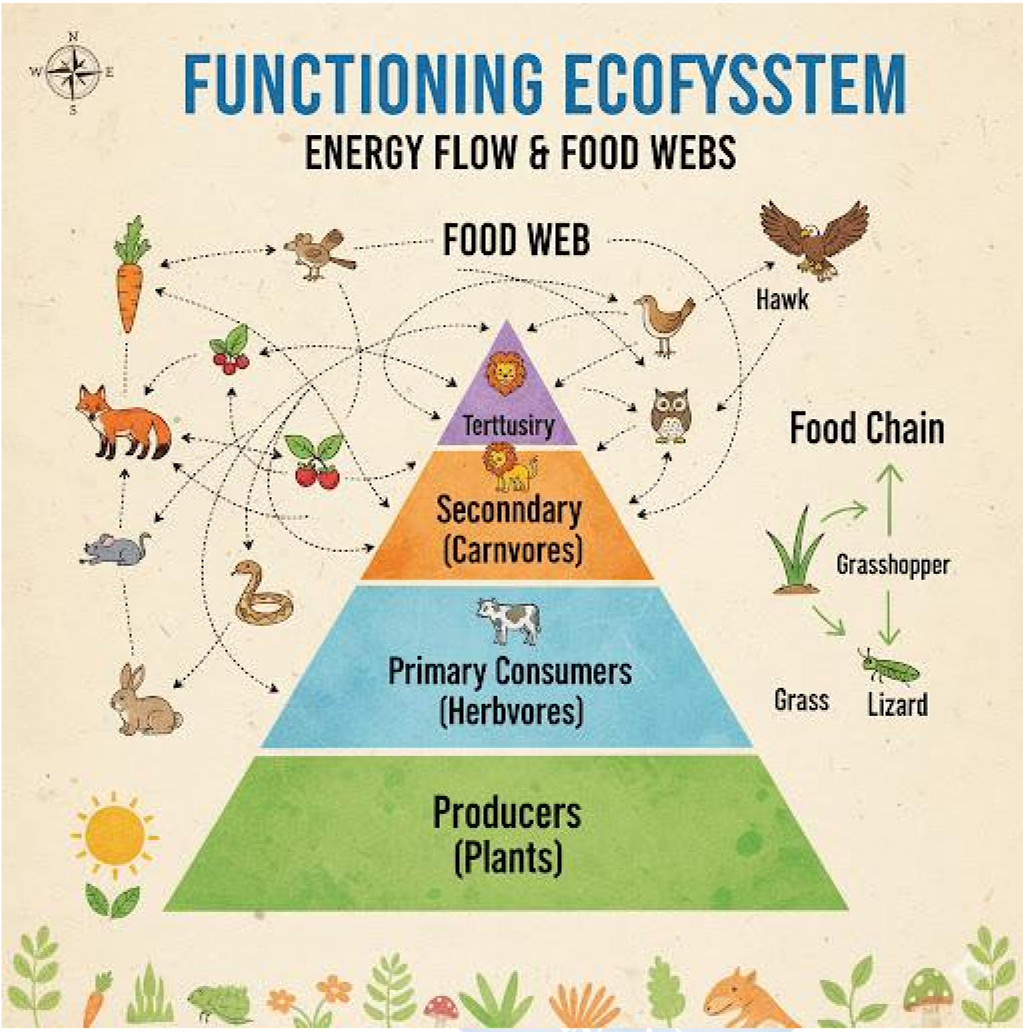
Evaluation:

1. Define population size and population density
2. Identify the methods of studying population
3. Identify factors that affect population
4. List instruments for measuring ecological factors
5. Describe the relationship between soil types and water holding capacity

Assignment:

1. Draw and label all the ecological instrument.

WEEK 6: FUNCTIONING ECOSYSTEM



Functioning Ecosystem

A functioning ecosystem involves the interplay of various components and processes that collectively maintain balance and sustain life. The following are aspects of a functioning ecosystem include:

1. Biodiversity: A diverse array of species contributes to ecosystem stability. Each species plays a unique role, and interactions among them support ecological processes.

2. Energy Flow: Ecosystems rely on the flow of energy through food chains and webs. Producers (plants) capture sunlight, and energy is transferred through consumers (herbivores, carnivores) and decomposers.
3. Nutrient Cycling: Nutrients such as carbon, nitrogen, and phosphorus cycle through the ecosystem. Decomposers break down organic matter, returning nutrients to the soil for reuse by plants.
4. Abiotic Factors: Physical and chemical components, like climate, soil, and water, influence the ecosystem. These abiotic factors shape the distribution and behavior of organisms.
5. Water Cycle: Movement of water through precipitation, evaporation, and transpiration is essential for maintaining water availability in ecosystems.

Roles of Autotrophs, Heterotrophs and decomposers

Autotrophs, heterotrophs, and decomposers play crucial roles in ecological systems, collectively contributing to the flow of energy and cycling of nutrients:

1. Autotrophs (Producers): Autotrophs are organisms capable of synthesizing their own food using sunlight or inorganic compounds. Most commonly, plants perform photosynthesis, converting solar energy into chemical energy stored in organic compounds. Autotrophs form the foundation of ecosystems by producing organic matter. They are essential for energy flow as they create the initial biomass that sustains all other trophic levels.

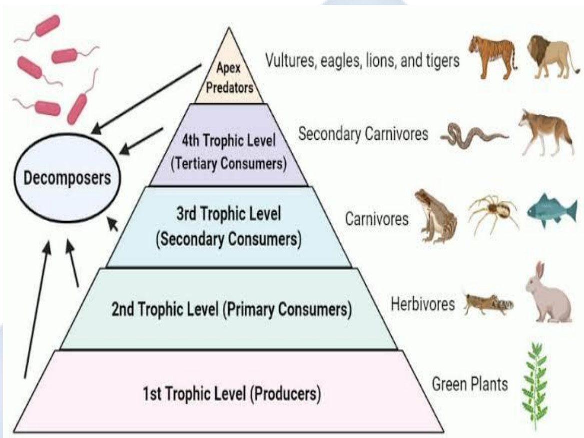
2. Heterotrophs (Consumers): Heterotrophs are organisms that obtain energy by consuming other organisms. They cannot produce their own food and depend on autotrophs or other heterotrophs for nutrients and energy. Heterotrophs are categorized into primary consumers (herbivores), secondary consumers (carnivores that feed on herbivores), and tertiary consumers (carnivores that feed on other carnivores).Heterotrophs are vital for the transfer of energy through trophic levels. They participate in the consumption and recycling of organic matter.

3. Decomposers: Decomposers, primarily bacteria and fungi, break down organic matter into simpler compounds. They play a crucial role in the recycling of nutrients by decomposing

dead organisms and organic waste. Decomposers secrete enzymes that break down complex organic molecules into simpler forms, releasing nutrients back into the environment.

Trophic level

A trophic level is a position in the food chain or web of an ecosystem, representing a group of organisms that share the same function in the transfer of energy and nutrients. There are typically three main trophic levels:



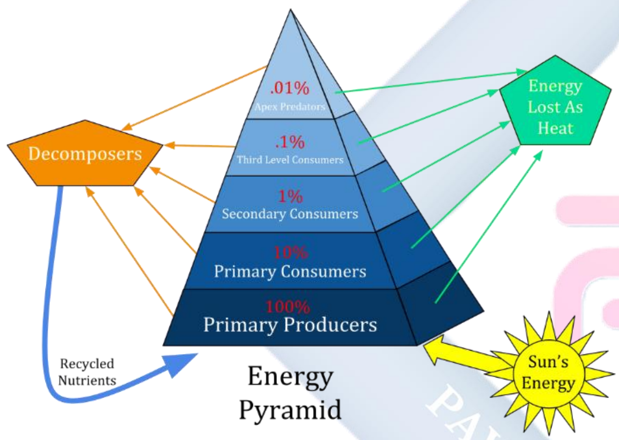
1. Primary Producers (Trophic Level 1): Primary producers, often autotrophs like plants, algae, and certain bacteria, are at the base of the food chain. They produce organic compounds through processes like photosynthesis, capturing energy from the sun or inorganic sources.
2. Consumers (Trophic Levels 2, 3, etc.):

- Primary Consumers (Herbivores - Trophic Level 2): Herbivores feed directly on primary producers (plants) and represent the second trophic level.
- Secondary Consumers (Carnivores - Trophic Level 3): Carnivores that feed on herbivores occupy the third trophic level.
- Tertiary Consumers (Carnivores - Trophic Level 4): Predators that consume other carnivores are at the fourth trophic level.

- Decomposers (Trophic Level Decomposer): Decomposers, such as bacteria and fungi, break down dead organic matter from all trophic levels. They play a vital role in nutrient recycling, releasing essential elements back into the environment.

Food chain and food web.

A food chain is a linear sequence showing the flow of energy from one organism to another. For example, grass is eaten by a herbivore, which is then consumed by a carnivore. In contrast, a food web is a more complex, interconnected network of multiple food chains. It illustrates how different species in an ecosystem are linked through their feeding relationships, providing a more realistic representation of energy flow.



Energy Flow along trophic level (Pyramid of number energy and biomas).

In an ecosystem, energy flows through trophic levels, representing different positions in the food chain. The energy pyramid, pyramid of numbers, and pyramid of biomass illustrate this flow.

1. Energy Pyramid: Shows the flow of energy from one trophic level to the next. It also shows how energy is lost as heat during metabolic processes, so each higher level receives less energy. Typically, the primary producers (plants) form the broad base, with energy decreasing as you move up to consumers.

2. Pyramid of Numbers: Represents the number of organisms at each trophic level. It may not always follow a pyramid shape; it can be inverted or irregular based on the size and

reproductive rates of organisms. In some ecosystems, a large number of small organisms (e.g., insects) can support fewer consumers (e.g., predators).

3. Pyramid of Biomass: Illustrates the total biomass (weight) of organisms at each trophic level. Biomass decreases as you move up the pyramid due to energy loss and the fact that higher trophic levels require more energy for maintenance. Like the energy pyramid, it generally has a pyramid shape.

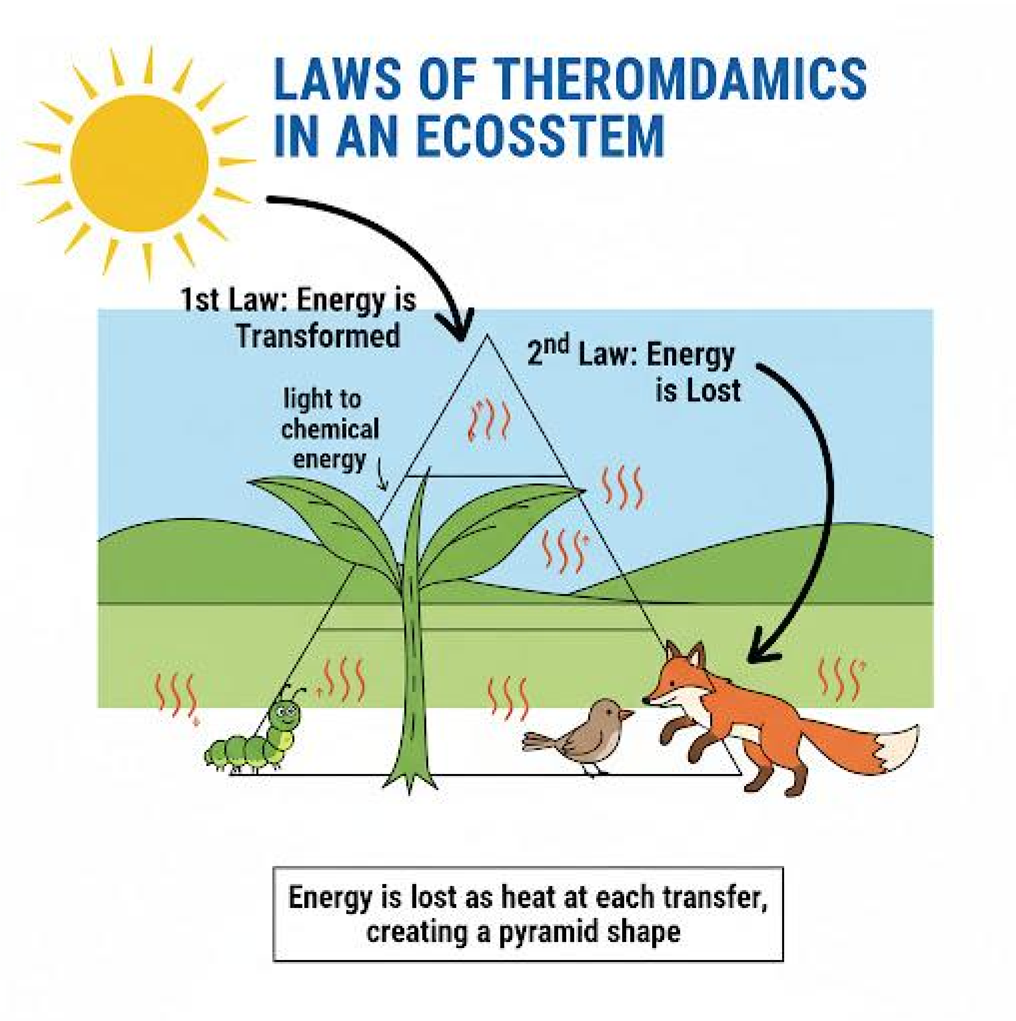
Evaluation: Identify 5 aspects of a functioning Ecosystem

2. Discuss the roles of autotrophs, heterotrophs and decomposers
3. What are trophic level?
4. Describe how energy flow along the trophic levels

Assignment:

Construct a food chain using the following organisms: toad, hawk,snake,lion.

WEEK 8: ENERGY TRANSFORMATION IN NATURE



Energy transformation in nature

Energy undergoes various transformations in nature. For example, sunlight is converted into chemical energy through photosynthesis in plants. This stored energy is then transferred through the food chain as organisms consume each other. Eventually, some energy dissipates as heat, contributing to Earth's overall heat budget. Additionally, natural processes like water cycle and weather patterns involve energy transformations. Understanding these transformations helps explain the dynamic balance within ecosystems.

Energy Loss in the ecosystem

Energy losses occur in ecosystems through various processes. As energy moves through trophic levels in a food chain, not all of it is transferred to the next level. Energy is lost as heat during metabolic processes, such as respiration, and as organisms excrete waste. Each trophic level receives only a fraction of the energy from the level below it.

The second law of thermodynamics dictates that energy transformations are never 100% efficient, leading to entropy and energy dissipation. This is why ecosystems have a pyramid-like structure, with fewer individuals at higher trophic levels.

NOTE: energy losses in ecosystems occur as heat, metabolic processes, and waste, contributing to the overall inefficiency of energy transfer between trophic levels.

Laws of thermodynamics

The laws of thermodynamics are fundamental principles that govern the behavior of energy in physical systems.

1. ***First Law of Thermodynamics (Conservation of Energy):***

The first law of thermodynamics states that energy cannot be created or destroyed in an isolated system. The total energy of a system remains constant; it can only change forms. In simple terms, it's the law of conservation of energy.

Application to ecological phenomena

The first law of thermodynamics has several applications in ecological phenomena:

i. Energy Flow in Ecosystems: The law explains how energy flows through ecosystems. Solar energy is captured by plants through photosynthesis, converted into chemical energy, and then transferred to herbivores, carnivores, and decomposers in a series of trophic levels. The total energy within the system remains constant, adhering to the first law.

ii. Metabolic Processes: Organisms within ecosystems undergo metabolic processes that involve the conversion of energy from food into usable forms. The first law is evident in these processes, as the energy obtained from consuming other organisms is used for growth, reproduction, and other life-sustaining activities.

iii. Energy Budgets: The first law helps ecologists analyze energy budgets within ecosystems. By accounting for the energy inputs (e.g., sunlight) and outputs (e.g., heat, work done), researchers can understand how energy is distributed and utilized by different components of the ecosystem.

iv. Conservation of Energy in Ecological Systems: The law reinforces the concept of energy conservation in ecological systems. Understanding that the total energy content remains constant allows ecologists to track energy transformations and predict the outcomes of various ecological processes.

Second Law of Thermodynamics**:** This law has two parts:

Entropy: The total entropy (measure of disorder or randomness) of an isolated system tends to increase over time. This implies that systems naturally move towards a state of greater disorder.

- Heat Flow: Heat energy tends to flow from hot to cold objects, and not the other way around spontaneously. This aspect relates to the irreversibility of certain processes.

Application to ecological phenomena

The second law of thermodynamics, particularly the concept of entropy, has several applications in ecological phenomena:

1. Entropy and Ecosystem Dynamics: The second law implies that natural processes tend to increase entropy, leading to greater disorder or randomness. In ecosystems, this is evident in the constant cycling of nutrients and the transformation of energy. Ecological systems evolve toward states of increased complexity and diversity, which aligns with the tendency towards higher entropy.
2. Energy Transfer Efficiency: The law underscores the limitations in energy transfer efficiency between trophic levels in food webs. As energy flows through the food chain, each transfer results in some energy being lost as heat. This inefficiency contributes to the

hierarchical structure of ecosystems, where energy becomes more dispersed and disorder increases.

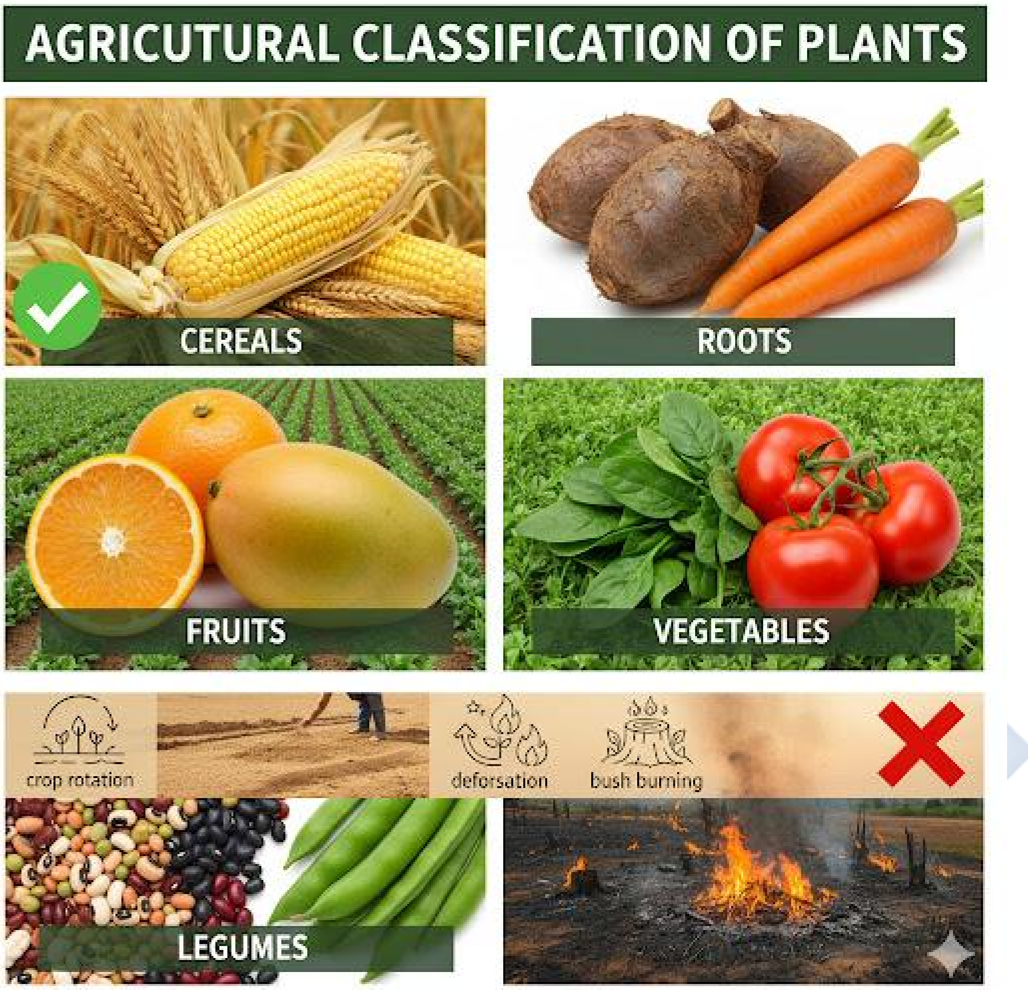
3. Decomposition and Nutrient Cycling: Decomposition of organic matter by microorganisms is a process driven by the second law. It involves the breakdown of complex organic compounds into simpler forms, increasing the disorder within the system. This process is essential for nutrient cycling and maintaining ecosystem function.
4. Climate and Heat Transfer: The second law influences the heat transfer processes in ecosystems. Heat tends to flow from warmer to cooler regions, influencing climate patterns and affecting the distribution of organisms within ecosystems.

Evaluation:

1. State the first law of thermodynamics and identify 3 of it's applications to ecological phenomena
2. Stat the second law of thermodynamics and identify its applications to ecological phenomena.
3. Describe how energy is lost in the ecosystem

Assignment: use the second law of thermodynamics and explain energy loss in the ecosystem**.**

WEEK 9: RELEVANCE OF BIOLOGY TO AGRICULTURE



Classification of plants

Botanical classification

The concept of binomial nomenclature has it that plant kingdom can be subdivided into divisions, classes, orders, families, general and species. This is based on their structures, functions and evolutionary trends. Plants are then generally classified into three brand groups

- Thallophytes (A)
- Bryophytes (B)

- Tracheophytes (C)

Tracheophytes are further grouped into pteridophytes (D) and spermatophytes (E). The Spermatophytes can also be grouped into gymnosperms (F) and angiosperms (G). The angiosperms are subdivided into dicot (H) and Monocot (I)

A & B are non- vascular plants, C refers to vascular plants. D refers to non-flowering plants; E refers to seed plants while G refers to flowering plants proper.

Agricultural classification

Agricultural classification of plants is based on:

1. The product obtained from the plants
2. The parts of the plant that is useful
3. The economic importance of the plants

Plants are therefore classified agriculturally into the following

1. **Crops and weeds:** - plants that are needed on the farm are called crops while other unwanted are called weeds.
2. **Food crops and cash crops**: - crops grown mainly for human consumption are called food crops e.g. maize, yam etc. Those grown mainly to earn money are cash crops e.g. cocoa, coffee
3. **Root crops:** - are plants which store mainly starch in edible underground stems or roots e.g. yam, cassava etc.
4. **Cereal crops:** - these are monocotyledonous plants of grass family, whose grains are eaten e.g. maize, millet, guinea corn, rice, wheat, etc. They are rich in carbohydrates.
5. **Fruits crops:-** are rich in vitamins and minerals. Fruits are also rich in sugar. These include oranges, mangoes, avocado peers, cashew etc. whose fruits are eaten
6. **Vegetable crops:** - are herbaceous plants whose vegetable (leafy) parts are eaten. They include spinach, lettuce, carrots, cabbage, okro, tomato, onion, pepper etc. They are also rich in vitamins and minerals.

7. **Legumes:** - are plants of beans family such as cowpea and groundnuts whose seeds are eaten. They are rich in proteins
8. **Spices:** Are plants whose parts are used for seasoning food such as pepper, curry, thyme and ginger.
9. **Latex plants**: Are plants that are grown for their useful latex (a milky fluid) e.g. rubber plants used for making natural rubber in the tropical countries
10. **Fibre Plants:** Are plants which produce fibre for the purpose of rope making, textile and bags production e.g. cotton, hemp etc.
11. **Beverage and drug plants:** - Are plants whose parts are taken as stimulants or drugs e.g. tea, coffee, cocoa and kola nut, quinine tree for medicine.
12. **Oil plant:** - Produce oil of economic value e.g. oil palm, sheanut, groundnut, coconut, castor oil plant and melon.

Plant classification based on life cycle and size

Based on life cycle (period or existence) plant can be classified as

1. Annual: - These are plants which complete their life cycle within one growing season or within a year e.g. Maize, Yam, Melon, cowpea, tomato etc.
2. Biennials: - These complete their life cycle within two years e.g. Banana, plantain, pineapple etc.
3. Perennials: - These persist over (more than) two years producing their yields every season e.g. orange, mango, oil palm cocoa etc.

Plant based on size fall into three categories;

1. Herbs are small plants with fleshy stem e.g. Spinach, waterleaf etc.
2. Shrubs are medium – sized plants with woody stem branch very close to the ground (soil) e.g. hibiscus
3. Trees are big plants with woody trunk, which branch at the top e.g. Iroko, Mahogany, Cashew, and Coffee etc.

Effects of agricultural activities on ecological system

The following agricultural or farming practices carried out by farmers have some consequences on the ecological system. These agricultural practices and their effects include;

Bush Burning: Bush burning involves the setting of fire in the bush to clear out the vegetation. Effects of bush burning include

- Destruction of the organic matter in the soil
- Atmosphere is polluted with smoke.
- Many of the micro-organisms are killed
- exposes the soil to erosion and leaching
- reduces the water holding capacity of the soil
- Bush burning leads to the extinction of some animals
- The ash produced by bush burning gives the soil a slightly alkaline nature

Overgrazing: Overgrazing is a situation where more animals than what can be supported on a particular pasture are put there to graze. It is a way of exceeding the carrying capacity of the soil. Overgrazing

- removes the vegetative cover of the soil
- exposes the soil to erosion
- destroys the soil structure
- More faeces are dropped on the soil which could improve the fertility of the soil.
- Weeds can eradicated from such lands
- It leads to compactness of the soil resulting from continuous trampling of animals.
- causes poor growth and regenerative capacity of vegetation

Tillage: Tillage is defined as the working, digging or breaking up of the soil in preparation for the planting of crops. Tillage encourages leaching

- helps to loosen the soil
- it enhances proper aeration of the soil
- tillage exposes the soil organisms and may kill some

- it changes the structure and texture of the soil
- tillage leads to changes in the ecology of the land
- Intensive tillage can lead to loss of soil fertility.
- It exposes the soil to erosion.

Deforestation: Deforestation is the continuous removal of forest stand (trees) either by bush burning or indiscriminate felling without replacing them. Deforestation

- It reduces water percolation due to absence of humus and dead leaves on the soil
- It reduces the amount of rainfall in the area
- Deforestation hinders micro-organisms activities in the soil
- It results in loss of nutrients through leaching and erosion
- It reduces wildlife population in the area concerned
- It reduces the humus content of the soil

Fertilizer application: This involves the application of certain chemicals or substances into the soil to improve its fertility. Effects of fertilizer application include

- It brings about the loss of organic matter or humus
- It deteriorates the structure of the soil
- Fertilizer increases the porosity of the soil
- It supplements nutrient content of the soil
- Excessive application of fertilizer can cause soil acidity
- The productive capacity of the soil is enhanced by the application of fertilizer
- It stimulates vegetative growth, hence it reduces soil erosion

Application of pesticides/herbicides: Pesticides are chemical substances which are used to destroy or kill pests while herbicides are also chemical substances in form of solution or gases capable of destroying weeds. Effects of pesticides application include

- It causes pollution of the environment.
- It affects or destroys other useful plants and animals.
- It reduces the population of the target insects or plants.

- Pesticides may leave undesirable residue in the environment.
- When such chemicals are washed into rivers or lakes, they can cause death of aquatic animals.

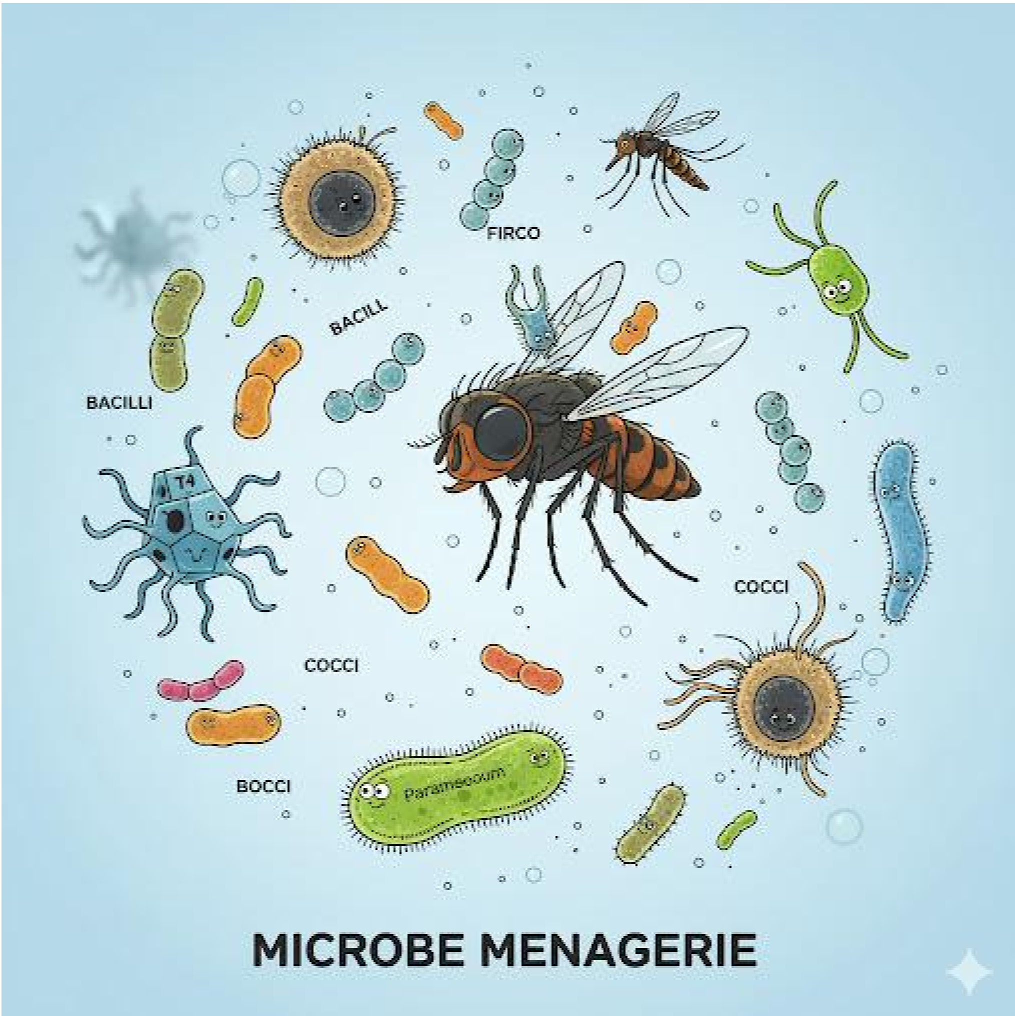
Evaluation

1. Based on size, classify water leaf, hibiscus and oil palm
2. Differentiate between shrubs and trees
3. Discuss the botanical classification (using example where appropriate)
4. In what ways are fruits and vegetable crops similar
5. Differentiate between root and cereals crops
6. Differentiate between annual and perennial crops giving two examples each.

Assignment

1. Which of the following is not an example of classification of plants (a) Herbs and shrubs (b) annual and perennials (c) Graminae and enphorbinosae (d) monocot and dicot
2. Plants can be classified based on all these except (a) botanical (b) size (c) agricultural use (d) planting season
3. Fruits crops are rich in (a) Vitamin and Minerals (b) Vitamins and protein (c) Mineral and carbohydrate (d) protein and carbohydrate
4. Spices include (a) Pepper and ginger (b) Palm oil and ginger (c) Lettuce and carrot (d) yam and maize
5. Which of these does not have negative effect on the ecological system (a) fertilizer application (b) crop rotation (c) tillage (d) bush burning

WEEK 10: MICROORGANISM AROUND US



Description and groups of microorganisms

Micro-organisms otherwise called microbes or germs can be defined as living things which cannot be seen with unaided eye but by the use of microscopes.

They exist almost everywhere, in water, air, soil, surface of objects, as well as on and within living organisms. They are carried by air currents from the earth's surface to the upper atmosphere. They occur most abundantly where there is food, moisture and adequate temperature for their growth.

It was the invention of microscope that opened the gateway to the world of these minute living organisms. The first person to discover microbes was a Dutch man called Anthony Leeuwenhoek (1632-1723). Using a simple microscope, he was astonished to discover that rain water that had been collected from pools was full of little organisms.

Groups of micro-organisms

Micro-organisms include all viruses, bacteria and the protists. Others are the cyanobacteria, certain fungi and algae.

- **Bacteria:** These are minute unicellular organisms or simple association of similar cells which multiply by binary fission. Most bacteria cells range between 0.2 μ-2μ in diameter and 0.0005mm-0.002mm long. Each bacterium cell has a cell wall with cytoplasm. There is no well-defined nucleus. Consequently, they are **prokaryotic organisms**.

There are different kinds of bacteria showing a range of shapes. Certain kinds of bacteria have long thread-like structures called **flagella** which assist in locomotion. Bacteria with spherical shape are referred to as **cocci** (singular-coccus). There are several forms as shown on the next page.

Streptococci - These are arranged in chains. They cause sore throat.

Staphylococci - These stick together to form irregular bunches. They cause boils.

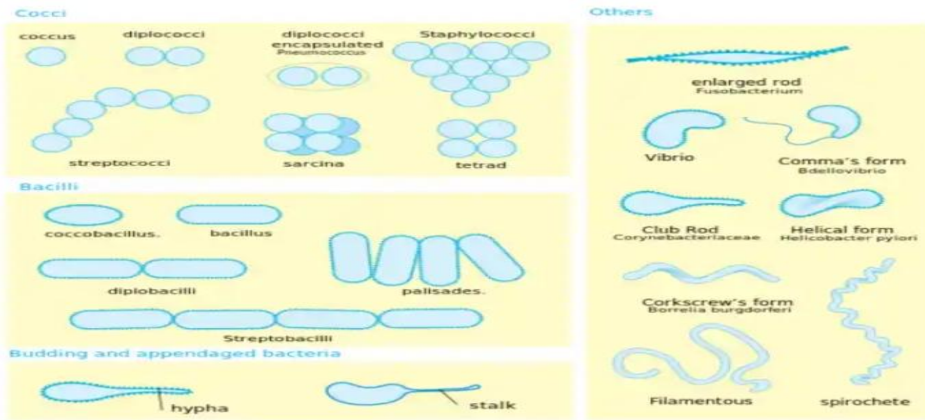
Diplococci - They occurs in pairs. E.g. pneumococci which causes pneumonia.

Bacilli - They is rod-shaped. They cause typhoid fever.

Spirilla (singular = spirillum) - These are rod-shaped bacteria twisted into a spiral shape.

Spirochaetes- These **are also s**piral in shape but are more flexible and slender with helically coiled structure e.g. *Treponemapallidum* which causes syphilis.

Vibrios - These are comma-shaped bacteria e.g. *Vibriocholera* which causes cholera.



- **VIRUSES:** Viruses are a large group of pathogens whose presence is felt only when they are in contact with living cells. They are very small and vary between 0.1µ-0.25 µ in diameter. The largest virus is less than one-fourth the size of typhoid bacterium. A virus consists of a nuclear material either DNA or RNA, enclosed within a protein coat. Outside living organisms they are like complex chemicals.

- **PROTISTS:** These are single-celled animals, most of which are only visible by means of microscope. They are common in fresh water and moist soils. Examples include *Euglena, Paramecium, Trypanosoma, Plasmodium,* etc.

- **FUNGI:** They are diversified in form. The blue and green growth on oranges, lemons, cheese and the white/grey growth on bread are usually signs of fungal infections. Fungi feed saprophytically. Examples of fungi include *Mucor, Rhizopus, Penicillium, Aspergillus*, etc.
- **ALGAE**: Most algae are unicellular and very small. They have chlorophyll. They occur abundantly in water, moist soils, bark of trees, stones, etc. Free floating microscopic algae are referred to as **phytoplanktons** and they form the major food of aquatic animals. Examples of unicellular algae include *Chlamydomonas and Protococcus*

Concept of culturing

A pre-requisite to studying microbes is their cultivation under laboratory conditions. Hence, it is important to know the nutrients and physical conditions needed by the organisms.

It is easier to grow bacteria, fungi, and algae in appropriate media. The material on or in which microbes grow in the laboratory is called **culture medium**. Some media are prepared from complex extracts of plant or animal tissues. A culture is the population of organisms cultivated in a medium.

If a culture contains only one living species of organism regardless of the number of individuals, it is said to be a **pure** or **axenic culture**. A culture which contains two or more species growing together is called a **mixed culture**.

An important medium used for growing microbes is **agar.** It is a dried polysaccharide extract of red algae which is used as a solidifying agent. It is not broken down by microbes.

Identification of micro-organisms

There are many ways of identifying micro-organisms around us. These include the use of microscopes stains of different types, types of colonies formed by the microbes, their food requirement and oxygen requirement of the organisms.

To determine the presence of microbes around us, suitable media are used to culture them in petri-dishes which are first sterilized by heating them in a pressure cooker, autoclave or oven.

Micro-organisms are not capable of growing in the air. The exposure of nutrient agar to the air will show the growth of different bacteria colonies in the air. Microbes commonly found in the air include viruses, bacteria, fungi, etc.

Microbes in aquatic habitat may be grouped into **natural water, soil and sewage microbes**. Examples of the first category include protists, algae, some fungi, bacteria, etc. Examples of the second group include *Rhizobium*, *Nitrosomonas*, and *Nitrobacter*. Examples of sewage microbes are *Entamoebahistolytica*, *Escherichia coli, etc.*

Microbes living in our bodies form normal population without causing any harm. However, under certain conditions, they may become dangerous. Pathogenic organisms cause diseases when body resistance is low or when normal microflora is de-established by the use of antibiotics. Any food item left unpreserved for a long time will be spoilt by the activities of microbes.

Pathogens enter the body through four main ways, namely: **air, food and water, contact** (direct or indirect), and **insect bites/cuts**.

Carriers of microorganism

Any agent that carries microbes from one place to another is called a carrier. Carriers can be living or non-living things. Non-living carriers include air, water, and food while animals (e.g. houseflies, mosquitoes, rats, cats, etc.) are the living carriers. Animal vectors carry pathogens either mechanically or biologically. In mechanical method, animals carry the pathogens on their bodies where they cannot grow or multiply. In biological method, the vector becomes infected by feeding on the body fluid of infected persons or animals.

Vector or Carrier	**Micro-organisms**	**Disease caused**
(i) Anopheles (female) mosquito	Plasmodium	Malaria fever
(ii) Tse-tse fly	Trypanosome	Sleeping sickness
(iii) Housefly	Vibro cholera	Cholera and typhoid fever
(iv) Aedes mosquito	Virus	Yellow fever

General evaluation

1. Give two examples each of the following microorganisms (i) fungi (ii) Bacteria (iii) Algae (iv) Protozoa
2. What do you understand by the word 'agar'
3. Describe ways by which microorganisms can be transmitted

4. State the vectors and the diseases caused by the following organisms (i) plasmodium (ii) trypanosome (iii) vibro-cholerea

Assignment

1. When bacteria are arranged in chains, they are called (a) spirilla(b) staphylococci (c) streptococci (d) bacilli
2. Viruses are considered to be living organisms because they (a) possess transmittable characters (b) move from one place to another (c) respond to stimulation (d) ingest food materials
3. Which of the following is not a protozoan? (a) paramecium (b) plasmodium (c) penicillium (d) Amoeba
4. Which of the following best describes a culture solution? (a) A population of micro-organisms cultivated in a medium (b) A population of weeds cultivated in a medium (c) Solution containing different chemicals (d) Solution containing dead organisms
5. Which of the following organisms is not a fungus? (a) Rhizopus (b) Plasmodium (c) Mucor (d) Aspergillus

THEORY

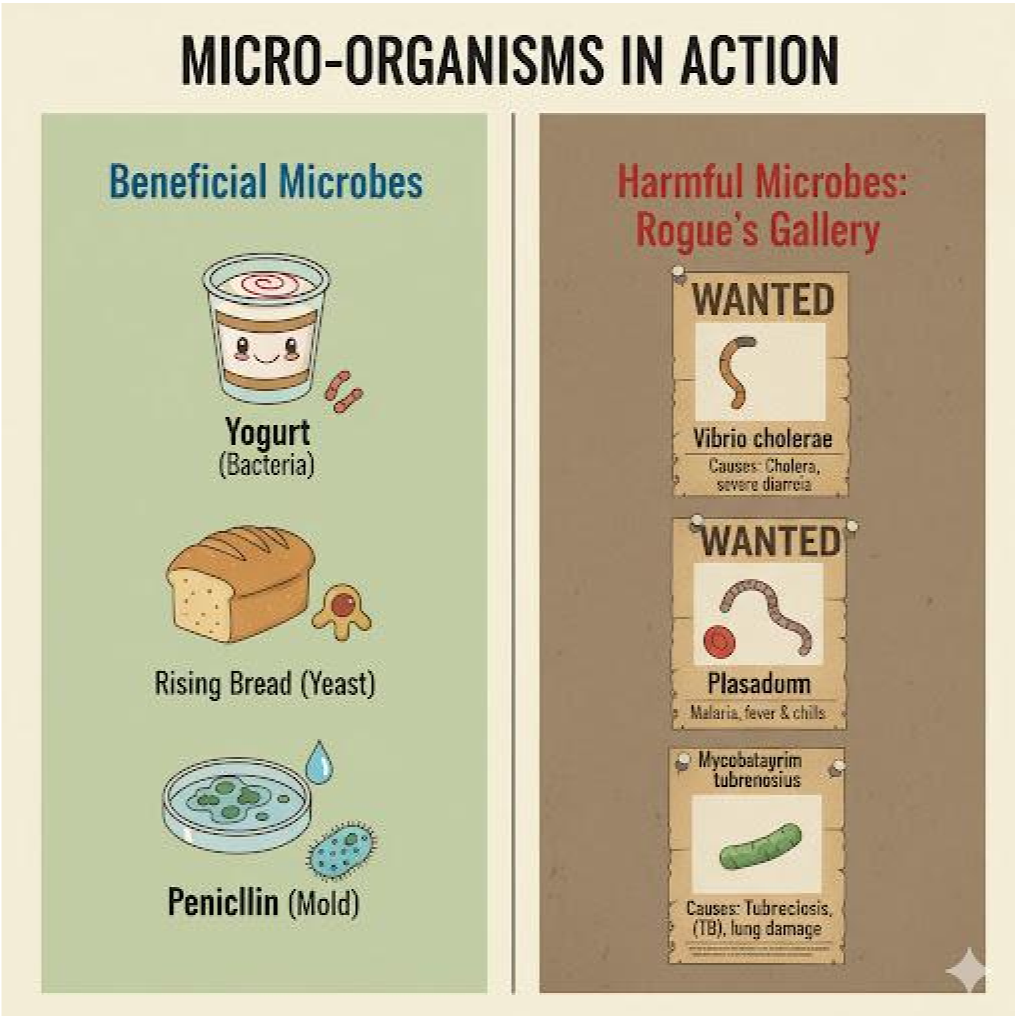
1. Differentiate between pathogens and vectors
2. Describe the structure of a virus

SS1 BIOLOGY LESSON NOTES

THIRD TERM

1. Micro-organisms in Action
2. Towards Better Health
3. Habitat
4. Aquatic Habitat (Estuarine and fresh water Habitat)
5. **Midterm examination**
6. Terrestrial habitat.
7. **Midterm break**
8. Terrestrial Habitat (Grassland)
9. Reproduction in Unicellular Organisms and Invertebrates
10. Reproduction in Unicellular Organisms and Invertebrates

WEEK 1: MICRO-ORGANISMS IN ACTION



Growth of micro-organisms

Bacteria reproduce by binary fission in which a single cell divides into two. This process is called asexual reproduction. The time interval required for the cell to divide into two is called **generation time.** This time varies from one organism to another. It strongly depends upon nutrient availability, temperature, gaseous requirement and pH. There are different phases in the growth of bacteria. These include the (i) lag phase (ii) logarithmic or exponential phase (iii) the stationary phase and (iv) decline or death. The growth of micro-organisms can be measured by using any of these methods:

1. Turbidity method.
2. Serial dilution method.
3. Squared transparent paper or cellophane method.

Beneficial and harmful effects of micro-organisms

Beneficial effects

1. Bacteria help to digest cellulose in herbivores.
2. In man, they synthesize vitamin K and B12
3. Bacteria and fungi are widely used in the synthesis of antibiotics
4. They are used to manufacture amino acids and vinegar
5. Bacteria are used to process milk into different tastes and flavours
6. They are used to decompose sewage into harmless inorganic compounds.
7. Microbial cultures are used to produce enzymes
8. Yeast is used as a leavening agent in baking industries.
9. Algae play important role in fertilizing the soil.
10. Bacteria are used to produce single-cells protein (SCP).

Harmful effects

1. Bacteria cause decay and spoilage of food items.
2. Materials like wood, paper, textiles, rubber and metals are destroyed by microbes.
3. They cause diseases of different types.

Diseases caused by

DISEASE	CAUSATIVE AGENT	SYMPTOMS	TRANSMISSION	CONTROL
Chickenpox	*Varicella Virus*	Fever, tiredness, and an itchy, blistery rash.	Contact	Isolate patient and use

				appropriate drugs.
Cholera	*Vibrio Cholerae*	• Watery diarrhea, vomiting, leg cramps.	flies, food , faeces, carriers	Personal hygiene.
Common Cold	*Rhino Virus*	Cough, runny nose, shivering, etc.	Contact	Take appropriate medications and avoid contact.
Dengue	*Dengue Virus (arbovirus)*	Severe headache, severe eye pain (behind eyes), joint pain, muscle and/or bone pain, rash, mild bleeding (e.g., nose or gum bleed, petechiae, or easy bruising),low white cell count.	Aedes Mosquito	Drain the water and clear the bushes around.
Diarrhoea	*Giardia intestinalis*	Frequent passing of watery faeces, cramps and pains in the abdomen (stomach), nausea and vomiting.	contaminated stools	Personal hygiene
Diphtheria	*Corynebacteri umdiptheriae (Bacteria)*	A sore throat, hoarseness, painful swallowing, swollen glands (enlarged lymph nodes) in your neck, thick, difficulty breathing or rapid	Contact	Use antibiotics and avoid contacts.

		breathing, nasal discharge, fever and chills, & malaise.		
Leprosy	*Mycobacterium leprae*	Disfiguring skin sores, lumps, or bumps (that do not go away after several weeks or months), loss of feeling in the arms and legs, muscle weakness.	Long and close contact	Use antibiotics and avoid contacts. Patients should be isolated.
Measles	*Measles virus* *(ParaMyxoVirus)*	A high temperature, sore eyes (conjunctivitis), runny nose, small white spots, harsh dry cough, going off food, tiredness, aches, pains, diarrhoea and/or vomiting.	Contact	Take appropriate medication and avoid contact.
Pneumonia	*Diplococcus pneumonia*	High fever, shaking chills, cough with phlegm (a slimy substance) which doesn't improve or worsens, shortness of breath, chest pain when you breathe or cough, suddenly feeling worse after a cold, etc.	Bacteria transmission by contact.	Use of antibiotics and avoidance of contact.
Poliomyelitis	*Polio Virus or Enterovirus*	Fever, sore throat, headache, vomiting,	houseflies, food and water	Good hygiene and appropriate

		fatigue, back pain or stiffness, neck pain or stiffness, pain or stiffness in the arms or legs, muscle weakness or tenderness, & meningitis.		medication.
Rabies	*Rhabdovirus*	Fever, cough, sore throat, etc.	Mad dog bites	Treat dogs and seek urgent medical attention in cases of bites.
Septic Sore Throat	*Streptococcus Bacteria*	Fever, nasal drainage, ***sore throat***, swollen glands, difficulty swallowing, and irritability.	Contact	Use of antibiotics and avoidance of contact.
Sleeping Sickness	*Tryponosoma Brucei*	• Anxiety, drowsiness during the day, fever, headache, • insomnia at night, mood changes, sleepiness, • sweating, swollen lymph nodes all over the body, • swollen, red, painful nodule at	reaches lymph nodes via transmission thru fly bites	Clear vegetation around, use insecticides and take appropriate medication.

		site of fly bite.		
Smallpox	*Variola Virus*	High fever, vomiting, fatigue, backache, a raised spotted rash, etc.	Contact	Take appropriate medication and avoid contact.
Tuberculosis	*Mycobacterium tuberculosis*	Malaise, weight loss, and night sweats.	Bacteria transmission by cough	Isolate patients and use appropriate antibiotics.
Tetanus	*Clostridium tetani*	Muscle spasms and breathing problems.	bacteria in soil through wounds	Treat wounds urgently.
Typhoid	*Salmonella Typhi*	• Abdominal tenderness, • agitation, bloody stools, chills, confusion, difficulty paying attention, fluctuating mood, nose bleeds, severe fatigue, weakness, etc.	Flies, food etc.	Treat water before drinking. Maintain personal hygiene. Use appropriate antibiotics.
Whooping cough	*Hameophilius Pertussis*	Cough, whooping sound during breathing, etc.	coughing and sneezing	Isolate patients and use appropriate antibiotics.
Influenza Flu	*Orthomixovirus*	fever and muscle aches, cold, runny nose, sore throat, etc.	Contact	Take appropriate medication and

				avoid contact.
Malaria	*Plasmodium*	Chills, headache, muscle aches, tiredness, nausea, vomiting and diarrhea.	Bite from female *Anopheles* mo squitoes	Use of drugs, insecticides, etc.

General evaluation

1. Mention the observable phases in the growth of microbes.
2. State five uses of microorganisms in industries.
3. Outline four general ways of controlling microorganisms.
4. Mention three microorganisms each that are involved in Nitrogen and Carbon cycle.
5. State three uses of microorganisms in (i) Agriculture (i) Medicine
6. Describe three ways of measuring growth of bacteria.
7. List five ways through which microorganisms can be transmitted

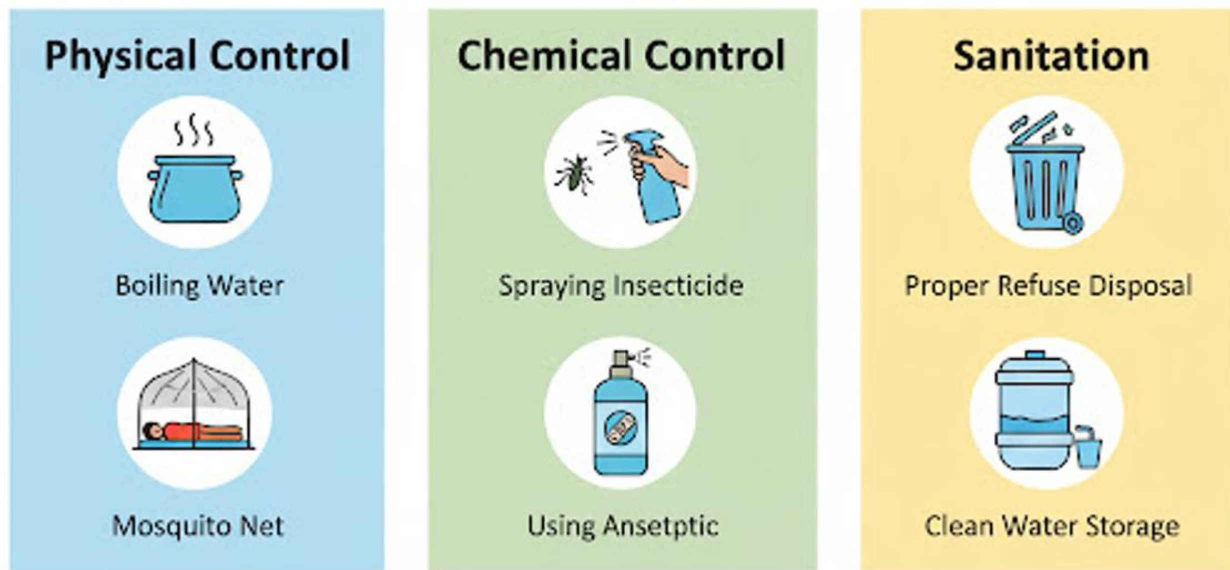
Assignment

1. The growth phase in bacteria in which cells divide steadily at a constant rate is called (a) Exponential phase (b) lag phase (c) stationary phase (d) decline phase
2. Which of the following microbes causes cholera? (a) Virus (b)Bacterium (c) Protozoan (d) Fungus
3. Growth of micro-organisms can be measured by the following methods except____________ method (a) serial dilution (b) turbidity (c) squared transparent paper (d) dry weight
4. The following practices contribute to the control of the spread of diseases except (a) sewage treatment with chemicals (b) proper sewage disposal (c) disinfecting the surrounding (d) using human faeces as manure
5. The vector of the trypanosome parasite is (a) housefly (b) tse-tse fly (c) mosquito (d) black fly

THEORY

1. Define the following phases in microorganisms growth (i) lag (ii) exponential (iii) stationary.
2. State four ways in which each of the following organisms are beneficial to humans. i. Bacteria ii. Fungi

WEEK 2: CONTROL OF HARMFUL MICRO-ORGANISMS



The control of harmful microbes include removal, inhibition of growth or killing by physical agents/processes and chemical agents or antibiotics. Some common methods of controlling harmful microorganisms in order to maintain good health include

1. **High and low temperature:** Boiling or heating of food, pasteurization of milk, sterilization of medical instruments and freezing of food to reduce the activities of microbes to barest minimum.
2. **Covering of food** to prevent vectors and pathogens in the air from coming in contact with the food.
3. **Antibiotics** such as amphiclox, ampicillin, penicillin, tetracycline, are drugs used to kill many bacteria causing diseases.
4. **Antiseptic** such as Dettol, Milton, chlorine water, medicated soap and hydrogen peroxide destroy micro-organisms while others prevent the multiplication of the micro-organisms.
5. **Disinfectants** are stronger antiseptic. Examples are sanitas, Lysol and izal. They are used to disinfect hospitals warehouses and public buildings. Antiseptics and disinfectant have to be diluted to render them gentle or mild to the skin

6. **High salinity (salting):** Salt is used to preserve food. When salt is applied to food items like fresh meat or fish, the micro-organisms are destroyed. The bacterial cells are plasmolysed due to the movement of water from the cells of the bacteria.
7. **Dehydration**: When foodstuffs such as fish and meat are dried, micro-organisms cannot thrive on them. Bacteria need water to survive. So dehydration prevents the survival of micro-organisms.
8. **Sanitation:** Keeping the body and the environment clean.
9. **Isolation of infected persons:** Persons suffering from infectious diseases such as tuberculosis and cholera must be isolated so as to prevent the spread of such diseases to other members of the community.
10. **Balanced diet:** Eating balanced diet everyday helps to promote good health and high body resistance to diseases.

Vectors and ways of controlling vectors

A vector is an animal which transmits disease-causing organisms (pathogens) from the victim of that disease to another person.

Control of mosquitoes

- Draining of swamps
- Clearing of bushes around houses
- Sleeping in a room protected by mosquito net.
- Spray swamps or rooms with insecticides.
- Spray oil on stagnant water.
- Using insect repellant on body.
- Use of drug.
- Burying broken pots and cans.

Control of Houseflies

- Spraying with insecticides
- Destruction of breeding spots

- Use of poison baits
- Closing of pit toilets
- Covering of food
- Keeping environment clean

Maintenance of good health

There are many ways of maintaining good health. These include:

- **Refuse disposal**: Refuse are solid waste materials discharged through human activities from homes and industries into the environment. Reckless refuse dump around dwelling places creates bad odour, provides breeding grounds for insects and rodents that spread diseases. Refuse disposal can be done through the following ways; Provision of dust bins in strategic locations, Burning of refuse in incinerators, Dumping them in isolated areas far from human habitation, Burying refuse in sanitary landfill.

- **Sewage disposal:** Sewage are waste water materials discharged from laundries, kitchen, toilets, bathrooms e.g. urine and faeces. Sewage disposal is done through the use of pit toilets where faeces and urine are passed into deep pits, the use of septic tanks where water is used to flush faeces and urine into a big tank dug in the ground, community treatment process where sewage from various homes are collected and treated before being discharged into the oceans or rivers.

- **Protection of water:** In view of various diseases which man contact because of drinking unclean water, water should be protected through the following ways: addition of alum to water, boiling of water before drinking, filtration of water on cooling, addition of chlorine to kill microscopic germs, storage of water in clean containers.

- **Protection of food:** The following methods of food protection are recommended: Keep food in refrigerators or deep freezers, boil or cook raw food properly before eating, there should be inspection of food meant for public consumption, washing of hands before and after eating of food, food should be preserved through canning, keep the environment where the food is prepared clean, avoid exposure of food to flies and other micro-organism.

Roles of health organizations

The administration of health services in Nigeria is achieved through the following approved organizations:

1. Ministry of Health
2. University Teaching Hospitals

Countries cooperate to tackle health issues. At international level, health control is organized top prevent the spread of diseases and also to provide aid to needy areas. This can be in the form of drugs, medical equipment, money, etc.

International Health Organizations include: World Health Organization (WHO), United Nations Children's Fund (UNICEF), International Red Cross (IRC), United Nations Educational, Scientific and Cultural Organization (UNESCO).

World Health Organization (WHO)

This is the world's principal agency for dealing with health and nutritional problems. It was established in 1964 and became operational in 1948. The functions of WHO are as follows:

1. It provides and assists national governments at their request to strengthen their health services.
2. It promotes and provides improved methods of training health, medical related professional experts for member countries.
3. It promotes cooperation among scientific and professional bodies for the improvement of health.
4. It cooperates with other organizations in the improvement of nutrition, sanitation, housing working conditions and other matters that relate to health.
5. It helps and promotes material and children's health care and welfare.
6. It produces medical publications.

7. It provides drugs and vaccines in cases of emergency

United Nations Children's Fund (UNICEF)

This is a body set up by the U.N. It performs the following functions:

1. It provides emergency needs for children in areas affected by diseases or famine.
2. It improves the nutritional condition of undernourished children.
3. It feeds and cares for disabled children.
4. It undertakes immunization programmes for children's diseases like measles, whooping cough, etc.
5. It ensures the provision of clothing and other needs for children

International Red Cross

This is a humanitarian organization whose functions are to serve humanity during peace and war times. During war, it performs the following functions:

1. Care for the injured.
2. Provision of emergency aid.
3. Negotiating for the exchange of the prisoners of war.
4. Evacuation of refugees.
5. Welfare of war prisoners.

United Nations Educational, Scientific and Cultural Organization (UNESCO)

Although, this is not directly a health organization, it assists health services indirectly by raising the educational standards of the people in developing countries.

Evaluation

- Describe at least five control of Harmful Microorganisms.
- Define and state ways of Controlling Vectors.
- Explain the maintenance of good health.

- State the roles of health organizations in disease control.

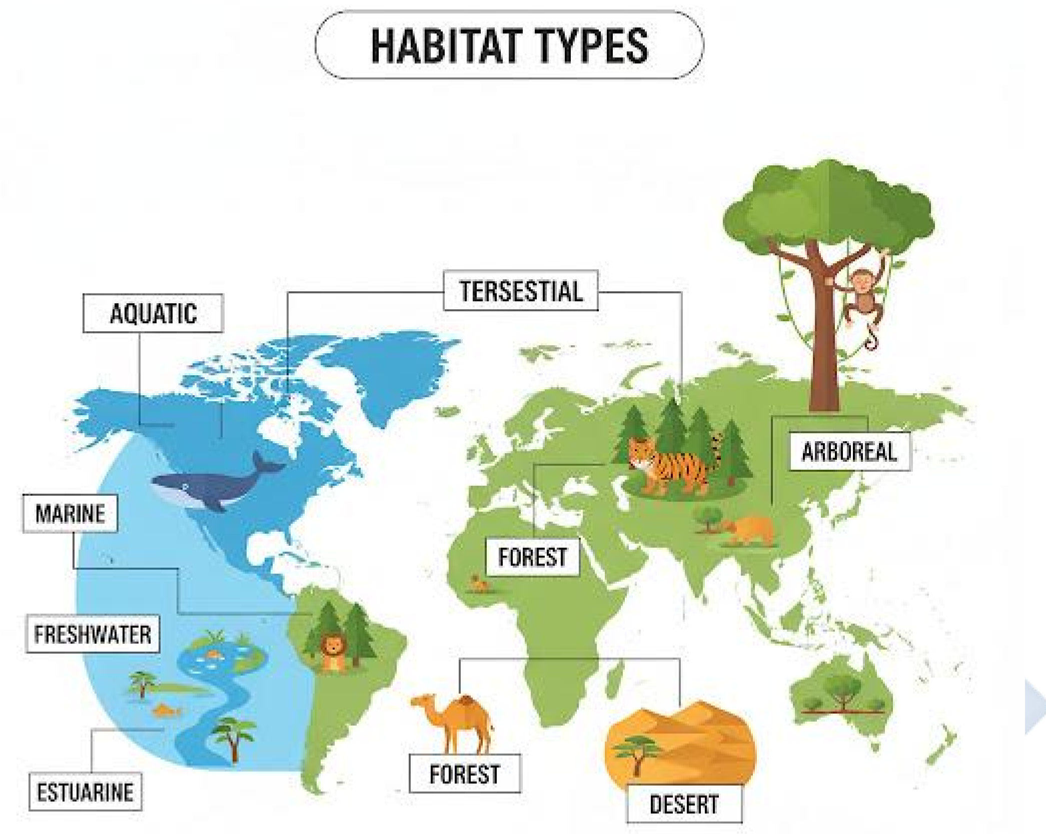
Assignment

1. A way of providing good health in a community is (a) control of diseases (b) sewage disposal (c) refuse disposal (d) all of the above
2. Which of these is not a vector? (a) Black fly (b) Snake (c) Dog (d) Housefly
3. The process of heating liquid food at a controlled temperature thereby enhancing its quality and destroying harmful micro-organisms (a) Pasteurization (b) Boiling (c) Frying (d) None of the above
4. An agent that stops the growth of fungi is called (a) fungistat (b) fungicide (c) germicide (d) none of the above
5. The process by which water is removed from bacteria cells which leads to the cell been plasmolysed is referred to as (a) drying (b) salting (c) dehydration (d) pasteurization

Theory

1. Name the causative organism and vector of Malaria
2. List five symptoms of Malaria
3. Mention five ways by which mosquito can be control and state the reason for each method

WEEK 3: MEANING AND TYPES OF HABITAT



Habitat is a place where organisms (plants, microorganisms and animals) are naturally found e. g. the habitat of tadpole is the bottom of fresh water ponds or streams

There are three main types of habitats, namely; aquatic habitat (in or around water), terrestrial habitat (in or on land) and arboreal habitat (in or on trees)

There are three kinds of aquatic habitat;

1. marine/salt water habitat e.g. ocean, seas
2. brackish water habitat (where salt and fresh water mix) e.g. delta, lagoon, bay
3. Fresh water habitat (contain little or no salt) e.g. lakes, rivers, streams.

Step 2: characteristics of Marine habitats

Characteristics of marine habitats are as follow:

1. The marine habitats constitute the largest habitat in the biosphere (70% of the earth's area)
2. They do not undergo sudden or rapid changes in physical factors such as temperature, PH and specific gravity. Hence they show the greatest stability of all habitats.
3. Chemical composition :- marine water consists of many kinds of dissolved ions including Na+, K+, Mg2+, Ca2+, Pb43-, I-, NO3- e. t. c.
4. Hydrogen (H+) concentration (PH): - salt water is alkaline in nature with PH of about 8.0 – 9.0 near the surface.
5. Salinity (salt concentration of water). The seawater has a high salinity. The average salinity of seawater is 35 parts per thousand.
6. Density of marine water is high. It is about 1.028 while that of fresh water is 1.0. This allows many organisms to float in it.
7. The temperature of the sea changes less quickly than that of the land. However, the temperature falls with increase in the depth of the sea.
8. Oxygen concentration is highest at the surface where the atmospheric oxygen dissolved in water. The concentration of oxygen decreases with depth.
9. Waves are temporary movement of surface water of the sea which occurs in any direction. They are caused by wind blowing against the surface of water. They also bring about the mixing of seawater. Waves can also beat against the shore and sometime caused it to be eroded.
10. Tides are alternate rise and fall of the surface of the sea at least twice daily. Tides are caused by water distribution resulting from the combined gravitational pull of the earth by the sun and moon.

Zonation of marine habitat

Horizontal zonation

The marine habitat is made up of the sea shore and open sea. The major zones of the marine habitat are generally as follow

1. Supratidal or splash zone is the exposed zone with occasional moisture being the area where water splashes when waves breaks at the shore.
2. Intertidal or neritic zone is the planktonic zone which is exposed at low tide or covered by water at high tide. This zone has high photosynthetic activities because of abundant sun shine. Water temperature fluctuates.
3. Subtidal or littoral zone is about 200m deep, constantly under water, with abundant sunlight and nutrient.
4. Benthic zone is about 500m deep with low light penetration and low nutrients. The water is dark, cold and with little oxygen. Hence, it is unfavourable for life.
5. Abyssal or palegic zone is about 7000m deep with low light penetration, low temperature and high pressure. The low light leads to low photosynthetic activities. Hence food production is primarily by chemosynthesis.
6. Hadal or aphotic zone is the deepest, over 7000m deep. This forms the floor or the bed of the ocean. No light penetration or photosynthesis

Vertical zonation

Based on light penetration or depth, the marine habitat can be zoned into three;

1. Euphotic zone is the area in direct contact with sun shine. Hence, there is enough light penetration for photosynthesis. Therefore producers, consumers and decomposer are all present.
2. Disphotic zone is a region of dim light. Light penetrate the water with low too intensity for photosynthesis to take place. Consumers and decomposers are found in this zone.
3. Aphotic zone is the bottom or bed of the seas and oceans. It is characterized by cold dark water without light penetration and very few living organisms.

Distribution of organisms and adaptation to marine habitat

- Organisms of the splash zone include periwinkles; crustaceans e.g. ghost crab, seaweeds and sargassum (algae).
- Those of intertidal zone include starfish, sea anemones, sponges, sea urchin, annelids, mollusca and barnacles.

- In the subtidal zone are snails, crabs, lobsters and crayfish.
- The benthic zone is unfavourable for life. The producers are absent, only few saprophytic animals are present.
- The neritic zone house plankton (microscopic floating organisms e.g. diatom, algae, protozoa, crustacean and worms) and nekton (e.g. fishes, crabs, prawns and whales).
- Oceanic water house sharks, croaker, sea cat fish, mackerel, bonga fish etc.

Adaptation of animals to marine habitat

Animals including barnacles, fishes, crustaceans e.t.c. found surviving in marine habitat do so with the following adaptive features;

- Barnacles have

  i. protection mantle for attachment to rock shore and water retention
  ii. Cilia for feeding.
  iii. Shell that prevents desiccation (drying up)

- Fishes possess

  i. Reduced or no kidney to retain urea in their body to cope with high salinity e.g. cartilaginous fishes like shark, dogfish etc.
  ii. Salt secreting glands in their gills or eyes for maintaining osmoregulation (salt balance) e.g. bony fishes like tilapia, herring etc.
  iii. Tube feet which enable them to hold on to rock shores and hard shell to prevent desiccation e.g. starfish, whales.

- Whale has

  i. fins for stability in water
  ii. An organ in front of the nostril for detecting pressure changes in water.
  iii. A thick layer of dermal fat insulation or food reservoir.

- Shrimps possess powerful claws for holding food or prey.

- Periwinkles possess lungs for breathing and foot for attachment.
- Crabs burrow fast into the mud to protect them against predators, strong waves or hide.

Adaptation of plants to marine habitat

Plants such as seaweeds, algae, sesuvium and diatoms are naturally found in marine habitat with the following adaptive features;

- Seaweeds have
    i. hold–fast for attachment.
    ii. mucillagenous cover to prevent desiccation.
    iii. Divided leaves or floating devices for buoyancy.
- Algae (e.g. sargassum) have
    i. chlorophyll for photosynthesis.
    ii. Small size or large surface area for floating in water.
- Planktons (e.g. diatoms) possess;
    i. air space in their tissues
    ii. Rhizoid for attachment to rocks
    iii. Air bladder for buoyancy (floating).

Examples of food chain in a marine habitat include

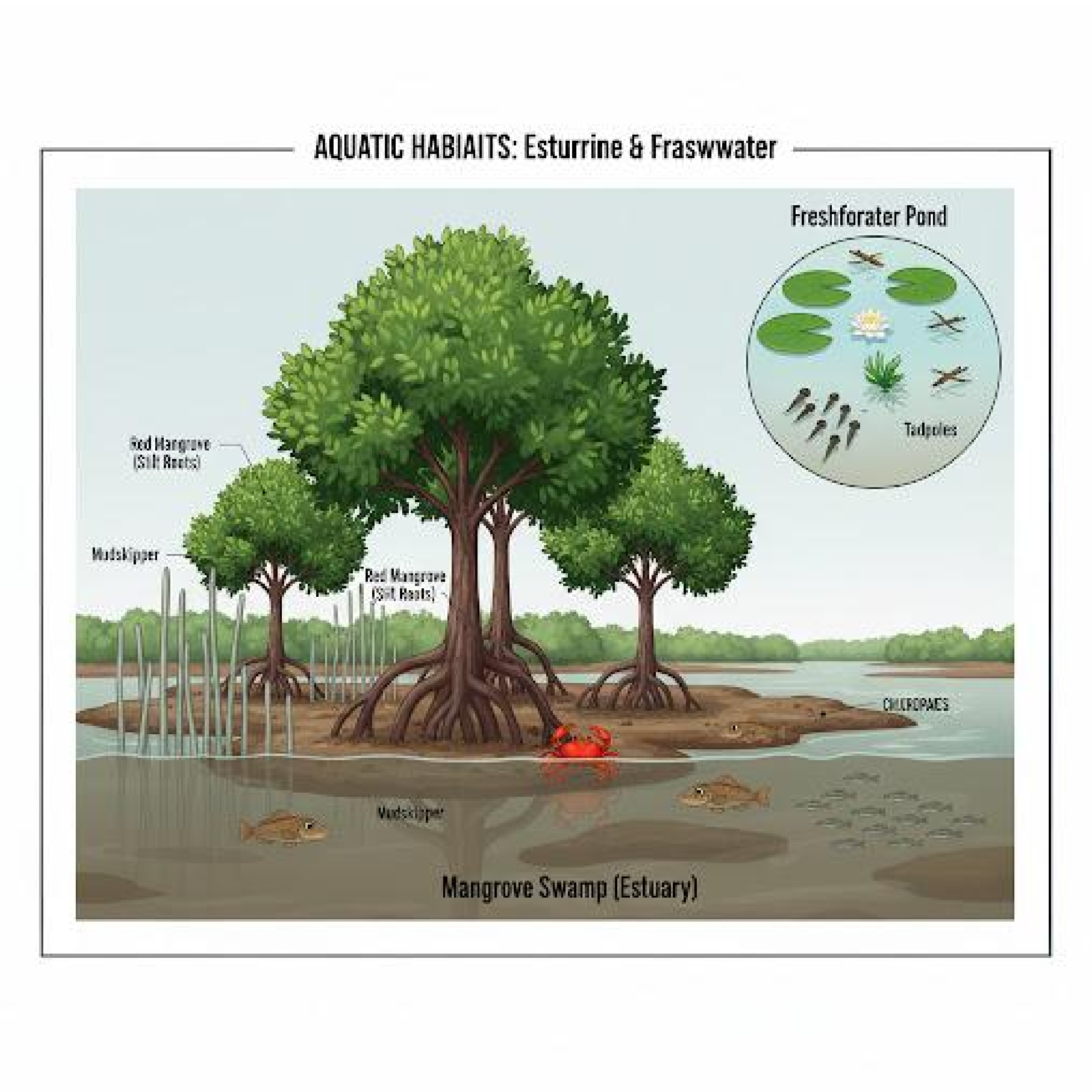
1. Diatom → crabs → tilapia
2. Diatom →zooplankton → tilapia →shark

Assignment

- Define and state the types of Habitat
- Describe the characteristics of Marine Habitat
- Explain the zones found in the marine habitat

- State the distribution of Organisms and Adaptations to Marine Habitat

WEEK 4: ESTUARINE AND FRESH WATER HABITAT



Meaning and characteristics of estuarine habitats (brackish water habitats)

An estuary is a body of water formed at the coast where fresh water flowing towards the sea mixes with sea (salt water) flowing inland. Estuarine habitats include deltas, lagoons and bays.

Characteristics of estuaries

1. The salinity fluctuates.
2. The specific gravity is less than that of the sea.

3. They have high turbidity due to frequent disturbances. Hence rate of photosynthesis and respiration by organisms reduces.
4. The water is shallow.
5. They have low diversity of species compared to marine habitat.
6. They have high level of nutrients
7. They have low oxygen content, hence anaerobic activities are common.

Step 2: Plants and animals distribution and adaptation in estuaries

Plants distribution and adaptation.

Plants found in estuaries include planktons, algae, red and white mangrove and they have the following adaptive features;

- Planktons (diatoms) have; i. air spaces in their tissues ii. Rhizoid for the attachment to rock shores iii. Air bladder for buoyancy
- Algae have: i. chlorophyll for photosynthesis ii. small size or large surface area for floating in water
- Red mangrove has; i. stilt roots with rootlets that have air-spaces for air conduction to the root tissues and support to prevent washing away of the plant by the tide ii. Seeds which germinate while they are still on the parent plant, thus preventing the carrying away of the seedlings by water current.
- White mangrove has pneumatophores (breathing roots) for gaseous exchange.

Animals' distribution and adaptation

Animals including mosquitoes, crustaceans, mollusca, worms, fishes etc. found in estuaries survive possessing the following features;

- Mosquito larvae and pupae possess breathing trumpets for gaseous exchange
- Crustaceans and water snails burrow into the mud against predators, strong waves or tides.
- Worms have strong protective and impermeable covering against high salinity.
- Mudskippers have fins for crawling on land and swimming in water.

- Fishes have fins for movement and swimming bladder for buoyancy.

Food chain in estuarine habitats

1. Detritus → worms → snails → birds
2. Diatoms → shrimps →fishes
3. Diatoms → small fish → sharks → man

Fresh water habitats and characteristics

This is a body of water formed mainly from inland waters and it contain very low or no salt. Fresh water is of two types based on its mobility;

1. Lotic fresh waters: - These are running waters flowing continuously in a specific direction e.g. rivers, springs, streams
2. Lentic fresh waters: - These are stagnant waters which do not flow e.g. lakes, ponds, puddles, swamps and dams

Characteristics of fresh water habitats

1. It contains little or no salt. Salinity is 5 parts per thousand i.e. 0.5%.
2. It is small in size.
3. Oxygen concentration is high, being available in all parts of the water body, especially at the surface.
4. The water is shallow, hence sunlight penetrate to the bottom.
5. The temperature varies with seasons and depth.
6. It has seasonal variation; decreasing or drying up in dry season and increasing in rainy season
7. Water currents affect distribution of organisms, salts and gases, especially in lotic fresh waters

Plants and animals distribution and adaptation in fresh water

1. Plants distribution and adaptation in fresh water

Plants of fresh water include water lily, spirogyra, water lettuce, water weeds etc. and they have the following adaptive features;

- Water lily has i. air bladder ii. Expanded tips and light weight which keep it afloat.
- Spirogyra has mucilaginous cover for protection
- Water lettuce has hairs in leaves to trap air and keep it afloat
- Water weed (elodea) has a long and flexible petiole for swinging with water currents.

2. Animals distribution and adaptation

Animals of fresh water habitats include protozoa, duck, pond skaters, hydra, fishes etc. their adaptive features include

- Protozoa have contractile vacuole for osmoregulation in water.
- Duck has webbed feet for locomotion and serrated beak for sieving food in water into its mouth.
- Hydra has slippery surface, hooks and suckers for attachment to water particles.
- Pond skaters has long legs for skating on water surface
- Fishes have swim bladders for buoyancy and gills for respiration

Food chain in fresh water habitats

1. Diatoms → fish fry →tilapia
2. Spirogyra → tad poles → carps → king fish
3. Algae → mosquito larva → small fish

Evaluation

- Define estuarine habitat and state their characteristic
- Describe the distribution of plants and animal in the estuarine habitat and also the food chain.
- Define fresh water habitat and state their characteristics
- Describe the distribution of plants and animals in the fresh water habitat and also the food chain present in the fresh water habitat.

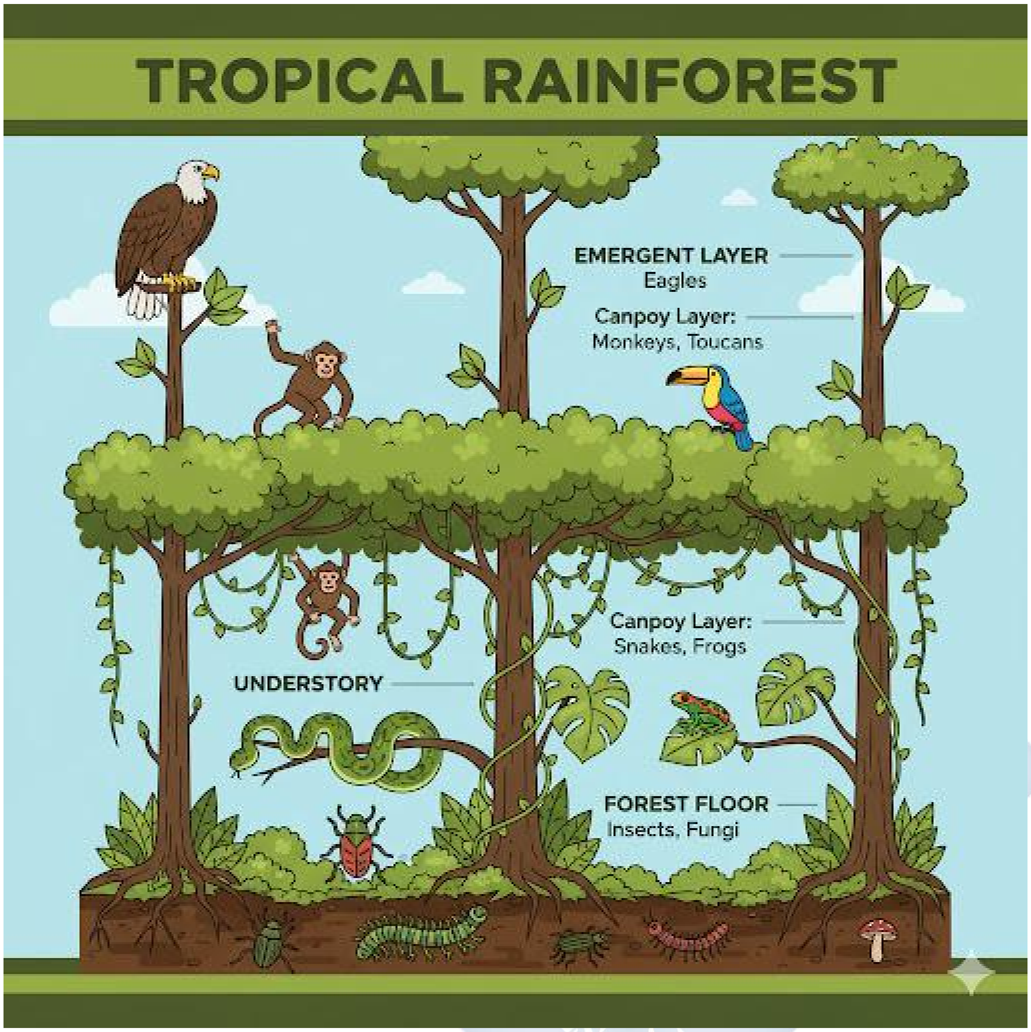
Assignment

1. Buoyancy in salt water is ensured by the following **except** A. divided leaves B. chlorophyll C. floating devices D. air bladder
2. The mucilaginous cover in sea weed and spirogyra is mainly for A. protection B. osmoregulation C. avoiding desiccation D. feeding
3. Which of the following is **not** a fresh water habitat? A. puddle B. swamp C. stream D. sea
4. Which of these is not an adaptive feature in a marine habitat? A. bladder for floating B. hold fast for attachment C. fur to prevent water loss D. rhizoid for attachment to rock
5. The following are characteristics of fresh water habitats **except** A. low salt content B. high salinity C. shallow water D. can be stagnant or running water

Theory

1. In a tabular form, state five differences between fresh water and a marine habitat
2. State three adaptive features each of plants and animals to fresh water habitat

WEEK 6: MEANING AND TYPES OF TERRESTRIAL HABITAT



Terrestrial habitats

Organisms of the land are called terrestrial organisms. They include plants and animals that are found living on the ground and under the ground.

Basically, terrestrial habitat is subdivided into four main parts, namely;

1. marsh
2. forest
3. grassland/ savanna

4. arid land/ desert

Marsh

Marsh is a low land, flooded in rainy season and usually waterlogged because of poor drainage. The vegetation is predominantly of grasses and shrubs. When trees grow in a marsh, it is called a swamp. Marsh is a transition between the aquatic habitat and terrestrial habitat.

Formation of a marsh

Marshes develop as a result of water overflowing its banks to accumulate on the adjoining coastal or low land area such as flood plains of rivers. This can be enhanced with extensive rainfall.

When ponds and lakes are filled up with soil and organic debris of plants, marshes can also be formed. Marshes formation is therefore a gradual process. Marshes can either be fresh water or salt water marshes.

Characteristics of a marsh

1. A marsh is lowland.
2. It is always flooded, wet and waterlogged.
3. It sometimes has pool of standing water.
4. It has a high relative humidity
5. Its water sometimes contain much decaying organisms
6. The water has a foul smell

Organisms of the Marsh

There are various plants and animals in this habitat. The plants include algae, grasses, water lettuce, water lilies, white and red mangrove, and raffia palms etc.

Animals found in the marsh include mangrove crabs, lagoon crabs, hermit crabs, mud-skippers, fishes, frogs, snakes, crocodiles, mammal etc.

Adaptive features of organisms of the marsh include

1. They must be able to tolerate the salinity of the soil or water
2. They have to tolerate low oxygen concentration in the soil or water

Plants of fresh water marsh have other adaptive features similar to those of fresh water habitat. Likewise the plants of salt water marsh.

Saprophytic organisms (e.g. bacteria) which live on dead organic matter in marshes have to adapt to anaerobic condition.

Food chains in marshes

1. Flowering plants → insects → frogs → crocodiles
2. Humus → earthworms → frogs → snakes

Forests

A forest is a community of plants in which trees species are dominant. There are different kinds of forest whose distribution is determined mainly by climatic factors such as temperature, rainfall and at times by soil elevation and man's activities such as farming, lumbering, bush burning, construction of roads and building.

The major type of forest in Nigeria is the rain forest

Characteristics of a forest

1. The forest is rich in epiphytes and climbers
2. The interior of the forest has high humidity, low light intensity and damp floor.
3. Presence of tall trees with canopies and existing in layers (stratified).
4. Trees are mesophytes with broad leaves.
5. The trees have buttress roots to support their heavy weight and height.

6. The trees have thin barks for gaseous exchange and transpiration.

Plants and animals distribution and adaptation the forest habitat

Plants distribution and adaptation

Forest plants (trees) include African walnut, mahogany, teak, obeche, iroko, oil palm, ferns (pteridophytes), bryophytes (mosses and liverwort), epiphytes (orchid), fungi and mistletoe etc. these plants adapt to life in the forest in the following ways;

- Possession of strong tap root systems and buttress roots.
- Possession of tall unbranched trunks
- Possession of broad leaves
- Epiphytes have mechanism (the aerial roots) storing water and absorbing moisture from air while growing on tree branches.
- Mistletoe (plant parasite) develops root system that can penetrate the stem of a plant withdrawing manufactured food directly from phloem vessels of the host plant.

Animals' distribution and adaptation

Most forest animals are arboreal (living on trees) and these include bats, monkeys, snakes, squirrels, birds, tree frogs, chameleons. Some live in the soil e.g. earthworms and beetles while others live among the litters on the ground e.g. millipedes, ants, snails.

These animals adapt to the forest in the following ways

- Monkeys have prehensile tails and long limbs for climbing and jumping.
- Bats modify their limbs into wings for flight
- Green snakes have protective colouration to camouflage
- Chameleon has prehensile tail and opposable digits for grasping as well as protective colouration to camouflage
- Apes moves in groups for protection, with high sense of sight
- Earthworms and snails have water permeable cuticle to reduce water loss and prevent desiccation.

- Birds have powerful wings for flight

Food chains in a forest

1. Green plants → grasshoppers → toads →hawks
2. Green plants → monkeys → lions

Evaluation

- Define and state the types of terrestrial habitats
- Explain marsh under the following headings; definition, mode of formation, organism found and food chain.
- Define Forest and state its characteristics
- Describe the plants and animal distribution and also food chain found in the forest habitats

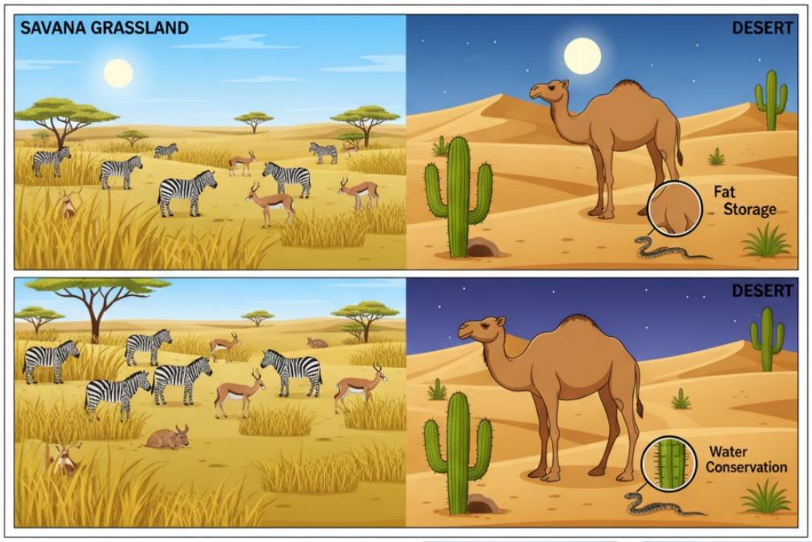
Assignment:

1. Marsh is a ……… flooded and waterlogged (a) highland (b) lowland (c) island (d) mountain
2. Marsh is described as a ……… when trees grow there (a) swamp (b) forest (c) puddle (d) desert
3. Dominant plants in the forest are (a) grasses (b) shrubs (c) trees (d) vegetables
4. An examples of forest plants' parasite is (a) orchid (b) mosses (c) liverwort (d) mistletoe
5. The plants of the forests are described as (a) hydrophytes (b) mesophytes (c) xerophytes (d) neophytes

Theory

1. What is a marsh? State two types of a marsh
2. State five unique features of a forest.

WEEK 8: GRASSLAND AND DESERT



Meaning and characteristics of a grassland [savanna]

This is a plant community in which grass species are dominant, but trees and shrubs may be present.

Characteristics of grassland

1. Temperature is usually high and sunshine is intense.
2. The relative humidity is low and rainfall scanty (60 – 150cm annual rainfall).
3. Abundant grassland with few short trees sparsely distributed
4. Bush fire is frequent and trees are fire resistant
5. Deciduous plants (plants that shed their leaves in dry seasons) are present.
6. Plants possess underground stems and deep roots to search out for water
7. Trees have modified leaves for adaptation to the environment

Step 2: Types of savanna

Basically, there are four major types of savanna in Nigeria, namely;

1. Southern guinea savanna

2. Northern guinea savanna
3. Sudan savanna
4. Sahel savanna

Southern guinea savanna is the largest biome in Nigeria

Plants and animals distribution and adaptation in grasslands

Plant adaptation and distribution

The grassland plants include acacia, elephant grass, guinea grass, spear grass, palms, baobab trees etc. their adaptive features include

- Trees have thick corky barks to resist severe fire
- Grasses with underground stems to escape fire and drought
- Leaves with waxy surface in addition to cuticle covering to reduce transpiration
- Reduced or small leaves to reduce transpiration
- Presence of curly leaves to conserve water
- Leaves fall (deciduous) in drying season to conserve water
- Baobab trees have broad and succulent leaves to conserve water

Animals' distribution and adaptation in a forest

Animals found in the forest include antelopes, elephants, giraffes, zebras, goats, cattle, grasshoppers, lizards, birds, lions, tigers, leopards, rats, snakes, grass cutters, kangaroos etc. They adapt to this habitat in the following ways

- Termites lived in air conditioned nests called anthills for cooling the animals.
- Rats burrow into the soil to avoid excessive heat and fire
- Zebras and giraffes can camouflage using their colours.
- Lions, tigers and leopards have powerful claws and teeth for attacking animals.
- Kangaroos have long legs to help them escape from danger and also have pocket of flesh to shield their young ones from hot weather and attack.
- Elephants and lion move in groups or herds to achieve strength in number

Food chains in grassland

There are several food chains due to numerous animals

1. Grass → grasshoppers→ lizards → snakes
2. Grass → grasshoppers → toads → birds
3. Grass → zebras → lions

Arid lands [deserts]

These are areas of very low rainfall and high evaporation rate. They are the driest habitats, receiving less than 25cm annual rainfall. Arid lands are of two types;

- Hot deserts e.g. Sahara desert (North Africa), Kalahari desert (South Africa)
- Cold deserts e.g. desert in North America

Characteristics of a desert

1. Water is very scarce
2. Temperature is very high by the day and very low by the night
3. Vegetation is very scanty
4. The soils are sandy or rocky
5. Strong winds occur frequently and sunshine very intense
6. Presence of drought resistance plants (xerophytes)

Plants and animals distribution and adaptation in deserts

Plants distribution and adaptation in desert

Deserts plants include thorny bushes, cacti, and scattered dwarf acacia, date palm, wiring grasses, baobab trees and euphorbia species. They adapt to this habitat in the following ways;

- Plants have thin leaves to reduce transpiration
- Cacti leafless have thorns to reduce transpiration and thick succulent stem to store water
- Acacia (drought resistant) has deep roots which absorb underground water

- Baobab tree has waxy leaves which can be hairy or needle shaped to reduce the rate of transpiration
- Wiring grass has narrow and slender leaves to reduce transpiration.

Animal distribution and adaptation in deserts

The deserts animals include camel, rodents, lizards, snakes, zebras, desert tortoise, grasshoppers, wasps, ants etc. They survive in the following ways;

- Most desert animals excrete solid wastes to conserve water.
- Kangaroos, rats remain in burrows during the day to avoid excessive heat
- Reptiles have scales to reduce water loss
- Camels can survive several days without drinking water. They can withstand a wide range of body temperature up to 40oc.
- Locusts have water-proof bodies and impervious cuticles

Food chains in arid lands

1. Plants → desert rats → snakes
2. Plants → ants → scorpions → snakes

Evaluation

- Define grass land and state its characteristics
- State the types of grassland
- Describe the plants and animals distribution and adaptation in a grassland and food chain.
- Define desert and state its characteristics
- Describe the plants and animals distribution and adaptation in a grassland and food chain.

Assignment

1. List the two types of arid land with examples.

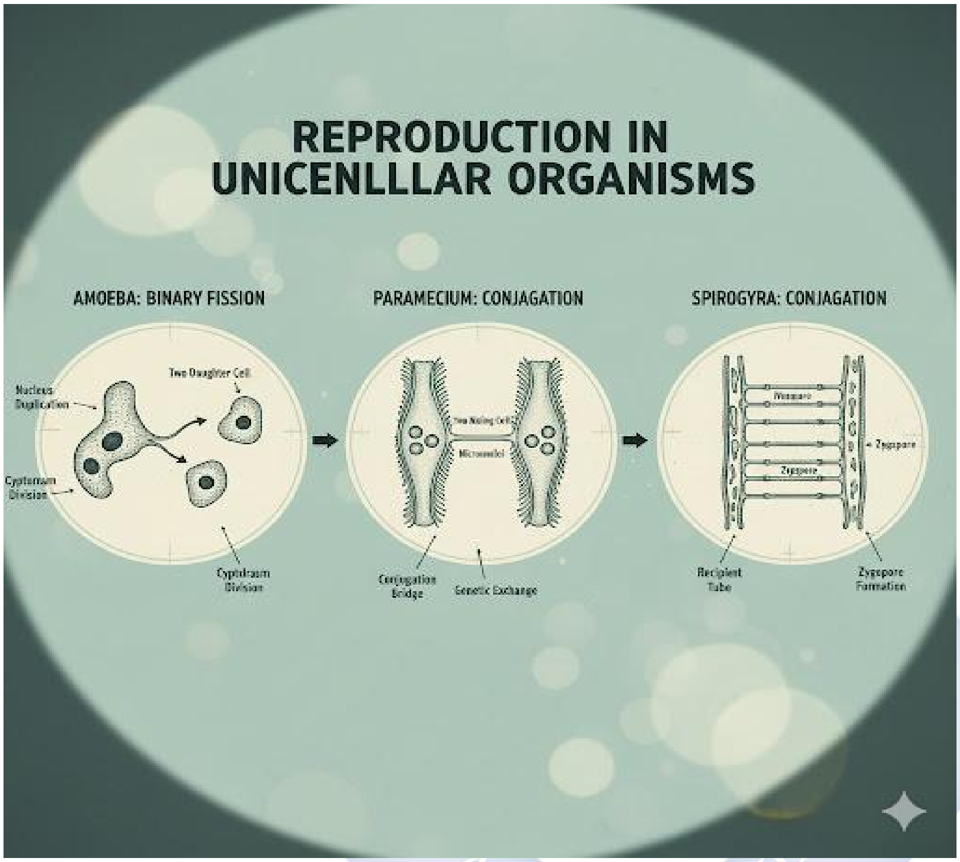
2. Mention four plants and four animals of the desert and explain how they adapt to life in this habitat.

3. List four characteristics each of (i) Trees of the tropical rain forest (ii) Trees / shrubs of the savanna.

4. State five adaptive features of animals that climb forest trees.

5. Mention four characteristics of a desert.

WEEK 9: REPRODUCTION IN UNICELLULAR ORGANISMS AND INVERTEBRATES



Meaning of reproduction:

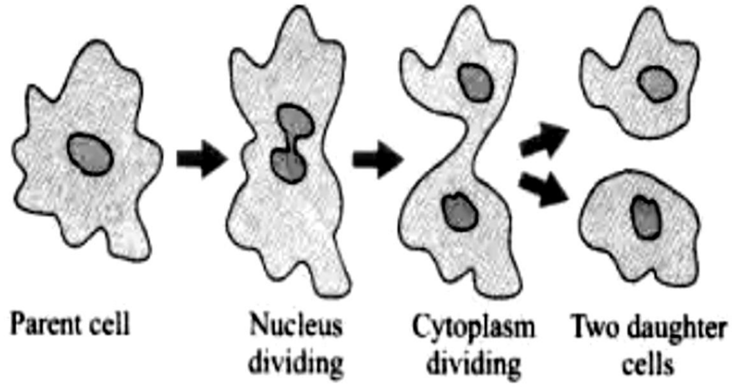
Reproduction is the process by which living organisms produce offspring (new individuals) of their type.

Types of Reproduction

1. Asexual reproduction – an individual produces an offspring by itself, that is, a single parent without production of gamete or fusion of nuclei. Closes (offspring which are identical to the parent) are produced. Asexual reproduction is common among single organisms, flowering plants and in organisms which can produce both sexually and asexually depending on certain

conditions (food availability, favorable environmental conditions for growth) the types of asexual reproduction include:

- **Binary fission**: The parent organisms simply divides into 2 or more parts by mitosis, each of which can exist by itself. If the cells divides into more than 2 cells, it is referred to as fission e.g. in bacteria and protists e.g. of Binary fission Amoeba, Paramecium.
- **Budding:** In this types, the offspring develops as an outgrowth of the parent. The bud may form an external or internal surface of the parents. Internal buds are formed in some sponges and are released when the parents dies. External buds occur in Hydra, coral polyps and yeast. These buds break off from the parent without causing any injury and lead an independent life.
- **Spore formation**: Spores are small unicellular bodies which are produced in large numbers. They are small light and dispersed by the wind. Spores are usually produced by fungi, mosses, ferns, algae, protists and bacteria.
- **Fragmentation** – a part of an organisms breaks up or fragments and give rise to a new organism. It occurs in algae, coelenterates and sponges e.g. planaria, spirogyra
- **Vegetative propagation** – occurs mainly in higher plants. A new plant grows from any portion of an old one other than the seeds e.g. stems and roots.
- **Patogenesis –** is a natural form of asexual reproduction in which growth and development of embryos occur without fertilization, that is developed from an unfertilized egg cell e.g. aphids, honey bee, ants, wasps. Some scorpions, some plants, nematodes, water flies, some mites, some fish, amphibians and reptiles.
- **Schizogony –** this is the asexual reproduction of a sporozoite by multiple fission within the body of the host giving rise to merozoites. This type is found in some protozoa especially parasitic sporozoans e.g. malaria parasite plasmodium



2. **Sexual reproduction**: Offspring are produced by the fusion of 2 different sex cells which usually come from 2 different parents.

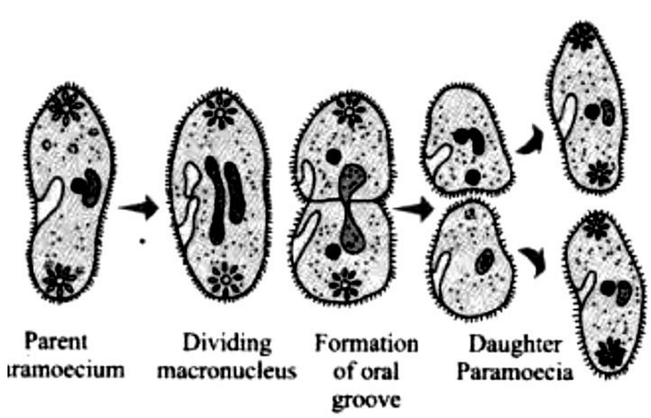
Reproduction in amoeba, paramecium and spirogyra

Amoeba

Amoeba reproduces asexually by binary fission and multiple fission (sporulation) during adverse condition.

In **binary fission** when an amoeba reaches full size, it stops moving and divides into two equal parts starting from the nucleus. This is followed by the division of the cytoplasm, after which two daughter amoebae are formed.

In **multiple fission** amoeba becomes rounded and secretes around itself a cyst. Inside the cyst, the nucleus divides several times. When conditions becomes favourable, the cyst burst; each nucleus surrounded by a part of the cytoplasm of the parent. In this way, very small amoebae are formed.



Reproduction in paramecium

Paramecium reproduces asexually by binary fission and sexually by conjugation.

Binary fission occurs under favourable conditions, the micronucleus divides into two equal halves by mitosis and each moves to the opposite side of the cell, the meganucleus elongates and the cytoplasm constricts after which two young paramecia are produced.

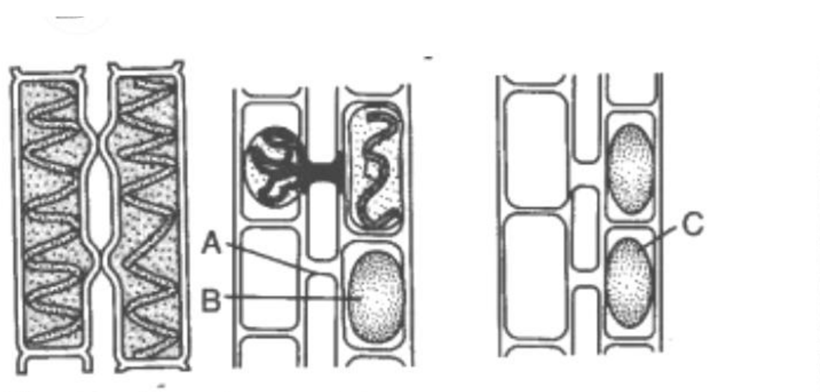
Sexual reproduction is by **conjugation** of two individuals of different lines of descent. Stages in conjugation include

1. Two matured paramecium come together and get fused by their oral grooves.
2. The micronucleus divides twice by meiosis and four nuclei are formed in each conjugant
3. The smaller micronuclei are exchanged between the two conjugants
4. The migratory micronucleus fuses with the stationary micronucleus in each conjugant to form a zygote
5. The zygote in each conjugant divides thrice to form eight nuclei
6. The ex-conjugant with four mega nuclei and four micronuclei divide to form four paramecia each having a mega nucleus and a micronucleus.

Reproduction in spirogyra

Spirogyra reproduces asexually by fragmentation and sexually by conjugation.

In fragmentation, when a filament reaches a certain length, parts of it break away and grow into new filaments



In conjugation

1. The cells of two filaments come to lie side by side and a conjugation tube is formed between them.
2. the cells in one filament act as the male gamete while the other act as the female
3. The male gamete passes through the conjugation tube to meet the female gamete and fuses to form a zygote.
4. The zygote secretes a resistant wall around itself and form a zygospore.
5. After a period of rest and favourable condition, the outer coats burst and a young filament grows out.

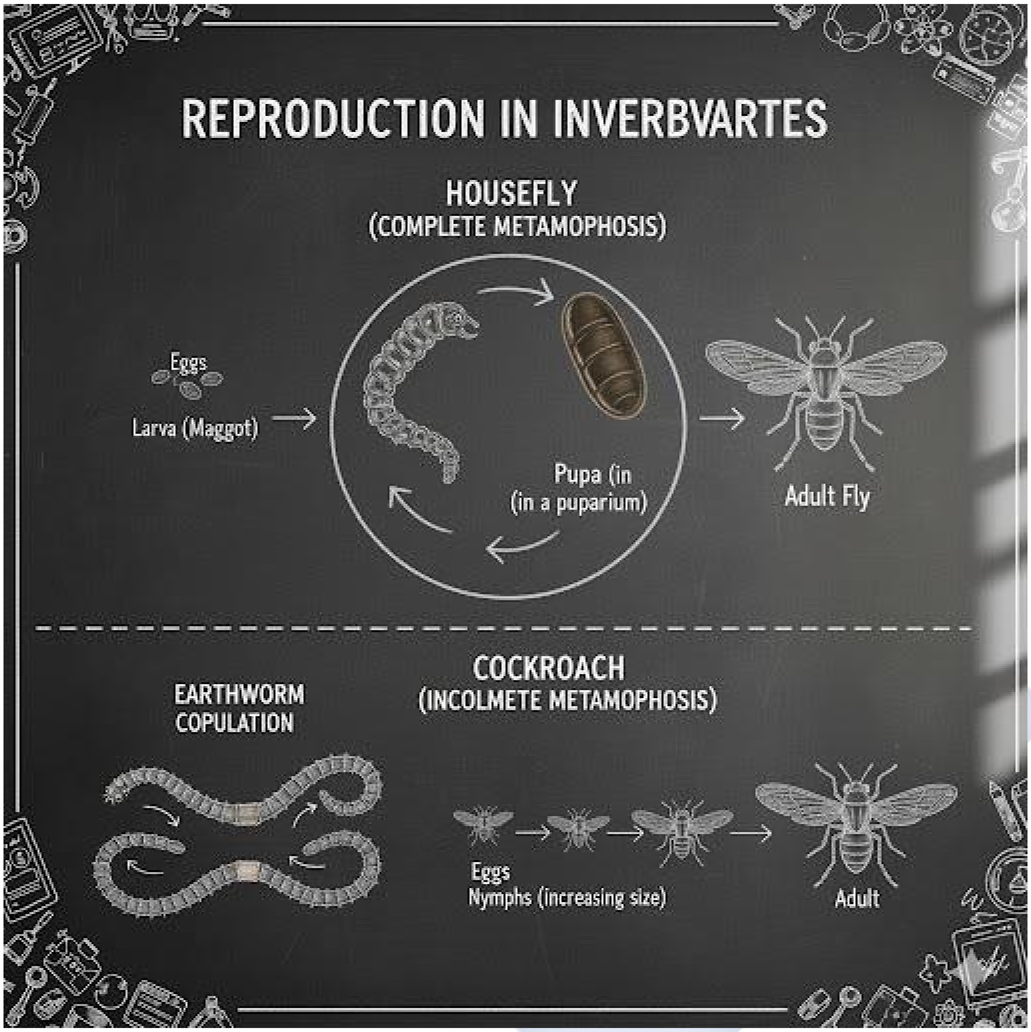
Evaluation:

1. Define reproduction
2. State the types of reproduction
3. Explain in details the types of asexual and sexual reproduction
4. Describe reproduction in Amoeba, Paramecium and spirogyra

Assignment

1. Conjugation occurs in the following organisms except A. spirogyra B. paramecium C. mucor D. hydra
2. The possession of the male and female reproductive parts by a single organism is termed A. oviparous B. meiosis C. hermaphroditism D. parthenogenesis
3. Internal fertilization takes place in the following organisms except A. earthworm B. tapeworm C. butterfly D. toad
4. The zygote of spirogyra secretes a resistant wall around itself known as A. membrane B. cyst C. clot D. zygospore
5. Paramecium reproduces asexually by A. binary fission B. meiosis C. mitosis D. budding

WEEK 10: REPRODUCTION IN UNICELLULAR ORGANISMS AND INVERTEBRATES



Reproduction in earthworm

Earthworms are hermaphrodites i. e. each has both male and female sex organs and therefore produces both male and female gametes. Reproduction is by sexual means.

Process of Copulation

Two worms to be engaged in copulation come to lie close together with their ventral surfaces touching. Copulation takes place at night outside the burrows. The reproductive organs of earthworms are anteriorly located. The worms lie in such position that the segments 9-15 of one worm are opposite the clitellum (segments 32-37) of the other and are held firmly by chaetae during copulation.

After copulation, the two worm separate. After few days eggs are laid and fertilized in a cocoon secreted by the clitellum. The development of the embryo takes place inside the cocoon and one worm hatches from a batch of eggs in one cocoon.

Step 2: Reproduction in cockroach

Sexual reproduction takes place in cockroaches and fertilization is internal. Male and female cockroaches mate and the male introduces sperm into the genital opening of the female. The sperms are then stored in the sperm pouch until the eggs are released from the two ovaries. As the eggs are released, they are fertilized by the stored sperm. Fertilized eggs are laid (about 10–16 eggs) in a horny egg case (ootheca) which the female carries in her abdominal pouch for some time and later deposit it in a safe dark place. After 30 – 100 days, the eggs hatch into nymphs which are wingless, small and whitish in colour.

The nymphs feed, grow, and become brown, moult about 13 times to become adults.

In the process of moulting, the wings first appear as wing pad and later develop into full grown wings. Cockroaches require 11 – 20 months to develop from eggs to imagos. Metamorphosis is incomplete.

Reproduction in housefly

Adult male and female mates and within two to three days fertilized eggs are laid. The laying of eggs takes place in the day light. Housefly undergoes complete metamorphosis.

2 – 7 batches of eggs (100 – 150 eggs in a batch) are laid by the female housefly in a moist dirty environment. The eggs hatch into white larvae in about 8 – 24 hours.

The larva called **maggot** has a segmented body. The head bears a pair of hook for tearing food and drawing the larva along. On the ventral surface of the segmented body lie spiny pads for movement. It has two pairs of spiracles for breathing. The larva moults several times and lasts for about 5 – 14 days after which it moves to a dry place to begin the pupal stage.

The maggot shortens; its skin becomes hard and brown forming the pupal case (puparium). It does not feed or move. Internal re-organisation takes place at this stage. In about 3 – 10 days, the young adult hatches out of the puparium.

The adult housefly called imago emerges from the puparium using a sac-like organ (ptilinum) to break it open. It moves to the surface of the dirt and flies away when the wings are dry.

Reproduction in snail

Reproduction in land snails is hermaphroditic and fertilization internal. The female snail has a fertilization pouch for sperm to travel into. The snails will transfer their spermatophores to a place called **epiphallus**. The epiphallus is part of the sperm duct to the penis to help put the spermatophores into place by using their flagellum. From here, sperm is travelled to the **bursa duct** where fertilization takes place. During snail development, there is a 180^{o} twist of the visceral mass that brings the anus and the mantle cavity forward to a position above the head. This process is known as **torsion.**

Evaluation:

- Explain reproduction in the following organisms

i. Earthworm
ii. Cockroach
iii. Housefly
iv. Snails

Assignment:

1. Describe sexual reproduction in tapeworm.
2. What are the advantages and disadvantages of sexual reproduction?